Quantum Information Theory



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These lecture notes are based on the *Quantum Information Theory* course given at EPFL for the academic year 2024/2025 by Prof. Zoë Holmes. In these notes, any Proposition/Lemma/Example/Proof/... with a $(*)$ are taken from the exercise sheets.

Chapter 1

State Space of Classical Systems

Consider a system S (e.g. S=ball) with property P and $P = \{\text{red}, \text{blue}, \text{green}\}$ its colour. If its colour is not known with certainty the state is just a description of the different colours:

$$\underline{r} = \Big(\mathbb{P}(\text{red}), \mathbb{P}(\text{blue}), \mathbb{P}(\text{green})\Big).$$

Example 1.0.1. *Consider the following states:*

$$\underline{r} = (1,0,0) \Longrightarrow \textit{ ball is red}, \quad \underline{r} = (1/3,1/3,1/3) \Longrightarrow \textit{ maximum uncertainty.}$$

$$\underline{r} = (1/2,1/2,0) \Longrightarrow \textit{ certain it is not green but no idea if it is red or blue.}$$

The states of perfect knowledge (zero uncertainty) are called pure states. We label them:

$$\underline{e_0} = (1,0,0), \quad \underline{e_1} = (0,1,0), \quad \underline{e_2} = (0,0,1).$$

Definition 1.0.2. *A **convex combination** of vectors $\underline{x}$ and $\underline{y}$ is given by, for $p \in [0,1]$:*

$$\underline{z} = p\underline{x} + (1-p)\underline{y}. \tag{1.1}$$

In fact, any state can be written as a convex combination of basis vectors:

$$\underline{r} = (r_0, r_1, r_2) = r_0\underline{e_0} + r_1\underline{e_1} + r_2\underline{e_2}, \tag{1.2}$$

where $r_0 + r_1 + r_2 = 1$ and $r_i \geq 0$. So, the state space of the system which has 3 possible properties is a triangle with the basis states as its vertices.
The state space of a 2-properties system is a line.
The state space of a 4-properties system is a tetrahedron.

Definition 1.0.3. *The **classical states** of a system S are given by $\underline{r} = (r_0, r_1, ..., r_{n-1})$ for some n, such that:*
(1) Non-negativity $\longrightarrow r_i \geq 0$.
(2) Normalisation $\longrightarrow \sum_i r_i = 1$.

The state space is an n-dimensional simplex over these states.

Definition 1.0.4. *Given a classical system S with a total of n pure states, any* ***classical measurement*** *$\mathcal{M}$ with d outcomes is described by a collection of vectors $(\underline{m}_0, \underline{m}_1, ..., \underline{m}_{d-1})$, such that:*
(1) Non-negativity $\longrightarrow$ For any k, the components of $\underline{m}_k$ are non-negative.
(2) Normalisation $\longrightarrow$ $\sum_k \underline{m}_k = (1, 1, ..., 1)$.
The probability of getting outcome k in the state $\underline{r}$ is given by:

$$\mathbb{P}(k) = \underline{m}_k \cdot \underline{r}, \tag{1.3}$$

where the vectors $\{\underline{m}_k\}$ are of length n.

It turns out that in order to respect probability theory, any possible transformation of the system must be presented by a linear transformation $\underline{r} \longrightarrow L\underline{r} = \underline{s}$. This linearity is not the same linearity as in classical mechanics, but it instead relates to respecting convex mixtures of states so that $L(p\underline{x} + (1-p)\underline{y}) = pL(\underline{x}) + (1-p)L(\underline{y})$. If this didn't hold then it turns out that it would be possible to signal faster than light!

Chapter 2

Quantum Theory

In this chapter, we adapt the notions of the previous section to a quantum theoretic framework. In particular: replace any vector $\underline{x}$ in classical theory by a Hermitian matrix X. Non-negativity is in terms of the eigenvalues of the matrices (i.e. $\text{eigs}(X) \geq 0$). For the normalisation, the $\sum_i x_i$ becomes $\text{Trace}(X) = 1$ with $(1,1,...,1) \longrightarrow \mathbb{I}$, where $\mathbb{I}$ is the identity matrix. Finally replace the dot product $\underline{x} \cdot \underline{y}$ by $\text{Trace}(XY)$.
We can now write down the general description of quantum theory.

Definition 2.0.1. *Given a quantum system S, a **quantum state** is any Hermitian matrix ρ such that:*
(1) Non-negativity $\longrightarrow$ $\text{eigs}(\rho) \geq 0$.
(2) Normalisation $\longrightarrow$ $\text{Tr}(\rho) = 1$.

The dimension of the quantum system S is d if ρ is a $d \times d$ matrix. We refer to ρ as the density matrix or the density operator for S.

Definition 2.0.2. *Given a quantum system S of dimension d, any general **quantum measurement** $\mathcal{M}$ on S with m outcomes is described by a set of matrices $\mathcal{M} = (M_0, M_1, ..., M_{m-1})$, such that:*
(1) Non-negativity $\longrightarrow$ For any $i = \{0,1,...,m-1\}$, $\text{eigs}(M_i) \geq 0$.
(2) Normalisation $\longrightarrow$ $\sum_i M_i = \mathbb{I}$,
where $\mathbb{I}$ is the identity matrix. Given a quantum system ρ, the probability of the k-th outcome of doing the measurement $\mathcal{M}$ on the state is given by:

$$\mathbb{P}(k) = \text{Tr}(M_k \rho). \tag{2.1}$$

Definition 2.0.3. *Given a quantum system S in a state ρ, any **quantum evolution** is described by a linear transformation $\rho \longrightarrow \mathcal{E}(\rho)$ such that $\sigma = \mathcal{E}(\rho)$ is always another valid quantum state.*

2.1 The State Space of a Qubit

A qubit corresponds to a quantum system of dimension 2. For this, satisfying $a + b = 1$ with 3 degrees of freedom, we are looking for ρ in the form:

$$\rho = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix} = \begin{pmatrix} a & c \\ c^* & 1-a \end{pmatrix}. \tag{2.2}$$

Now, any 2×2 Hermitian matrix can be expanded in a Pauli basis $\{\mathbb{I},\sigma_x,\sigma_y,\sigma_z\}$:

$$\mathbb{I}=\begin{pmatrix}1&0\\0&1\end{pmatrix},\quad \sigma_x=\begin{pmatrix}0&1\\1&0\end{pmatrix},\quad \sigma_y=\begin{pmatrix}0&-i\\i&0\end{pmatrix},\quad \sigma_z=\begin{pmatrix}1&0\\0&-1\end{pmatrix}. \tag{2.3}$$

Therefore, in this basis:

$$\rho=a_0(\mathbb{I}+\underline{r}\cdot\underline{\sigma})=a_0(\mathbb{I}+x\sigma_x+y\sigma_y+z\sigma_z). \tag{2.4}$$

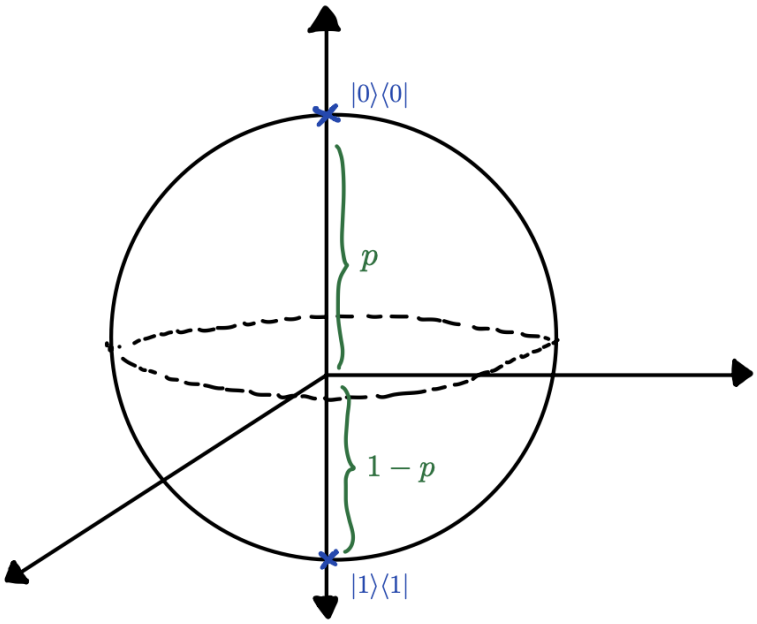
For the normalisation condition, we require:

$$\mathrm{Tr}(\rho)=a_0\mathrm{Tr}(\mathbb{I})=2a_0=1\Longrightarrow a_0=\frac{1}{2}. \tag{2.5}$$

Therefore, any state ρ is written in the form:

$$\rho=\frac{1}{2}\begin{pmatrix}1+z&x-iy\\x+iy&1-z\end{pmatrix}, \tag{2.6}$$

where the eigenvalues of ρ are $\mathrm{eigs}(\rho)=\frac{1}{2}\left(1\pm\sqrt{x^2+y^2+z^2}\right)=\frac{1}{2}\left(1\pm|\underline{r}|\right)\geq 0$. This implies in particular that $|\underline{r}|\leq 1$. Now, any $\underline{r}$ satisfying $|\underline{r}|\leq 1$ specifies a unique quantum state. Therefore, the state space of a 2-level quantum system is a ball of unit radius.



Example 2.1.1. *For* $\underline{r}=\begin{pmatrix}0\\0\\\pm 1\end{pmatrix}$ *we get* $\rho_\pm=\frac{1}{2}\left(\begin{pmatrix}1&0\\0&1\end{pmatrix}\pm\begin{pmatrix}1&0\\0&-1\end{pmatrix}\right)$.

$$\rho_+=\begin{pmatrix}1&0\\0&0\end{pmatrix}=\begin{pmatrix}1\\0\end{pmatrix}\otimes\begin{pmatrix}1&0\end{pmatrix}=|0\rangle\langle 0|\quad or,\quad \rho_-=\begin{pmatrix}0&0\\0&1\end{pmatrix}=\begin{pmatrix}0\\1\end{pmatrix}\otimes\begin{pmatrix}0&1\end{pmatrix}=|1\rangle\langle 1|. \tag{2.7}$$

Therefore:

$$\rho_p^{classical}=p|0\rangle\langle 0|+(1-p)|1\rangle\langle 1|=\begin{pmatrix}p&0\\0&1-p\end{pmatrix}. \tag{2.8}$$

Given any two states σ_0 represented $\underline{r}_0$ and σ_1 represented $\underline{r}_1$ we can always form:

$$\rho=p\sigma_0+(1-p)\sigma_1\qquad \text{for, } 0\leq p\leq 1. \tag{2.9}$$

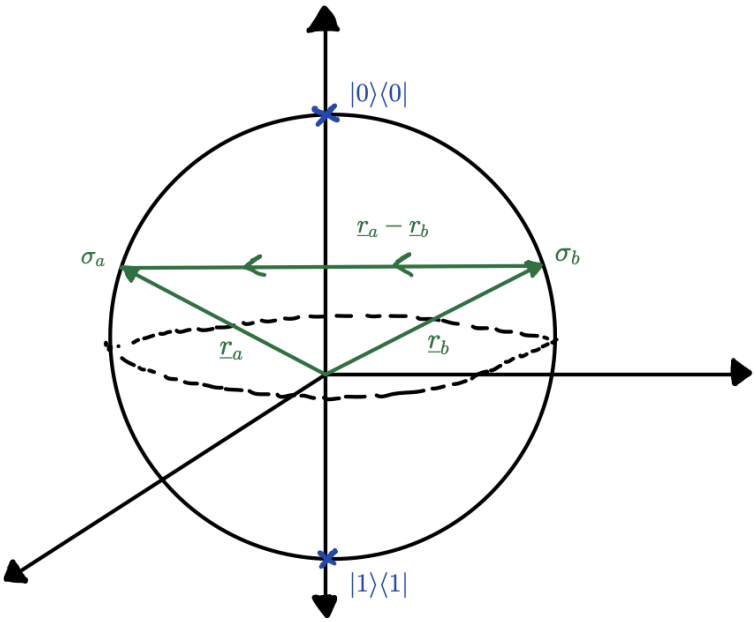
This is called an ensemble decomposition. Operationally, it can be understood as preparing state σ_0 with probability p and σ_1 with probability $1-p$. In terms of block vectors it yields:

$$\sigma_a = \frac{1}{2}\left(\mathbb{I} + \underline{r}_a \cdot \underline{\sigma}\right) \quad \text{and,} \quad \sigma_b = \frac{1}{2}\left(\mathbb{I} + \underline{r}_b \cdot \underline{\sigma}\right). \tag{2.10}$$

Then:

$$\rho = \frac{1}{2}\left(p\mathbb{I} + (1-p)\mathbb{I} + \left(p\underline{r}_a + (1-p)\underline{r}_b\right)\cdot\underline{\sigma}\right) = \frac{1}{2}\left(\mathbb{I} + \underline{r'}\cdot\underline{\sigma}\right), \tag{2.11}$$

where $\underline{r'} = p\underline{r}_a + (1-p)\underline{r}_b = \underline{r}_b + p\left(\underline{r}_a - \underline{r}_b\right)$.



2.2 Pure and Mixed States

Pure states are those that cannot be formed by mixing two other states.

Proposition 2.2.1. *ρ is a pure state if and only if $\rho \neq p\sigma_a + (1-p)\sigma_b$ for any two different states σ_a and σ_b, where $0 < p < 1$.*

Any state that is not pure is said to be mixed. Pure states must therefore live on the edge of the Bloch sphere with $|\underline{r}| = 1$. And one can in fact show that $\rho = \frac{1}{2}(\mathbb{I} - \underline{r}\cdot\underline{\sigma}) \equiv |\psi\rangle\langle\psi|$ with $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|\alpha|^2 + |\beta|^2 = 1$.

Proposition 2.2.2. *A state ρ is pure if and only if $\rho = \rho^2$ (i.e. $\mathrm{Tr}(\rho^2) = \mathrm{Tr}(\rho) = 1$).*

2.3 Unitary Evolutions

We have encountered quantum evolutions before in the context of Schrödinger's equation. Indeed, for an equation of the form:

$$i\hbar\frac{\partial|\psi\rangle}{\partial t} = H|\psi\rangle, \tag{2.12}$$

it gives us the so-called (unitary) *evolution operator* $U = e^{-iHt}$:

$$|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle. \tag{2.13}$$

This is furthermore a reversible evolution $U^\dagger(U|\psi\rangle) = |\psi\rangle$ and also length preserving $\langle\phi|\phi\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle$ for $|\phi\rangle = U|\psi\rangle$. In quantum information we often abstract away from the underlying Hamiltonian and think entirely in terms of unitary operations applied to quantum states.

2.3.1 Single Qubit Gates

We first consider simple classical gates that may be adapted to a single qubit. The *identity gate* taking $|0\rangle \longrightarrow |0\rangle$ and $|1\rangle \longrightarrow |1\rangle$ is just given by $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. We can also mention the *NOT gate* taking $|0\rangle \longrightarrow |1\rangle$ and $|1\rangle \longrightarrow |0\rangle$ yielding the transformation $\text{NOT} = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. We may now mention a pure quantum gate called the *Hadamard gate* given by:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{2.14}$$

This transformation acts on basis states as follows:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \big(|0\rangle + |1\rangle \big) =: |+\rangle . \tag{2.15}$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \big(|0\rangle - |1\rangle \big) =: |-\rangle . \tag{2.16}$$

It is helpful to know how a unitary on a single qubit affects a state on a Bloch sphere. Let $\rho = \frac{1}{2}\big(\mathbb{I} + \underline{r} \cdot \underline{\sigma}\big)$ and consider $\rho_{\text{out}} = U\rho U^\dagger = \frac{1}{2}\big(\mathbb{I} + U(\underline{r} \cdot \underline{\sigma})U^\dagger\big) \equiv \frac{1}{2}\big(\mathbb{I} + \underline{s} \cdot \underline{\sigma}\big)$.

Proposition 2.3.1. *For an orthogonal matrix O (i.e. $OO^T = O^T O = \mathbb{I}$), $\underline{s} = O\underline{r}$.*

Proof. To show that O is orthogonal we just need to prove that $|\underline{r}| = |\underline{s}|$. To do so, we note the following:

$$\text{Tr}\left(\big(\underline{s} \cdot \underline{\sigma} \big)^2 \right) = \text{Tr}\left(\left(\sum_i s_i \sigma_i \right)^2 \right) = \text{Tr}\left(\sum_{ij} s_i s_j \sigma_i \sigma_j \right) = \sum_{ij} s_i s_j \delta_{ij} = \sum_i s_i^2 = |\underline{s}|^2.$$

Now we remark that:

$$\begin{aligned} |\underline{s}|^2 = \text{Tr}\left(\big(\underline{s} \cdot \underline{\sigma} \big)^2 \right) = \text{Tr}\left(\big(U\underline{r} \cdot \underline{\sigma} U^\dagger \big)^2 \right) = \text{Tr}\left(U \big(\underline{r} \cdot \underline{\sigma} \big)^2 U^\dagger \right) = \text{Tr}\left(UU^\dagger \big(\underline{r} \cdot \underline{\sigma} \big)^2 \right) = \text{Tr}\left(\big(\underline{r} \cdot \underline{\sigma} \big)^2 \right) \\ = \sum_{ij} \text{Tr}\left(r_i r_j \sigma_i \sigma_j \right) \\ = \sum_{ij} r_i r_j \delta_{ij} = \sum_i r_i^2 = |\underline{r}|^2. \end{aligned}$$

In conclusion we have that $|\underline{r}| = |\underline{s}|$, implying that the matrix O is orthogonal.

□

To get a better handle on the nature of this rotation we may note that any unitary on a qubit can be written as follows:

$$U = e^{i\theta \underline{n} \cdot \underline{\sigma}}. \tag{2.17}$$

This is equivalent to the unitary:

$$U = e^{-i\theta \underline{n} \cdot \underline{\sigma}} = \cos(\theta)\mathbb{I} - i\sin(\theta)\underline{n} \cdot \underline{\sigma}. \tag{2.18}$$

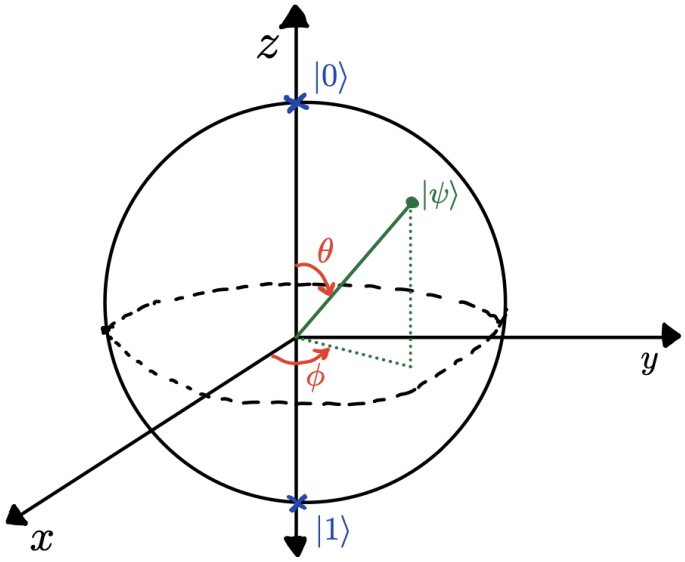
The relation (2.17) can be understood by noting that any unitary can be written as $U = e^{-iHt}$ where H can always be expanded in the Pauli basis (H is a Hamiltonian).
Now, let us look at the effect of evolving $|0\rangle\langle 0|$ under $e^{-i\theta\sigma_x}$:

$$e^{-i\theta\sigma_x}|0\rangle = \cos(\theta)|0\rangle - i\sin(\theta)\sigma_x|0\rangle = \cos(\theta)|0\rangle - i\sin(\theta)|1\rangle. \tag{2.19}$$

Proposition 2.3.2. *We may represent a state $|\psi\rangle$ on the Bloch sphere by:*

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \tag{2.20}$$

where the range of the latitude is given by $0 \leq \theta \leq \pi$ and the longitude by $0 \leq \phi < 2\pi$.



2.3.2 Two-Qubit Gates

In this subsection we present two gates for the study of two qubits.

- We can simply apply multiple copies of a gate onto systems in parallel. For example:

$$X^{\otimes 2}|0\rangle^{\otimes 2} := (X\otimes X)(|0\rangle\otimes|0\rangle) = X|0\rangle\otimes X|0\rangle = |1\rangle\otimes|1\rangle. \tag{2.21}$$

- We may consider the CNOT gate, which is the fact of applying a NOT gate to the second qubit if and only if the first qubit is in state $|1\rangle$. We have the following transformations: $\mathrm{CNOT}|00\rangle = |00\rangle$, $\mathrm{CNOT}|01\rangle = |01\rangle$, $\mathrm{CNOT}|10\rangle = |11\rangle$ and $\mathrm{CNOT}|11\rangle = |10\rangle$. Its explicit matrix form is given by:

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{2.22}$$

The interesting fact about this gate is that it can generate entanglement. For example we may consider the following sequence of gates:

$$(\mathrm{CNOT})(H\otimes\mathbb{I})|00\rangle = \mathrm{CNOT}|+0\rangle = \mathrm{CNOT}\frac{1}{\sqrt{2}}(|00\rangle+|10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle+|11\rangle). \tag{2.23}$$

This final result are the so-called *Bell states* $\left|\phi^{\pm}\right\rangle = \frac{1}{\sqrt{2}}\big(\,|00\rangle \pm |11\rangle\,\big)$ which corresponds to qubits perfectly correlated each with probability $\frac{1}{2}$. We also have $\left|\psi^{\pm}\right\rangle = \frac{1}{\sqrt{2}}\big(\,|01\rangle \pm |10\rangle\,\big)$ which corresponds to qubits perfectly anti-correlated each with probability $\frac{1}{2}$.

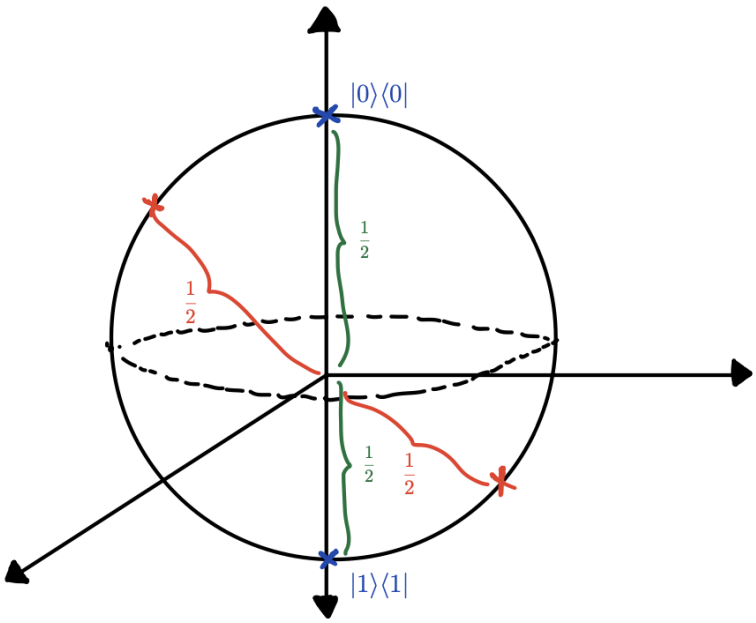
Chapter 3

Ensembles and Purifications of Quantum States

In this section, we first introduce a famous paradox concerning the superposition of two states. We will then follow on the study of multiple quantum systems along with the definition of the partial trace. We finally end on some topics of purifications of a mixed quantum state.

3.1 Ensemble Ambiguity Paradox

Consider the equal mixture of two states, $\rho_+ = \frac{1}{2}(\mathbb{I} + \underline{r}\cdot\underline{\sigma})$ and $\rho_- = \frac{1}{2}(\mathbb{I} - \underline{r}\cdot\underline{\sigma})$, and set $\rho = \frac{1}{2}\rho_+ + \frac{1}{2}\rho_- = \frac{1}{2}\mathbb{I}$. This last result is the same for any $\underline{\sigma}$. On the Bloch sphere this gives:



The ambiguity comes from the definition of seemingly two different states. Define $\rho = \frac{1}{2}\mathbb{I} = \frac{1}{2}\big(|0\rangle\langle 0| + |1\rangle\langle 1|\big)$ and $\rho = \frac{1}{2}\mathbb{I} = \frac{1}{2}\big(|+\rangle\langle +| + |-\rangle\langle -|\big)$, with $|\pm\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle \pm |1\rangle\big)$. Assume that the states are $|0\rangle =$ *alive cat* and $|1\rangle =$ *dead cat*. Then, the state $\frac{1}{2}\big(|0\rangle\langle 0| + |1\rangle\langle 1|\big)$ corresponds to the cat being alive with probability $\frac{1}{2}$ and dead with probability $\frac{1}{2}$. However, the state $\frac{1}{2}\big(|+\rangle\langle +| + |-\rangle\langle -|\big)$ corresponds to the superposition of being dead and alive at the same time. Overall, both are equal yet their preparations are very different in nature and in difficulty (much easier to be dead or alive, rather than being in both states at the same time).

3.2 Multiple Quantum Systems and the Partial Trace

In this section, we seek to understand what it means to have a state space of an n-qubits system. In our case, we will consider a d-dimensional system with $d = 2^n$. We will consider arbitrary $d \times d$ complex matrices that have $2d^2$ parameters, and also arbitrary $d \times d$ Hermitian matrices with $d + 2\left(\frac{d^2-d}{2}\right) = d^2$ parameters. With the extra fact that $\mathrm{Tr}(\rho) = 1$ it implies that ρ has $(d^2 - 1)$ degrees of freedom. Therefore, the state space is $(d^2 - 1)$-dimensional.
Often it is helpful to think of a higher dimensional system as the composition of lower dimensional ones:

$$\begin{matrix}\rho \ d-\text{dim.} \\ 2^n-\text{dim.}\end{matrix} \longrightarrow \rho_{AB}, \tag{3.1}$$

where A is $\{d_A, 2^{n_A}\}$-dimensional and B is $\{d_B, 2^{n_B}\}$-dimensional, with $d_A + d_B = d$ and $n_A + n_B = n$. In fact the story usually goes the other way, we identify subsystems then consider a joint system:

$$\rho_A, \rho_B \longrightarrow \rho_A \otimes \rho_B = \rho_{AB}. \tag{3.2}$$

Now, we may consider states that capture classical correlations and are of the form:

$$\rho_{AB} = \sum_i p_i \rho_{A_i} \otimes \rho_{B_i}, \tag{3.3}$$

with p_i being the associated probability of preparing A in state ρ_{A_i} and preparing B in state ρ_{B_i}.

Definition 3.2.1. *An **entangled state** is a state ρ_{AB} such that $\rho_{AB} \neq \sum_i p_i \rho_{A_i} \otimes \rho_{B_i}$ for some states ρ_{A_i}, ρ_{B_i} with $0 < p_i < 1$.*

Example 3.2.2. *The state $\rho_{AB} = \left|\phi^+\right\rangle\left\langle\phi^+\right|$ is an entangled state, where $\left|\phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$.*

Now, suppose you have ρ_{AB} and want to understand properties of A and B separately. In other words, say we want an object that quantifies all measurement outcomes on A alone $\longrightarrow$ this is captured by reduced states. Before formally defining the notion of reduced states, we first define the partial trace.

Definition 3.2.3. *The **partial trace** $\mathrm{Tr}_B(\cdot)$ is defined as:*

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) := \sum_k \left(\mathbb{I}_A \otimes \langle k|_B\right)\rho_{AB}\left(\mathbb{I}_A \otimes |k\rangle_B\right). \tag{3.4}$$

Definition 3.2.4. *For any bipartite state ρ_{AB}, the **reduced state** on A is formed by taking the partial trace $\rho_A = \mathrm{Tr}_B(\rho_{AB})$.*

Let's now see why that is. It turns out that the partial trace is the unique operation which gives rise to the correct observable quantities for subsystems. Suppose we are only able/interested in measuring A, i.e. $\widetilde{M}_{AB} = M_A \otimes \mathbb{I}_B$. We then want a state ρ_A satisfying $\mathrm{Tr}(M_A \rho_A) = \mathrm{Tr}\left(\widetilde{M}_{AB}\rho_{AB}\right)$.

Proposition 3.2.5. *The unique mathematical entity satisfying $\mathrm{Tr}(M_A\rho_A) = \mathrm{Tr}\left(\widetilde{M}_{AB}\rho_{AB}\right)$ is $\rho_A = \mathrm{Tr}_B(\rho_{AB})$.*

Proof. By direct calculation, we have:

$$\begin{aligned}\mathrm{Tr}_A\left(M_A \mathrm{Tr}_B(\rho_{AB})\right) &= \mathrm{Tr}_A\left(M_A \sum_k \big(\mathbb{I}\otimes\langle k|\big)\rho_{AB}\big(\mathbb{I}\otimes|k\rangle\big)\right)\\ &= \sum_k \mathrm{Tr}_A\Big(M_A\big(\mathbb{I}\otimes\langle k|\big)\rho_{AB}\big(\mathbb{I}\otimes|k\rangle\big)\Big)\\ &= \sum_{k,k'} \big\langle kk'\big|\left(M_A\otimes\mathbb{I}\right)\rho_{AB}\big|kk'\big\rangle\\ &= \mathrm{Tr}\left(\widetilde{M}_{AB}\rho_{AB}\right),\end{aligned}$$

where the second to the third step is a result of simply expanding the trace. The proof that the partial trace is the unique object for this condition is quite technical, and it can be widely found in the general literature.

□

Intuition behind the partial trace: Consider a d_A-dimensional system, and expand M_A with a spectral decomposition:

$$M_A = \sum_{i=1}^{d_A} \lambda_i |\lambda_i\rangle\langle\lambda_i|. \tag{3.5}$$

Now consider the average of M_A to be:

$$\langle M_A\rangle = \sum_{i=1}^{d_A}\lambda_i \mathbb{P}(A=\lambda_i) = \sum_{i=1}^{d_A}\sum_{j=1}^{d_B}\lambda_i\mathbb{P}(A=\lambda_i, B=j), \tag{3.6}$$

where $\mathbb{P}(A=\lambda_i, B=j)$ is the joint probability of finding A in state $|\lambda_i\rangle$ and B in some state $|j\rangle$. In other words:

$$\mathbb{P}(A=\lambda_i, B=j) = \langle\lambda_i, j|\,\rho_{AB}\,|\lambda_i, j\rangle. \tag{3.7}$$

Hence:

$$\begin{aligned}\langle M_A\rangle = \sum_{i=1}^{d_A}\sum_{j=1}^{d_B}\lambda_i\langle\lambda_i,j|\rho_{AB}|\lambda_i,j\rangle &= \sum_{i=1}^{d_A}\lambda_i\langle\lambda_i|_A \underbrace{\left(\sum_{j=1}^{d_B}\big(\mathbb{I}\otimes\langle j|\big)\rho_{AB}\big(\mathbb{I}\otimes|j\rangle\big)\right)}_{\rho_A=\mathrm{Tr}_B(\rho_{AB})}|\lambda_i\rangle_A\\ &= \sum_{i=1}^{d_A}\lambda_i\langle\lambda_i|\rho_A|\lambda_i\rangle = \mathrm{Tr}(M_A\rho_A). \qquad \left(=\mathrm{Tr}\left(\sum_i\lambda_i|\lambda_i\rangle\langle\lambda_i|\rho_A\right)\right)\end{aligned}$$

Recap: In the end, the reduced state ρ_A that captures all local information about system A can be computed using the partial trace. Writing $|ij\rangle = |i\rangle_A\otimes|j\rangle_B$ and $\rho_{AB} = \sum\alpha_{ij,i'j'}|ij\rangle\langle i'j'|$, recalling also that $\langle k|j\rangle = \delta_{jk}$, let us review one last time how it works:

$$\begin{aligned}\rho_A = \mathrm{Tr}_B(\rho_{AB}) &= \sum_k\big(\mathbb{I}_A\otimes\langle k|_B\big)\rho_{AB}\big(\mathbb{I}_A\otimes|k\rangle_B\big)\\ &= \sum_k\sum_{ij,i'j'}\alpha_{ij,i'j'}\big(\mathbb{I}_A\otimes\langle k|_B\big)|ij\rangle\big\langle i'j'\big|\big(\mathbb{I}_A\otimes|k\rangle_B\big)\\ &= \sum_{ij,i'j'}\alpha_{ij,i'j'}|i\rangle\big\langle i'\big|\underbrace{\sum_k\langle k|\big(|j\rangle\big\langle j'\big|\big)|k\rangle}_{=\mathrm{Tr}_B\left(|j\rangle\langle j'|_B\right)}\\ &= \sum_{ij,i'j'}\alpha_{ij,i'j'}|i\rangle\big\langle i'\big|\underbrace{\mathrm{Tr}_B\big(|j\rangle\big\langle j'\big|_B\big)}_{\langle j'|j\rangle}.\end{aligned}$$

Example 3.2.6. *Consider the state* $\rho = |\phi^+\rangle\langle\phi^+| = \frac{1}{2}\big(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|\big)$ *where* $|\phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$ *then the reduced state is:*

$$\rho_A = \mathrm{Tr}_B\big(|\phi^+\rangle\langle\phi^+|\big) = \frac{1}{2}\Big(|0\rangle\langle 0|\underbrace{\mathrm{Tr}\big(|0\rangle\langle 0|\big)}_{\langle 0|0\rangle=1} + |1\rangle\langle 0|\underbrace{\mathrm{Tr}\big(|1\rangle\langle 0|\big)}_{\langle 0|1\rangle=0} + |1\rangle\langle 1|\underbrace{\mathrm{Tr}\big(|1\rangle\langle 1|\big)}_{\langle 1|1\rangle=1} + |0\rangle\langle 1|\underbrace{\mathrm{Tr}\big(|0\rangle\langle 1|\big)}_{\langle 1|0\rangle=0}\Big)$$
$$= \frac{1}{2}\mathbb{I}.$$

Example 3.2.7. *Consider the state* $\rho_{AB} = \frac{1}{4}\mathbb{I} = \frac{1}{4}\big(|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + |11\rangle\langle 11|\big)$*, then the reduced state is:*

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \frac{1}{4}\Big(|0\rangle\langle 0| + |0\rangle\langle 0| + |1\rangle\langle 1| + |1\rangle\langle 1|\Big) = \frac{1}{2}\mathbb{I}.$$

There are many global system states on A and B consistent with the same reduced state on A.

3.3 Purifications of a Mixed Quantum State

Consider a generic mixed state ρ_A of dimension d_A. There exists a state $|\psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ such that:

$$\mathrm{Tr}_B\big(|\psi\rangle_{AB}\langle\psi|_{AB}\big) = \rho_A. \tag{3.8}$$

Proposition 3.3.1. *Given any state* ρ_A *and considering its eigen-decomposition:*

$$\rho_A = \sum_{k=0}^{d_A-1} \lambda_k |\lambda_k\rangle\langle\lambda_k|, \tag{3.9}$$

a purification is given by:

$$|\psi\rangle_{AB} = \sum_{k=0}^{d_A-1} \sqrt{\lambda_k}\,|\lambda_k\rangle \otimes |k\rangle. \tag{3.10}$$

Proof. By direct calculation, we obtain:

$$\mathrm{Tr}_B\big(|\psi\rangle_{AB}\langle\psi|_{AB}\big) = \mathrm{Tr}_B\left(\sum_{k,k'} \sqrt{\lambda_k}\sqrt{\lambda_{k'}}\,|\lambda_k k\rangle\langle\lambda_{k'}k'|\right) = \sum_{k,k'} \sqrt{\lambda_k}\sqrt{\lambda_{k'}}\,|\lambda_k\rangle\langle\lambda_{k'}|\Big(Tr\big(|k\rangle\langle k'|\big)\Big)$$
$$= \sum_{k,k'} \sqrt{\lambda_k}\sqrt{\lambda_{k'}}\,|\lambda_k\rangle\langle\lambda_{k'}|\,\delta_{kk'}$$
$$= \sum_k \lambda_k |\psi_k\rangle\langle\psi_k| = \rho_A.$$

□

Example 3.3.2. • *The purification of the state* $\rho_A = \frac{1}{2}\big(|0\rangle\langle 0| + |1\rangle\langle 1|\big)$ *is given by* $\rho_A = \mathrm{Tr}_B\big(|\phi^+\rangle\langle\phi^+|\big)$*, where* $|\phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$*.*

• *The purification of the state* $\rho_A = \frac{1}{4}|0\rangle\langle 0| + \frac{3}{4}|1\rangle\langle 1|$ *is given by* $\rho_A = \mathrm{Tr}_B\big(|\psi\rangle\langle\psi|\big)$*, where* $|\psi\rangle = \frac{1}{\sqrt{4}}|00\rangle + \sqrt{\frac{3}{4}}|11\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$*. In fact, the state* $\frac{1}{2}|0\psi\rangle + \frac{\sqrt{3}}{2}|1\phi\rangle$ *is such that* $\langle\psi|\phi\rangle = 0$*.*

- *The purification of the state* $\rho_A = \frac{1}{4}|+\rangle\langle+| + \frac{3}{4}|-\rangle\langle-|$ *is given by* $\rho_A = \mathrm{Tr}_B\big(|\psi\rangle_{AB}\langle\psi|_{AB}\big)$*, where* $|\psi\rangle_{AB} = \frac{1}{\sqrt{4}}|+0\rangle + \sqrt{\frac{3}{4}}|-1\rangle = \frac{1}{2}|+0\rangle + \frac{\sqrt{3}}{2}|-1\rangle$*. To see this, diagonalise the state* ρ_A *with:*

$$\sum_{i=1}^{d_A} \lambda_i |\lambda_i\rangle\langle\lambda_i| . \tag{3.11}$$

To then obtain the purification, apply the previous proposition:

$$|\psi\rangle = \sum_{j=1}^{d_A} \sqrt{\lambda_j}\,|\lambda_j\rangle \otimes |j\rangle . \tag{3.12}$$

One important fact is that purifications are <u>not</u> unique. Consider:

$$|\phi\rangle_{AB} = \big(\mathbb{I}_A \otimes U_B\big)|\psi\rangle_{AB} . \tag{3.13}$$

Since the trace is unitary invariant then ρ_A remains unchanged: $\mathrm{Tr}\big(U|k\rangle\langle k'|U^\dagger\big) = \delta_{kk'}$. Or, intuitively, it does not matter in what state B is in. Concerning the dimension of the purifying system, a system of dimension d_A suffices, however a system of rank r (number of non-zero eigenvalues in ρ) is necessary.

Example 3.3.3. *Consider the state* $\rho = \frac{1}{3}\big(|0\rangle\langle0| + |1\rangle\langle1| + |+\rangle\langle+|\big)$*. This state may admit the following two purifications:* $|\psi\rangle_{AB} = \frac{1}{\sqrt{3}}\big(|00\rangle + |11\rangle + |+2\rangle\big)$ *and* $|\phi\rangle_{AB} = \sum_j \sqrt{\lambda_j}\,|\lambda_j\rangle \otimes |j\rangle$*. However, these two purifications do <u>not</u> have the same dimension!*

Now, we can get the state $\rho = \frac{1}{2}\mathbb{I}$ in two ways. Either we toss a coin and then prepare $|0\rangle$ if it lands on heads and $|1\rangle$ if it lands on tails. This will indeed give $\rho_A = \frac{1}{2}\mathbb{I}$. Or, we can prepare the Bell state $\big|\phi^+\big\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$ and throw B away. This will also yield $\rho_A = \frac{1}{2}\mathbb{I}$. Operationally, these two operations are indistinguishable, however they are computationally different. Indeed, the first preparation is a <u>proper</u> mixture and the second one is an <u>improper</u> mixture.

Chapter 4

Quantum Measurements

In this chapter, we will look at several examples of quantum measurements. We first define the notion of a measurement before moving on to a-priori conditions on the possible measurements $\{M_i\}$.
A measurement $\mathcal{M} = (M_1, ..., M_L)$ is a list of individual measurements $\{M_i\}$ that are $d \times d$ matrices where ρ is also d-dimensional. In this vector, L is the number of possible answers. The associated probability of making a particular measurement is given by $\mathbb{P}(k) = \mathrm{Tr}(\rho M_k)$. As usual, we have the following conditions:

- Normalisation $\longrightarrow \sum_k \mathbb{P}(k) = \sum_k \mathrm{Tr}(\rho M_k) = \mathrm{Tr}\left(\rho \sum_k M_k\right) = 1 \Longrightarrow \sum_k M_k = \mathbb{I}$.
- Positivity $\longrightarrow \mathrm{eigs}(M_k) \geq 0$.

The previous vector $\mathcal{M}$ along with the above conditions are called **Positive Operator Value Measurements**, or more compactly **POVMs**. There are, however, some limitations to these measurements: (i) it doesn't say anything about post measurement state, (ii) it doesn't say anything about how you realise the measurement, (iii) it doesn't necessarily associate outcome with a number.

4.1 Projective Measurements

For a projective measurement, consider $\mathcal{M}$ as follows:

$$\mathcal{M} = \left\{M_1 = |+\rangle\langle+|, M_2 = |-\rangle\langle-|\right\}. \tag{4.1}$$

Since the two states $|+\rangle\langle+|$, $|-\rangle\langle-|$ have eigenvalues 1 and 0 respectively, and also $|+\rangle\langle+| + |-\rangle\langle-| = \mathbb{I}$, then $\mathcal{M}$ is a valid POVM. Projective measurements are also orthogonal:

$$M_i M_j = \delta_{ij} M_i. \tag{4.2}$$

Hence we must always have d projectors in $\mathcal{M}$. If we further associate each outcome with a numerical value λ_i, we can then associate measurements with Hermitian operators in the form:

$$\langle \mathcal{O} \rangle = \sum_i \lambda_i M_i. \tag{4.3}$$

Then, we regain the standard quantum measurement account with:

$$\langle \mathcal{O} \rangle = \mathrm{Tr}(\rho \mathcal{O}) = \sum_i \mathrm{Tr}(\rho M_i)\lambda_i := \sum_i p_i \lambda_i. \tag{4.4}$$

4.2 State Discrimination

Let us introduce this measurement with a setup. Suppose Alice gives Bob a state $|0\rangle$ or $|+\rangle$ and Alice wants to do a measurement to determine which one she gets. Now it is impossible for Bob to determine this with certainty. But with a POVM, we can do a measurement which gives the correct answer some of the time and is inconclusive other times. Consider the measurement:

$$\mathcal{M} = (M_1, M_2, M_3) = \left(\frac{\sqrt{2}}{1+\sqrt{2}} |1\rangle\langle 1|, \frac{\sqrt{2}}{1+\sqrt{2}} |-\rangle\langle -|, \mathbb{I} - M_1 - M_2 \right). \tag{4.5}$$

The last measurement M_3 provides the normalisation condition, and we note that all the eigenvalues of the measurements are positive. Hence this is a valid POVM. In conclusion, we can deduce that we were in state $|+\rangle$, or in state $|0\rangle$, or that the measurement was inconclusive.

4.3 Non-Rank 1 POVM

A Non-rank 1 POVM takes, for example, a measurement given by:

$$\mathcal{M} = (M_1, M_2) = \left(|\psi^-\rangle\langle\psi^-|, |\phi^+\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-| + |\psi^+\rangle\langle\psi^+| \right). \tag{4.6}$$

The first measurement M_1 has eigenvalues $\{1,0\}$ and the second measurement has eigenvalues $\{1,1,1,0\}$, hence all eigenvalues are positive. Furthermore it holds that $M_1 + M_2 = \mathbb{I}$. Hence $\mathcal{M}$ is a valid POVM. And in fact, this measurement answers the question: "Is the system in the singlet $|\psi^-\rangle$ state?".

Chapter 5

Quantum Channels

In this chapter, we present the notions and requirements of quantum channels, where we first present the so-called Kraus-representation. We will then delve into several examples of channels, and finally end on the statement of the Naimark's Dilation Theorem. We first present two requirements $(*)$ for a quantum channel:

- The outputs of a quantum state $\sigma = \mathcal{E}(\rho)$ is also a state, i.e. $\mathrm{Tr}(\sigma) = 1$ and $\sigma \geq 0$ with $\mathrm{eigs}(\sigma) \geq 0$.
- We require linearity: $\mathcal{E}(p\rho + (1-p)\sigma) = p\mathcal{E}(\rho) + (1-p)\mathcal{E}(\sigma)$.

Proposition 5.0.1 (Kraus Representation)**.** *The two requirements* $(*)$ *above are satisfied if:*

$$\mathcal{E}(\rho) = \sum_i A_i \rho A_i^\dagger, \quad \text{such that } \sum_i A_i^\dagger A_i = \mathbb{I}. \tag{5.1}$$

The operators $\{A_i\}$ *are called Kraus Operators.*

Proof. We first start with linearity:

$$\begin{aligned}\mathcal{E}(p\rho + (1-p)\sigma) = \sum_i A_i(p\rho + (1-p)\sigma)A_i^\dagger &= p\sum_i A_i \rho A_i^\dagger + (1-p)\sum_i A_i \sigma A_i^\dagger \\ &= p\mathcal{E}(\rho) + (1-p)\mathcal{E}(\sigma).\end{aligned}$$

We now move on to the first condition, with first the calculation involving the trace:

$$\mathrm{Tr}(\mathcal{E}(\rho)) = \mathrm{Tr}\left(\sum_i A_i \rho A_i^\dagger\right) = \sum_i \mathrm{Tr}\big(A_i \rho A_i^\dagger\big) = \sum_i \mathrm{Tr}\big(A_i A_i^\dagger \rho\big) = \mathrm{Tr}(\rho).$$

We finally end on the positivity condition. First note that $\mathrm{eigs}(\rho) \geq 0$ if and only if $\langle\phi|\,\sigma\,|\phi\rangle \geq 0$ for all $|\phi\rangle$. To see this, write $\sigma = \sum_\lambda \lambda\,|\lambda\rangle\langle\lambda|$. Then:

$$\langle\phi|\,\sigma\,|\phi\rangle = \sum_\lambda \lambda\,\langle\phi|\lambda\rangle\langle\lambda|\phi\rangle = \sum_\lambda \lambda |\langle\phi|\lambda\rangle|^2 \geq 0 \iff \text{all } \lambda \geq 0.$$

Now, let ρ be positive with $\rho = \sum_\lambda \lambda\,|\lambda\rangle\langle\lambda|$. Then we find:

$$\langle\phi|\,\mathcal{E}(\rho)\,|\phi\rangle = \langle\phi|\sum_i A_i \rho A_i^\dagger\,|\phi\rangle = \sum_\lambda \lambda \sum_i \langle\phi|\,A_i\,|\lambda\rangle\,\langle\lambda|\,A_i^\dagger\,|\phi\rangle = \sum_\lambda \lambda \sum_i |\langle\phi|\,A_i\,|\lambda\rangle|^2 \geq 0.$$

□

5.1 Unitary Dynamics, Convex Combination of Unitaries and Isometric Dynamics

In this section, we present three (simple) examples of channels where we will look at unitary dynamics, convex combinations of unitaries and finally isometric dynamics in that order.

• Considering a unitary matrix U satisfying $UU^\dagger = U^\dagger U = \mathbb{I}$, a **unitary dynamic** is a quantum channel that takes the form:

$$\mathcal{E}(\rho) = U\rho U^\dagger. \tag{5.2}$$

Proposition 5.1.1. *A unitary dynamic has a Kraus operator given by $A_i = U$.*

Proof. We simply check the Kraus condition:

$$\sum_i A_i^\dagger A_i = U^\dagger U = \mathbb{I}. \tag{5.3}$$

□

• Still considering a unitary matrix U satisfying $UU^\dagger = U^\dagger U = \mathbb{I}$ along with a probability $0 \leq p_i \leq 1$, a **convex combination of unitaries** is a quantum channel that has the following form:

$$\mathcal{E}(\rho) = \sum_i p_i U_i \rho U_i^\dagger. \tag{5.4}$$

This channel simply lets us apply U_i with probability p_i.

Proposition 5.1.2. *A convex combination of unitaries has a Kraus operator given by $A_i = \sqrt{p_i}U_i$.*

Proof. We simply check the Kraus condition:

$$\sum_i A_i^\dagger A_i = \sum_i p_i U_i^\dagger U_i = \sum_i p_i \mathbb{I} = \mathbb{I}. \tag{5.5}$$

□

• This last type of dynamic is similar to the unitary dynamic but without the unitary assumption of U. Hence, a **isometric dynamic** is a quantum channel that has the following form:

$$\mathcal{E}(\rho) = U\rho U^\dagger, \quad \text{with } U^\dagger U = \mathbb{I} \text{ however } UU^\dagger \neq \mathbb{I}. \tag{5.6}$$

We remark that $\mathcal{E}(\rho)$ is a map between different dimensional systems. It is also not difficult to see that an isometric dynamic will have the same Kraus operator as a unitary dynamic. Note that, from a 2-dimensional system $\{|0\rangle, |1\rangle\}$, it maps states to a 4-dimensional system $\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$ as follows:

$$U|0\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |2\rangle\big) \text{ and, } U|1\rangle = \frac{1}{\sqrt{2}}\big(|1\rangle + |3\rangle\big). \tag{5.7}$$

Note also that, although U maps between different subspaces, it is still norm and orthogonality preserving:

$$\langle 0|U^\dagger U|0\rangle = \langle 1|U^\dagger U|1 = 1\rangle \text{ and, } \langle 0|U^\dagger U|1\rangle = 0. \tag{5.8}$$

5.2 Completely Dephasing Channel

In this section, we present the **completely dephasing channel** as the quantum channel defined by:

$$\mathcal{E}(\rho) = |0\rangle\langle 0|\rho|0\rangle\langle 0| + |1\rangle\langle 1|\rho|1\rangle\langle 1| = \langle 0|\rho|0\rangle|0\rangle\langle 0| + \langle 1|\rho|1\rangle|1\rangle\langle 1|. \tag{5.9}$$

Proposition 5.2.1. *A completely dephasing channel has a Kraus operator given by $A_0 = |0\rangle\langle 0|$ and $A_1 = |1\rangle\langle 1|$.*

Proof. We simply check the Kraus condition:

$$\sum_i A_i^\dagger A_i = |0\rangle\langle 0|0\rangle\langle 0| + |1\rangle\langle 1|1\rangle\langle 1| = |0\rangle\langle 0| + |1\rangle\langle 1| = \mathbb{I}. \tag{5.10}$$

□

Let us now look at the action of this map on a general state ρ in the form:

$$\rho = \begin{pmatrix} \langle 0|\rho|0\rangle & \langle 0|\rho|1\rangle \\ \langle 1|\rho|0\rangle & \langle 1|\rho|1\rangle \end{pmatrix}. \tag{5.11}$$

Then, it is not a difficult calculation to check that the map yields:

$$\mathcal{E}(\rho) = \begin{pmatrix} \langle 0|\rho|0\rangle & 0 \\ 0 & \langle 1|\rho|1\rangle \end{pmatrix}. \tag{5.12}$$

In conclusion, the completely dephasing channel kills off the off-diagonal of the state ρ.

5.3 Completely Dephasing Map

Given some fixed state τ and for all states ρ, the **completely dephasing map** is defined as:

$$\mathcal{E}(\rho) = \tau. \tag{5.13}$$

Proposition 5.3.1. *A completely dephasing map has a Kraus operator given by $A_{ij} = \sqrt{\lambda_i}|\lambda_i\rangle\langle j|$, and is furthermore a well-defined quantum channel.*

Proof. We first check the Kraus condition:

$$\sum_{ij} A_{ij}^\dagger A_{ij} = \sum_{ij} \lambda_i |j\rangle\langle\lambda_i|\lambda_i\rangle\langle j| = \underbrace{\sum_i \lambda_i}_{=1} \sum_j |j\rangle\langle j| = \mathbb{I}. \tag{5.14}$$

Checking linearity is immediate. We now verify that the output is a genuine state. To this end, consider the spectral decomposition of $\tau = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$, and by a simple calculation we find:

$$\mathcal{E}(\rho) = \sum_{ij} A_{ij}\rho A_{ij}^\dagger = \sum_{ij} \lambda_i |\lambda_i\rangle\langle j|\rho|j\rangle\langle\lambda_i| = \sum_i \lambda_i |\lambda_i\rangle \left(\sum_j \langle j|\rho|j\rangle\right) \langle\lambda_i| = \underbrace{\mathrm{Tr}(\rho)}_{=1} \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i| = \tau.$$

□

5.4 Partial Trace

Given a state ρ_{AB}, the **partial trace** is given by the map:

$$\mathcal{E}(\rho_{AB}) = \mathrm{Tr}_B(\rho_{AB}) = \rho_A. \tag{5.15}$$

Comparing the definition of the partial trace and the one of a general Kraus operator, it is not difficult to spot the corresponding operator for the partial trace:

$$\mathrm{Tr}_B(\rho) = \sum_k \underbrace{\big(\mathbb{I}\otimes\langle k|\big)}_{:=A_k}\rho\big(\mathbb{I}\otimes|k\rangle\big). \tag{5.16}$$

Proposition 5.4.1. *The partial trace has a Kraus operator given by* $A_k = \mathbb{I}\otimes\langle k|$.

Proof. We simply check the Kraus condition:

$$\sum_k A_k^\dagger A_k = \sum_k \big(\mathbb{I}\otimes|k\rangle\big)\big(\mathbb{I}\otimes\langle k|\big) = \mathbb{I}\otimes\sum_k |k\rangle\langle k| = \mathbb{I}\otimes\mathbb{I} = \mathbb{I}. \tag{5.17}$$

□

5.5 State Preparation

In this section we look at the preparation of a quantum state. This is a little unusual because the input system is *trivial*, while the output system is some non-trivial quantum system A. The trivial system is simply a $d=1$ scenario, with no degrees of freedom with a Hilbert space of $\mathbb{C}$. The only *state* in question is the number 1. The state preparation quantum channel is then the mapping $1 \longrightarrow \rho$ for some fixed ρ.

Proposition 5.5.1. *The state preparation has a Kraus operator given by* $A_k = \sqrt{\lambda_k}\,|\lambda_k\rangle$.

Proof. We simply check the Kraus condition:

$$\sum_k A_k^\dagger A_k = \sum_k \lambda_k \langle\lambda_k|\lambda_k\rangle = \sum_k \lambda_k \mathbb{I} = \mathbb{I}. \tag{5.18}$$

□

Considering the spectral decomposition of ρ, the map is given by:

$$\mathcal{E}(1) = \sum_k A_k A_k^\dagger = \sum_k \lambda_k\,|\lambda_k\rangle\langle\lambda_k| = \rho. \tag{5.19}$$

Therefore A_i should be viewed as a mapping $A_i : \mathbb{C} \longrightarrow \mathcal{H}_A$ from $\mathbb{C}$ into some non-trivial Hilbert space $\mathcal{H}_A$.

5.6 Measure-Record-and-Update Channel

Given a state ρ, the **measure-record-and-update** channel is given by the map:

$$\mathcal{E}(\rho) = \sum_k |k\rangle\langle k| \otimes B_k \rho B_k^\dagger. \tag{5.20}$$

In this case, ρ is the system being measured, $|k\rangle\langle k|$ is the classical register of measurement outcome and B_k is the update rule. For future reference, we set $x = |k\rangle\langle k|$ and $s = B_k \rho B_k^\dagger$.

Proposition 5.6.1. *The measure-record-and-update channel has a Kraus operator given by* $A_k = |k\rangle \otimes B_k$.

Proof. Provided that B_k is also a Kraus operator, we check the Kraus condition:

$$\sum_k A_k^\dagger A_k = \sum_k \langle k|k\rangle B_k^\dagger B_k = \sum_k B_k^\dagger B_k = \mathbb{I}. \tag{5.21}$$

Then, if B_k are Kraus operators then so does A_k.

□

Let us now look at the output state on the classical register. Set $M_k := B_k^\dagger B_k$ and we find:

$$\mathrm{Tr}_s(\mathcal{E}(\rho)) = \mathrm{Tr}_s\left(\sum_k |k\rangle\langle k| \otimes B_k \rho B_k^\dagger\right) = \sum_k |k\rangle\langle k| \,\mathrm{Tr}\big(B_k \rho B_k^\dagger\big) = \sum_k \mathrm{Tr}\big(M_k \rho\big)\,|k\rangle\langle k| = \sum_k \mathbb{P}(k)\,|k\rangle\langle k|.$$

We now observe that the $\{M_k\}$ are in the form of a POVM. Let us then verify that it is the case:
- Since the B_ks are Kraus operators, then $\sum_k M_k = \sum_k B_k^\dagger B_k = \mathbb{I}$.
- For all $|\phi\rangle$, we have positive definiteness given by $\langle\phi| B_k^\dagger B_k |\phi\rangle = \|B_k|\phi\rangle\|^2 \geq 0$.

Let us now look at the output state if we trace out the classical register:

$$\mathrm{Tr}_x(\mathcal{E}(\rho)) = \mathrm{Tr}_x\left(\sum_k |k\rangle\langle k| \otimes B_k \rho B_k^\dagger\right) = \sum_k \underbrace{\mathrm{Tr}\big(|k\rangle\langle k|\big)}_{=1} B_k \rho B_k^\dagger = \sum_k B_k \rho B_k^\dagger.$$

This means that if we look at the state of the system conditioned on determining measurement outcome $|x\rangle\langle x|$ we have $\rho \longrightarrow \mathcal{E}(\rho)$ and get x by finding:

$$|x\rangle\langle x| \otimes B_x \rho B_x^\dagger \;\text{ with probability }\; \mathrm{Tr}\big(M_x\rho\big) = \mathrm{Tr}\big(B_x^\dagger B_x \rho\big). \tag{5.22}$$

Note that the state $|x\rangle\langle x| \otimes B_x \rho B_x^\dagger$ is not normalised, where for a proper normalisation we must impose:

$$\rho \longrightarrow |x\rangle\langle x| \otimes \frac{B_x \rho B_x^\dagger}{\mathrm{Tr}\big(M_x\rho\big)}. \tag{5.23}$$

Remark 5.6.2. *This update rule is not necessary to a measurement. It also <u>not</u> unique. Indeed, say* $\widetilde{B_k} = U B_k$ *for some unitary matrix* U. *Then:*

$$\widetilde{B_k}^\dagger \widetilde{B_k} = B_k^\dagger U^\dagger U B_k = B_k^\dagger B_k \Longrightarrow \widetilde{M_k} = M_k. \tag{5.24}$$

Theorem 5.6.3 (Naimark's Dilation Theorem)**.** *If* $\{M_i\}_{i=1}^N$ *is a POVM acting on a Hilbert space of dimension* d_A *then there exists a projective measurement* $\{\mathcal{M}_i\}_{i=1}^N$ *acting on a space of dimension* d'_A *and an isometry* $V : \mathcal{H}_A \longrightarrow \mathcal{H}'_A$ *such that, for all* i, $M_i = V^\dagger \mathcal{M}_i V$.

Intuitively, we need to use measurement-record-and-update channel. Indeed, we need to introduce a classical register that records the outcome and performs a projective measurement on that register. More formally, we

need to consider $V = \sum_k |k\rangle \otimes B_k$, $V^\dagger = \sum_k \langle k| \otimes B_k^\dagger$ and $\mathcal{M}_k = |k\rangle\langle k| \otimes \mathbb{I}$. We can also look at B_k as $\sqrt{M_k}$.

$$\mathrm{Tr}\big(V\rho V^\dagger M_k\big) = \sum_{ij} \mathrm{Tr}\Big[\rho\big(\langle j| \otimes \sqrt{M_j}\big)\big(|k\rangle\langle k| \otimes \mathbb{I}\big)\big(|i\rangle \otimes \sqrt{M_i}\big)\Big] = \mathrm{Tr}\big(\rho M_k\big) = \mathbb{P}(k). \tag{5.25}$$

5.7 Stinespring Dilation

Considering a system and its environment, and then ignoring the later (i.e. tracing it out) induces a non-unitary channel on the system.

Example 5.7.1. *Suppose we start with $\rho_S \otimes |0\rangle\langle 0|_E$ and evolve under $U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1$. Then:*

$$U\big(\rho_S \otimes |0\rangle\langle 0|_E\big)U^\dagger = \rho_S^{00}|0\rangle\langle 0| \otimes U_0|0\rangle\langle 0|U_0^\dagger + \rho_S^{11}|1\rangle\langle 1| \otimes U_1|0\rangle\langle 0|U_1^\dagger + \rho_S^{01}|0\rangle\langle 1| \otimes U_0|0\rangle\langle 0|U_1^\dagger + \rho_S^{10}|1\rangle\langle 0| \otimes U_1|0\rangle\langle 0|U_0^\dagger.$$

And now, tracing out yields:

$$\rho_s = \mathrm{Tr}_E\Big(U\big(\rho_S \otimes |0\rangle\langle 0|_E\big)U^\dagger\Big) = \rho_s^{00}|0\rangle\langle 0| + \rho_s^{11}|1\rangle\langle 1| + c\rho_s^{01}|0\rangle\langle 1| + c^*\rho_s^{10}|1\rangle\langle 0|.$$

Above, $c := \langle 0|U_1^\dagger U_0|0\rangle \equiv \langle \psi_1|\psi_0\rangle$, and for simplicity assume $c \in \mathbb{R}$, then:

$$\begin{aligned}\rho_s &= p\rho + (1-p)z\rho z \\ &= p\rho + (1-p)\big(\rho_s^{00}|0\rangle\langle 0| + \rho_s^{11}|1\rangle\langle 1| - \rho_s^{01}|0\rangle\langle 1| - \rho_s^{10}|1\rangle\langle 0|\big) \\ &= \rho_s^{00}|0\rangle\langle 0| + \rho_s^{11}|1\rangle\langle 1| + \underbrace{(2p-1)}_{=c}\big(\rho_s^{01}|0\rangle\langle 1| + \rho_s^{10}|1\rangle\langle 0|\big).\end{aligned}$$

This is known as the non-completely dephasing channel, and it kills off coherence turning a quantum state into a classical mixture. Note that if $c = 0 \iff p = \frac{1}{2}$, we end up with a perfect classical mixture (i.e. the completely dephasing channel).

Similarly to how tracing out part of an entangled state leads to a mixed state and any mixed state can be purified into a part of a larger entangled system, any quantum channel can be written in terms of a unitary acting on an enlarged system.

Theorem 5.7.2 (Stinespring Dilation)**.** *For a unitary U, any quantum channel can be written in the form:*

$$\mathcal{E}(\rho) = \mathrm{Tr}_E\Big(U\big(\rho \otimes |0\rangle\langle 0|\big)U^\dagger\Big). \tag{5.26}$$

Proof. Let $\mathcal{E}(\rho) = \sum_i A_i \rho A_i^\dagger$ such that $\sum_i A_i^\dagger A_i = \mathbb{I}$. For all ψ we need:

$$U|0\rangle|\psi\rangle = \sum_i |i\rangle A_i|\psi\rangle. \tag{5.27}$$

Let us see why this is the case:

$$|0\rangle|\psi\rangle = \begin{pmatrix}1\\0\\\vdots\end{pmatrix} \otimes |\psi\rangle = \begin{pmatrix}|\psi\rangle\\0\\\vdots\end{pmatrix} \quad \text{and,} \quad |1\rangle|\psi\rangle = \begin{pmatrix}0\\|\psi\rangle\\0\\\vdots\end{pmatrix}. \tag{5.28}$$

Therefore, we can write:

$$U|0\rangle|\psi\rangle = \begin{pmatrix}A_0 & \vdots\\A_1 & \text{Pick remaining entries}\\A_2 & \text{to make unitary.}\\\vdots & \vdots\end{pmatrix}\begin{pmatrix}|\psi\rangle\\0\\0\\\vdots\end{pmatrix} = \begin{pmatrix}A_0|\psi\rangle\\A_1|\psi\rangle\\A_2|\psi\rangle\\\vdots\end{pmatrix} = \sum_i |i\rangle A_i|\psi\rangle. \tag{5.29}$$

Now, if it is the case that we can write $U|0\rangle|\psi\rangle$ as $\sum_i |i\rangle A_i|\psi\rangle$ then:

$$\mathrm{Tr}_E\left(U\big(|0\rangle\langle 0|\otimes|\psi\rangle\langle\psi|\big)U^\dagger\right) = \sum_{i,j}\mathrm{Tr}_E\left(\mathbb{I}\otimes A_i\big(|i\rangle\langle j|\otimes|\psi\rangle\langle\psi|\big)\mathbb{I}\otimes A_j^\dagger\right) = \sum_i A_i\rho A_i^\dagger = \mathcal{E}(\rho). \tag{5.30}$$

□

5.8 The Choi Representation

We first present a simple but powerful trick with the notion of vectorization. Say $A = \sum_{ij} a_{ij}|i\rangle\langle j|$, then the vectorized version of A is given by:

$$|\mathrm{vec}(A)\rangle = \sum_{ij} a_{ij}|ij\rangle. \tag{5.31}$$

Example 5.8.1. • *For a* 2×2 *matrix* $\begin{pmatrix}1&0\\0&1\end{pmatrix}$, *its vectorized version is* $|00\rangle + |11\rangle$.

• *For a* 2×2 *matrix* $\begin{pmatrix}0&a\\b&0\end{pmatrix}$, *its vectorized version is* $a|01\rangle + b|10\rangle$.

One of the very important identity related to vectorization is the following:

Proposition 5.8.2 (Central Vectorization Identity)**.**

$$|\mathrm{vec}(AXB)\rangle = A\otimes B^T|\mathrm{vec}(X)\rangle. \tag{5.32}$$

Proof. Using a direct computation and setting $C = AXB$ yields:

$$\begin{aligned}A\otimes B^T|\mathrm{vec}(X)\rangle &= \sum_{i,j,i',j'} A_{ij}|i\rangle\langle j|\left(B_{i'j'}|i'\rangle\langle j'|\right)^T X_{kk'}|kk'\rangle\\ &= \sum_{i,j,i',j'} A_{ij}|i\rangle\langle j|\left(B_{i'j'}|j'\rangle\langle i'|\right)X_{kk'}|kk'\rangle\\ &= \sum_{i,j,i',j'} A_{ij}B_{i'j'}X_{kk'}|ij'\rangle\delta_{jk}\delta_{i'k'} = \sum_{i,j'} A_{ik}X_{ki'}B_{k'j'}|ij'\rangle =: \sum_{i,j'} C_{ij'}|ij'\rangle = |\mathrm{vec}(C)\rangle.\end{aligned}$$

□

With the above identity, we can now show an even more powerful result that will be very useful for computations.

Proposition 5.8.3.

$$\left(U^\dagger \otimes \mathbb{I}\right)\left|\phi^+\right\rangle = (\mathbb{I}\otimes U^*)\left|\phi^+\right\rangle. \tag{5.33}$$

Proof. For a d-dimensional system, we use that $\left|\phi^+\right\rangle = \frac{1}{\sqrt{d}}\sum_i |ii\rangle = \frac{1}{\sqrt{d}}|\mathrm{vec}(\mathbb{I})\rangle$. Now, using the central vectorization identity twice yields:

$$\begin{aligned}\left(U^\dagger \otimes \mathbb{I}\right)\left|\phi^+\right\rangle = \frac{1}{\sqrt{d}}\left(U^\dagger\otimes\mathbb{I}\right)|\mathrm{vec}(\mathbb{I})\rangle = \frac{1}{\sqrt{d}}\left|\mathrm{vec}\left(U^\dagger\mathbb{I}\cdot\mathbb{I}^T\right)\right\rangle &= \frac{1}{\sqrt{d}}\left|\mathrm{vec}\left(U^\dagger\right)\right\rangle \\ &= \frac{1}{\sqrt{d}}\left(\mathbb{I}\otimes\left(U^\dagger\right)^T\right)|\mathrm{vec}(\mathbb{I})\rangle \\ &= \frac{1}{\sqrt{d}}(\mathbb{I}\otimes U^*)|\mathrm{vec}(\mathbb{I})\rangle \\ &= (\mathbb{I}\otimes U^*)\left|\phi^+\right\rangle.\end{aligned}$$

□

Example 5.8.4. *This trick can be used to reduce circuit depths:*

$$\left(UV^\dagger\otimes\mathbb{I}\right)\left|\phi^+\right\rangle = (U\otimes\mathbb{I})\left(V^\dagger\otimes\mathbb{I}\right)\left|\phi^+\right\rangle = (U\otimes V^*)\left|\phi^+\right\rangle. \tag{5.34}$$

Assuming U and V are of depth D, we have reduced the global depth by a half.

Proposition 5.8.5 (The Choi-Jamiolkowski Representation)**.** *For any quantum channel $\mathcal{E}$, we define the Choi state associated to the channel as:*

$$J(\mathcal{E}) := (\mathcal{E}\otimes\mathbb{I})|\mathrm{vec}(\mathbb{I})\rangle\langle\mathrm{vec}(\mathbb{I})|. \tag{5.35}$$

The Choi state uniquely specifies the quantum channel. If $\mathcal{E}$ has a Kraus representation $\{A_i\}$ then the Choi state is given by:

$$J(\mathcal{E}) = \sum_i |\mathrm{vec}(A_i)\rangle\langle\mathrm{vec}(A_i)|. \tag{5.36}$$

- This definition can be seen both with and without the factor of dimension included. Indeed, in place of $|\mathrm{vec}(\mathbb{I})\rangle$ one could consider $|\mathrm{vec}(\mathbb{I}\otimes\mathbb{I})\rangle = \sum_i |ii\rangle \propto \left|\phi^+\right\rangle$.

- To see the final claim of the proposition, we may use the central vectorization identity and find:

$$J(\mathcal{E}) = \sum_i \left(A_i\otimes\mathbb{I}\right)|\mathrm{vec}(\mathbb{I})\rangle\langle\mathrm{vec}(\mathbb{I})|\left(A_i^\dagger\otimes\mathbb{I}\right) = \sum_i |\mathrm{vec}(A_i)\rangle\langle\mathrm{vec}(A_i)|. \tag{5.37}$$

Example 5.8.6. *Consider $\mathcal{E}(\rho)$ to be the completely dephasing channel on a single qubit, $A_0 = |0\rangle\langle 0|$ and $A_1 = |1\rangle\langle 1|$. Then it holds that $|\mathrm{vec}(\mathbb{I})\rangle = |00\rangle + |11\rangle$. Hence the Choi representation is given by:*

$$\begin{aligned}J(\mathcal{E}) &= \left(\mathcal{E}\otimes\mathbb{I}\right)\left(|00\rangle\langle 00| + |11\rangle\langle 11| + |00\rangle\langle 11| + |11\rangle\langle 00|\right) \\ &= \mathcal{E}\left(|0\rangle\langle 0|\right)\otimes|0\rangle\langle 0| + \mathcal{E}\left(|1\rangle\langle 1|\right)\otimes|1\rangle\langle 1| + \underbrace{\mathcal{E}\left(|0\rangle\langle 1|\right)}_{=0}\otimes|0\rangle\langle 1| + \underbrace{\mathcal{E}\left(|1\rangle\langle 0|\right)}_{=0}\otimes|1\rangle\langle 0| \\ &= |00\rangle\langle 00| + |11\rangle\langle 11| = \left|\mathrm{vec}\left(|0\rangle\langle 0|\right)\right\rangle\left\langle\mathrm{vec}\left(|0\rangle\langle 0|\right)\right| + \left|\mathrm{vec}\left(|1\rangle\langle 1|\right)\right\rangle\left\langle\mathrm{vec}\left(|1\rangle\langle 1|\right)\right| \\ &= |\mathrm{vec}(A_0)\rangle\langle\mathrm{vec}(A_0)| + |\mathrm{vec}(A_1)\rangle\langle\mathrm{vec}(A_1)|.\end{aligned}$$

Overall, the point of the Choi representation is that it often ends up being easier mathematically/computationally. The idea is to work in terms of states rather than with channels (e.g. in algorithms for learning/simulating channels). Remark that it also provides a way of finding the Kraus operators for a channel.

General recipe for finding Kraus operators

(1) Take $\mathcal{E}$ and compute $J(\mathcal{E})$.

(2) Find the eigendecomposition of $J(\mathcal{E})$:

$$J(\mathcal{E}) = \sum_k \lambda_k |e_k\rangle\langle e_k| . \tag{5.38}$$

(3) Find the matrix A_k such that:

$$|\mathrm{vec}(A_k)\rangle = \sqrt{\lambda_k}\,|e_k\rangle . \tag{5.39}$$

(4) The matrices $\{A_k\}$ are the smallest set of Kraus operators for $\mathcal{E}$.

Note that finding the eigendecomposition of the Choi state always yields the minimum set of Kraus operators.

Example 5.8.7. *Let us find the Kraus operators of the channel* $\mathcal{E}(X) = \frac{1}{3}\big(\mathrm{Tr}(X)\mathbb{I} + X^T\big)$. *We first start by finding* $J(\mathcal{E})$. *Recall that* $\mathrm{Tr}\big(|i\rangle\langle j|\big) = \delta_{ij}$, *and then:*

$$J(\mathcal{E}) = \big(\mathcal{E}\otimes\mathbb{I}\big)\left(\sum_{ij}|ii\rangle\langle jj|\right) = \sum_{ij}\mathcal{E}\big(|i\rangle\langle j|\big)\otimes|i\rangle\langle j| = \sum_{ij}\frac{1}{3}\Big(\mathrm{Tr}\big(|i\rangle\langle j|\big)\mathbb{I} + |i\rangle\langle j|^T\Big)\otimes|i\rangle\langle j|$$
$$= \frac{1}{3}\left(\sum_i \mathbb{I}\otimes|i\rangle\langle i| + \sum_{ij}|ji\rangle\langle ij|\right) = \frac{1}{3}\big(\mathbb{I} + \mathrm{SWAP}\big).$$

Now, to proceed forward we may either find the eigendecomposition of $\mathbb{I}+\mathrm{SWAP}$ *or one may remark that:*

$$\mathbb{I} = |\phi^+\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-| + |\psi^+\rangle\langle\psi^+| + |\psi^-\rangle\langle\psi^-| . \tag{5.40}$$

$$\mathrm{SWAP} = |\phi^+\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-| + |\psi^+\rangle\langle\psi^+| - |\psi^-\rangle\langle\psi^-| . \tag{5.41}$$

Hence:

$$J(\mathcal{E}) = \frac{2}{3}\big(|\phi^+\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-| + |\psi^+\rangle\langle\psi^+|\big) = \frac{2}{3}\big(|00\rangle\langle 00| + |11\rangle\langle 11| + |\psi^+\rangle\langle\psi^+|\big). \tag{5.42}$$

We now have two different possible choices in Kraus operators.

Option 1:

$$|\mathrm{vec}(A_0)\rangle = \sqrt{\frac{2}{3}}|\phi^+\rangle = \frac{1}{\sqrt{3}}\big(|00\rangle + |11\rangle\big) \Longrightarrow A_0 = \frac{1}{\sqrt{3}}\mathbb{I}. \tag{5.43}$$

$$|\mathrm{vec}(A_1)\rangle = \sqrt{\frac{2}{3}}|\phi^-\rangle = \frac{1}{\sqrt{3}}\big(|00\rangle - |11\rangle\big) \Longrightarrow A_1 = \frac{1}{\sqrt{3}}Z. \tag{5.44}$$

$$|\mathrm{vec}(A_2)\rangle = \sqrt{\frac{2}{3}}|\psi^+\rangle = \frac{1}{\sqrt{3}}\big(|01\rangle + |10\rangle\big) \Longrightarrow A_2 = \frac{1}{\sqrt{3}}X. \tag{5.45}$$

It is not difficult to see that the $\{A_k\}$ *are valid Kraus operators since:*

$$\sum_i A_i^\dagger A_i = \sum_i \frac{1}{3}\mathbb{I} = \mathbb{I}. \tag{5.46}$$

Option 2:

$$|\text{vec}(A_0)\rangle = \sqrt{\frac{2}{3}}\,|00\rangle \Longrightarrow A_0 = \sqrt{\frac{2}{3}}\,|0\rangle\langle 0|\,. \tag{5.47}$$

$$|\text{vec}(A_1)\rangle = \sqrt{\frac{2}{3}}\,|11\rangle \Longrightarrow A_1 = \sqrt{\frac{2}{3}}\,|1\rangle\langle 1|\,. \tag{5.48}$$

$$|\text{vec}(A_2)\rangle = \sqrt{\frac{2}{3}}\,|\psi^+\rangle \Longrightarrow A_2 = \frac{1}{\sqrt{3}}X. \tag{5.49}$$

Again, it is not difficult to see that the $\{A_k\}$ are valid Kraus operators since:

$$\sum_i A_i^\dagger A_i = \frac{2}{3}\mathbb{I} + \frac{1}{3}\mathbb{I} = \mathbb{I}. \tag{5.50}$$

5.8.1 Unitary Mixing Freedom in Kraus Representations

We saw previously, for instance in Example (5.8.7), that multiple sets of Kraus operators can be used to describe the same channel. This can be understood within the Choi representation. We have:

$$J(\mathcal{E}) = \sum_i |\text{vec}(A_i)\rangle\langle\text{vec}(A_i)|\,. \tag{5.51}$$

But remember, this is just a state and we can write it in its eigendecomposition or just as an ensemble decomposition. In other words, we have:

$$J(\mathcal{E}) = \sum_i |\text{vec}(A_i)\rangle\langle\text{vec}(A_i)| = \underbrace{\sum_k \lambda_k\,|\lambda_k\rangle\langle\lambda_k|}_{(*)} = \underbrace{\sum_i p_i\,|\phi_i\rangle\langle\phi_i|}_{(**)}\,. \tag{5.52}$$

As we have seen at the beginning of the course, $(*)$ and $(**)$ are related by $\sqrt{p_i}\,|\phi_i\rangle = \sum_k U_{ik}\sqrt{\lambda_k}\,|\lambda_k\rangle$, for U_{ik} elements of a unitary/isometry. In particular, $(*)$ comes from $|\text{vec}(A_i)\rangle = \sqrt{\lambda_i}\,|\lambda_i\rangle$ and as a result we have the following relationship for $(**)$:

$$|\text{vec}(B_i)\rangle = \sqrt{p_i}\,|\phi_i\rangle = \sum_k U_{ik}\sqrt{\lambda_k}\,|\lambda_k\rangle = \sum_k U_{ik}\,|\text{vec}(A_k)\rangle\,. \tag{5.53}$$

Hence this yields:

$$\sum_{jj'}[B_i]_{jj'}\,|jj'\rangle = \sum_{jj'}\sum_k U_{ik}[A_k]_{jj'}\,|jj'\rangle \Longrightarrow [B_i]_{jj'} = \left[\sum_k U_{ik}A_k\right]_{jj'} \Longrightarrow B_i = \sum_k U_{ik}A_k. \tag{5.54}$$

Therefore, different possible sets of Kraus operators are related by unitary/isometric mixing.

5.9 The Representer Theorem

In this section, we point out a very important detail that was left out (intentionally) at the start of this course. We started with an operational definition of channels, that is we stated that they had to satisfy the following requirements:
(1) Linearity: $\mathcal{E}\big(p\rho + (1-p)\sigma\big) = p\mathcal{E}(\rho) + (1-p)\mathcal{E}(\sigma)$.
(2) Output states need to be real states: (i) $\text{Tr}\big(\mathcal{E}(\rho)\big) = 1$ (normalization), (ii) $\mathcal{E}(\rho) = \mathcal{E}(\rho)^\dagger$ (Hermitian), and (iii) $\text{eigs}\big(\mathcal{E}(\rho)\big) \geq 0$ (positivity/non-negativity).

If condition (2.iii) holds then the operation $\mathcal{E}$ is said to be positive. Furthermore, the conditions in (2) are actually not enough in the general case. For the map to be physical we also need the possibility to apply the map to only part of a state (i.e. to one subsystem) and still get a genuine state out. In other words, we need $\big(\mathcal{E}_A \otimes \mathbb{I}_B\big)\big(\rho_{AB}\big)$ to also be a valid quantum state (this operation is to apply $\mathcal{E}$ to A and do nothing to B).

Definition 5.9.1. *$\mathcal{E}$ is said to be **completely positive** if* $\operatorname{eigs}\big(\mathcal{E}\otimes\mathbb{I}\big)(\rho) \geq 0$ *for all ρ.*

Let us now look at an example where there is a difference between positivity and complete positivity.

Example 5.9.2 (Transpose Operator)**.** *This is a classic example of an operation that is positive but not completely positive. If* $\operatorname{eigs}(\rho) \geq 0$ *then we also have that* $\operatorname{eigs}(\rho^T) \geq 0$*. This is easy to see for a single qubit. Set $\rho = \frac{1}{2}\big(\mathbb{I} + \underline{r}\cdot\underline{\sigma}\big)$ then:*

$$\rho^T = \frac{1}{2}\big(\mathbb{I} + \underline{r}\cdot\underline{\sigma}^T\big) = \frac{1}{2}\big(\mathbb{I} + \widetilde{\underline{r}}\cdot\underline{\sigma}\big), \tag{5.55}$$

where $\underline{\sigma} = \begin{pmatrix}\sigma_x & -\sigma_y & \sigma_z\end{pmatrix}$ and $\underline{r}^T = \begin{pmatrix}r_x & -r_y & r_z\end{pmatrix}$. This is still a valid state on the Bloch sphere. However, the eigenvalues of $\mathcal{E}_T \otimes \mathbb{I}(\rho)$ may not be positive. Indeed, for example:

$$\begin{aligned}
\big(\mathcal{E}_T \otimes \mathbb{I}\big)\big(\left|\phi^+\right\rangle\left\langle\phi^+\right|\big) &=: \left|\phi^+\right\rangle\left\langle\phi^+\right|^{T_A} \\
&= \big(\mathcal{E}_T \otimes \mathbb{I}\big)\left(\frac{1}{2}\big(\left|00\right\rangle\left\langle 00\right| + \left|00\right\rangle\left\langle 11\right| + \left|11\right\rangle\left\langle 00\right| + \left|11\right\rangle\left\langle 11\right|\big)\right) \\
&= \frac{1}{2}\big(\left|00\right\rangle\left\langle 00\right| + \left|10\right\rangle\left\langle 01\right| + \left|01\right\rangle\left\langle 10\right| + \left|11\right\rangle\left\langle 11\right|\big) \\
&= \frac{1}{2}\big(\left|\phi^+\right\rangle\left\langle\phi^+\right| + \left|\phi^-\right\rangle\left\langle\phi^-\right| + \left|\psi^+\right\rangle\left\langle\psi^+\right| - \left|\psi^-\right\rangle\left\langle\psi^-\right|\big).
\end{aligned}$$

Above, in the first line, the notation T_A is to denote the so-called partial transpose and the channel $\mathcal{E}_T$ modifies the states as follows: $\left|0\right\rangle\left\langle 0\right| \longrightarrow \left|0\right\rangle\left\langle 0\right|$, $\left|0\right\rangle\left\langle 1\right| \longrightarrow \left|1\right\rangle\left\langle 0\right|$, $\left|1\right\rangle\left\langle 0\right| \longrightarrow \left|0\right\rangle\left\langle 1\right|$ and $\left|1\right\rangle\left\langle 1\right| \longrightarrow \left|1\right\rangle\left\langle 1\right|$. The eigenvalues are:

$$\operatorname{eigs}\Big(\left|\phi^+\right\rangle\left\langle\phi^+\right|^{T_A}\Big) = \left\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right\}. \tag{5.56}$$

Since we have a negative eigenvalue, $\left|\phi^+\right\rangle\left\langle\phi^+\right|^{T_A}$ is not positive, implying that $\mathcal{E}(X) = X^T$ is positive but not completely positive.

Theorem 5.9.3 (Representer Theorem)**.** *A channel $\mathcal{E}$ is:*
(i) $\mathcal{E}\big(\alpha\rho + \beta\sigma\big) = \alpha\mathcal{E}(\rho) + \beta\mathcal{E}(\sigma)$ *- linear,*
(ii) $\operatorname{eigs}\big(\mathcal{E}\otimes\mathbb{I}\big)(\rho) \geq 0$ *- completely positive,*
(iii) $\operatorname{Tr}\big(\big(\mathcal{E}\otimes\mathbb{I}\big)(\rho)\big) = 1$ *- trace preserving, if and only if* $\mathcal{E}(\rho) = \sum_i A_i \rho A_i^\dagger$ *with* $\sum_i A_i^\dagger A_i = \mathbb{I}$*.*

Proof. ($\Longleftarrow$) Consider the spectral decomposition of ρ to be $\rho = \sum_\lambda \lambda \left|\lambda\right\rangle\left\langle\lambda\right|$. Linearity is easy to check. For complete positivity we have:

$$\left\langle\psi\right| \sum_i \big(A_i \otimes \mathbb{I}\big)\rho\big(A_i^\dagger \otimes \mathbb{I}\big)\left|\psi\right\rangle = \sum_i \sum_\lambda \lambda \big|\left\langle\psi\right| A_i \otimes \mathbb{I} \left|\psi\right\rangle\big|^2 \geq 0. \tag{5.57}$$

Finally, for the trace preserving property we have:

$$\operatorname{Tr}\big(\big(\mathcal{E}\otimes\mathbb{I}\big)(\rho)\big) = \sum_i \operatorname{Tr}\Big(\big(A_i \otimes \mathbb{I}\big)\rho\big(A_i^\dagger \otimes \mathbb{I}\big)\Big) = \sum_i \operatorname{Tr}\big(\big(A_i^\dagger A_i \otimes \mathbb{I}\big)\rho\big) = \operatorname{Tr}(\rho) = 1. \tag{5.58}$$

($\Longrightarrow$) We use Choi representation as a tool. Consider $\sigma = \big(\mathbb{I}_R \otimes \mathcal{E}_Q\big)\Big(\sum_{ij} |ii\rangle\langle jj|_{RQ}\Big)$. Assuming we have complete positiveness, as σ is a quantum state we can write $\sigma = \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k|$. Now, define a map $A_k |\psi\rangle_Q = \sqrt{\lambda_k}\,\langle\psi^*|_R |\lambda_k\rangle_{RQ}$, where of course $\sqrt{\lambda_k}$ is positive. We now remark that:

$$\begin{aligned}\sum_k A_k |\psi\rangle\langle\psi|_Q A_k^\dagger = \sum_k \lambda_k \langle\psi^*|_R |\lambda_k\rangle\langle\lambda_k|_{RQ} |\psi^*\rangle_R &= \langle\psi^*|_R \sigma |\psi^*\rangle_R \\ &= \sum_{ij}\sum_{i'j'} \psi_{i'}^* \psi_{j'} \langle i'|_R |i\rangle\langle j| \otimes \mathcal{E}\big(|i\rangle\langle j|\big)|j'\rangle_R \\ &= \sum_{ij} \psi_i^* \psi_j \mathcal{E}\big(|i\rangle\langle j|\big) = \mathcal{E}\big(|\psi\rangle\langle\psi|\big).\end{aligned}$$

That is, we have found a set of operators such that $\sum_k A_k \rho A_k^\dagger = \mathcal{E}(\rho)$. Furthermore, for all ρ:

$$\mathrm{Tr}\big(\mathcal{E}(\rho)\big) = 1 \Longrightarrow \mathrm{Tr}\left(\sum_k A_k \rho A_k^\dagger\right) = \mathrm{Tr}\left(\sum_k A_k^\dagger A_k \rho\right) = 1 \Longrightarrow \sum_k A_k^\dagger A_k = \mathbb{I}. \tag{5.59}$$

Thus, we have found well-defined Kraus operators, hence concluding the proof.

□

Chapter 6

Shot Noise

In this chapter, we delve into the statistical and probabilistic description of measurement strategies. In particular we state the estimator for a measurement strategy, we will show whether a measurement strategy is biased or unbiased. Finally we will use Chebyshev's inequality and Hoeffding's inequality to bound the convergence of a measurement strategy.

6.1 Estimators

Say we have an observable $M = \sum_{m=1}^{d} \lambda_m |\lambda_m\rangle \langle\lambda_m|$ and we want to estimate $\overline{M} = \mathrm{Tr}(M\rho)$ where ρ is some state. The strategy will be to measure ρ in the eigenbasis of M and perform an average. In particular we consider the estimator:

$$\widehat{M}_N = \frac{1}{N} \sum_{s=1}^{N} \lambda_{m(s)}, \tag{6.1}$$

where we have that:

$$\begin{aligned}
\mathbb{E}\left(\widehat{M}_N\right) = \frac{1}{N} \sum_{s=1}^{N} \mathbb{E}\left(\lambda_{m(s)}\right) = \frac{1}{N} \sum_{s=1}^{N} \sum_{m(s)} \lambda_{m(s)} \mathbb{P}\left(\lambda_{m(s)}\right) &= \frac{1}{N} \sum_{s=1}^{N} \sum_{m(s)} \lambda_{m(s)} \left\langle \lambda_{m(s)} \middle| \rho \middle| \lambda_{m(s)} \right\rangle \\
&= \frac{1}{N} \sum_{s=1}^{N} \sum_{m(s)} \lambda_{m(s)} \mathrm{Tr}\left(\rho \middle| \lambda_{m(s)} \right\rangle \left\langle \lambda_{m(s)} \middle| \right) \\
&= \frac{1}{N} \sum_{s=1}^{N} \mathrm{Tr}(\rho M) = \mathrm{Tr}(\rho M).
\end{aligned}$$

Thus we obtain $\mathbb{E}\left(\widehat{M}_N\right) = \mathrm{Tr}(\rho M)$ which corresponds to an unbiased estimator. Let us now look at the variance $\mathrm{Var}\left(\widehat{M}_N\right) = \mathbb{E}\left(\widehat{M}_N^2\right) - \mathbb{E}\left(\widehat{M}_N\right)^2$, where:

$$\begin{aligned}
\mathbb{E}\left(\widehat{M}_N^2\right) = \mathbb{E}\left(\frac{1}{N^2} \sum_{s,t=1}^{N} \lambda_{m(s)} \lambda_{m(t)}\right) &= \mathbb{E}\left(\frac{1}{N^2}\left(\sum_{s=1}^{N} \lambda_{m(s)}^2 + \sum_{\substack{s,t=1 \\ s \neq t}}^{N} \lambda_{m(s)} \lambda_{m(t)}\right)\right) \\
&= \frac{1}{N^2}\left(\sum_{s=1}^{N} \mathbb{E}\left(\lambda_{m(s)}^2\right) + \sum_{\substack{s,t=1 \\ s \neq t}}^{N} \mathbb{E}\left(\lambda_{m(s)} \lambda_{m(t)}\right)\right).
\end{aligned}$$

We now use independence to find:

$$\mathbb{E}\left(\widehat{M}_N^2\right) \overset{\text{indep.}}{=} \frac{1}{N^2}\left(\sum_{s=1}^{N}\mathrm{Tr}(\rho M^2)+\sum_{\substack{s,t=1\\ s\neq t}}^{N}\mathbb{E}\left(\lambda_{m(s)}\right)\mathbb{E}\left(\lambda_{m(t)}\right)\right) = \frac{1}{N^2}\left(N\cdot\mathrm{Tr}(\rho M^2)+\sum_{\substack{s,t=1\\ s\neq t}}^{N}\mathrm{Tr}(\rho M)^2\right)$$
$$= \frac{1}{N^2}\left(N\cdot\mathrm{Tr}(\rho M^2)+N(N-1)\mathrm{Tr}(\rho M)^2\right)$$
$$= \mathrm{Tr}(\rho M)^2+\frac{1}{N}\left(\mathrm{Tr}(\rho M^2)-\mathrm{Tr}(\rho M)^2\right).$$

Thus, $\mathrm{Var}\left(\widehat{M}_N\right) = \frac{1}{N}\left(\mathrm{Tr}(\rho M^2)-\mathrm{Tr}(\rho M)^2\right) \equiv \frac{1}{N}\mathrm{Var}_\rho(M)$. In particular, $\epsilon := \sqrt{\mathrm{Var}\left(\widehat{M}_N\right)}$ is a measure of estimate precision. Hence the number of shots must satisfy:

$$\epsilon^2 = \frac{1}{N}\mathrm{Var}_\rho(M) \Longrightarrow N = \frac{1}{\epsilon^2}\mathrm{Var}_\rho(M). \tag{6.2}$$

Example 6.1.1 (Estimating Fidelity)**.** *Consider two ways of estimating fidelity (i.e. overlaps) between two states* $|\langle\psi|\phi\rangle|^2$*.*
Case A
Consider $|\psi\rangle = U_\psi|0\rangle$ *and* $|\phi\rangle = U_\phi|0\rangle$ *then the fidelity is given by:*

$$|\langle\psi|\phi\rangle|^2 = |\langle 0|U_\psi^\dagger U_\phi|0\rangle|^2 = \langle 0|U_\psi^\dagger U_\phi|0\rangle\langle 0|U_\phi^\dagger U_\psi|0\rangle = \langle\psi|U_\phi|0\rangle\langle 0|U_\phi^\dagger|\psi\rangle = \mathrm{Tr}\Big(\underbrace{|0\rangle\langle 0|}_{M}\underbrace{U_\phi^\dagger|\psi\rangle\langle\psi|U_\phi}_{\rho}\Big).$$

Thus in this first case the number of shots is given by:

$$N_A = \frac{1}{\epsilon^2}\left(\mathrm{Tr}(M^2\rho)-\mathrm{Tr}(\rho M)^2\right) = \frac{1}{\epsilon^2}\left(\mathrm{Tr}\left(|0\rangle\langle 0|^2\rho\right)-\mathrm{Tr}(\rho M)^2\right) = \frac{1}{\epsilon^2}\left(\mathrm{Tr}(|0\rangle\langle 0|\rho)-\mathrm{Tr}(M\rho)^2\right)$$
$$= \frac{1}{\epsilon^2}\left(|\langle\psi|\phi\rangle|^2-|\langle\psi|\phi\rangle|^4\right).$$

Case B
Consider the following:

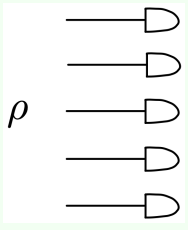
$$(*) = \langle\psi|\otimes\langle\phi|\,\mathrm{SWAP}\,|\psi\rangle\otimes|\phi\rangle = \left(\langle\psi|\otimes\langle\phi|\right)\left(|\phi\rangle\otimes|\psi\rangle\right) = |\langle\psi|\phi\rangle|^2 = \mathrm{Tr}\left(|0\rangle\langle 0|U_\phi^\dagger|\psi\rangle\langle\psi|U_\phi\right).$$

Since $\mathrm{SWAP} = |\phi^+\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-| + |\psi^+\rangle\langle\psi^+| - |\psi^-\rangle\langle\psi^-|$*, measuring* SWAP *consists of measuring the Bell basis. Note that* $(*) = \mathrm{Tr}\big(\mathrm{SWAP}\,|\psi\phi\rangle\langle\psi\phi|\big)$*. Now we find:*

$$N_B = \frac{1}{\epsilon^2}\left(\mathrm{Tr}(\mathrm{SWAP}^2\rho)-\mathrm{Tr}(\rho M)^2\right) = \frac{1}{\epsilon^2}\left(1-|\langle\psi|\phi\rangle|^4\right).$$

As $|\langle\psi|\phi\rangle|^2 \leq 1$ *we need a less number of shots in A than in B, i.e.* $N_A \leq N_B$*.*

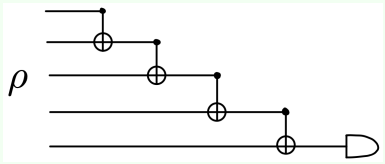
Example 6.1.2 (Marco's Problem)**.** *Two different ways of measuring* $\langle\psi|\sigma_Z^{\otimes n}|\psi\rangle$*:*
Case A



Here, each branch represents a measurement in the computational basis. We have:

$$\left\langle Z^{\otimes n}\right\rangle_{\rho} = \mathrm{Tr}\left(\rho Z^{\otimes n}\right) = (-1)^{\#\{\text{of 1 in } \rho\}}. \tag{6.3}$$

Case B



Consider the state ρ' to be the state right before the measurement. For an entry state $\rho = |i_1, i_2, ..., i_n\rangle$ applying CNOTs successively yields the state $\rho' = |i_1 \oplus i_2 \oplus ... \oplus i_n\rangle = |0\rangle \mathbb{1}_{\{\text{even \# of 1}\}} + |1\rangle \mathbb{1}_{\{\text{odd \# of 1}\}}$. Hence:

$$\langle \mathbb{I} \otimes Z\rangle_{\rho} = \mathrm{Tr}(\rho \mathbb{I} \otimes Z) \Longrightarrow (\rho')^{\dagger} Z \rho' = (-1)^{\#\{\text{of 1 in } \rho\}}. \tag{6.4}$$

Overall these are equivalent representations. Let us look at which case converges faster. For $M_A = Z^{\otimes n}$ and $M_B = \mathbb{I} \otimes Z$ this yields $M_A^2 = M_B^2 = \mathbb{I}$. Thus $\mathrm{Tr}(M_A^2 \rho) = \mathrm{Tr}(\mathbb{I}\rho) = 1$ and $\mathrm{Tr}(M_B^2 \rho) = \mathrm{Tr}(\mathbb{I}\rho) = 1$, meaning that the variances are the same.

6.2 Probability Bounds

In this section we present well-known probability inequalities and look at bounds for the convergence of a measurement strategy. In particular given a random variable Z with expectation $\mathbb{E}(Z)$ we want to know how likely it is that Z is close to $\mathbb{E}(Z)$. In our case, the random variable is the estimator and $\mathbb{E}(Z) = \mathbb{E}\left(\widehat{M}_N\right) = \mathrm{Tr}(\rho M)$.

Proposition 6.2.1 (Markov's Inequality)**.** *Let $t \in \mathbb{R}_{\geq 0}$ and $Z \geq 0$ then:*

$$\mathbb{P}(Z \geq t) \leq \frac{\mathbb{E}(Z)}{t}. \tag{6.5}$$

Proof. We decompose the expectation as:

$$\mathbb{E}(Z) = \mathbb{E}(Z|Z \geq t)\mathbb{P}(Z \geq t) + \underbrace{\mathbb{E}(Z|Z < t)}_{\geq 0 \text{ as } Z \geq 0}\underbrace{\mathbb{P}(Z < t)}_{\geq 0} \geq \mathbb{E}(Z|Z \geq t)\mathbb{P}(Z \geq t) \geq t\mathbb{P}(Z \geq t). \tag{6.6}$$

□

Proposition 6.2.2 (Chebyshev's Inequality)**.** *Let Z be a random variable then:*

$$\mathbb{P}\big(|Z - \mathbb{E}(Z)| \geq t\big) \leq \frac{\mathrm{Var}(Z)}{t^2}. \tag{6.7}$$

Proof. We simply apply Markov's Inequality to find:

$$\mathbb{P}\big(|Z - \mathbb{E}(Z)| \geq t\big) = \mathbb{P}\big((Z - \mathbb{E}(Z))^2 \geq t^2\big) \leq \frac{\mathbb{E}\left[(Z - \mathbb{E}(Z))^2\right]}{t^2} = \frac{\mathrm{Var}(Z)}{t^2}. \tag{6.8}$$

□

This inequality provides a bound on how fast estimators converge:

$$\mathbb{P}\Big(\big|\widehat{M}_N - \mathrm{Tr}(\rho M)\big| \geq t\Big) \leq \frac{\mathrm{Var}\big(\widehat{M}_N\big)}{t^2} = \frac{\mathrm{Var}_\rho(M)}{Nt^2}. \tag{6.9}$$

Suppose now we want an estimator to yield $\mathrm{Tr}(\rho M) \pm \epsilon$ but allows a δ failure probability. Hence we want the number of shots to satisfy the following inequality:

$$\frac{\mathrm{Var}_\rho(M)}{N\epsilon^2} \leq \delta \Longrightarrow N \geq \frac{\mathrm{Var}_\rho(M)}{\delta\epsilon^2}. \tag{6.10}$$

Proposition 6.2.3 (Hoeffding's Inequality)**.** *Let $S_N = \sum_{i=1}^N X_i$ be a sum of i.i.d. random variables $\{X_i\}$ such that $a_i \leq X_i \leq b_i$. Then:*

$$\mathbb{P}\big(|S_N - \mathbb{E}(S_N)| \geq t\big) \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^N (b_i - a_i)^2}\right). \tag{6.11}$$

In our context, $\widehat{M}_N = \frac{1}{N}\sum_{s=1}^N \lambda_{m(s)} = \sum_{s=1}^N \frac{\lambda_{m(s)}}{N} \equiv \sum_{s=1}^N X_s$ and then $S_N = \frac{1}{N}\sum_{s=1}^N \lambda_{m(s)} = \widehat{M}_N$ with $\frac{\lambda_{\min}}{N} \leq \frac{\lambda_{m(s)}}{N} \leq \frac{\lambda_{\max}}{N}$. We consider also:

$$\sum_{i=1}^N (b_i - a_i)^2 = \sum_{s=1}^N \left(\frac{\lambda_{\max} - \lambda_{\min}}{N}\right)^2 := \sum_{s=1}^N \frac{\Delta\lambda^2}{N}, \tag{6.12}$$

where $\Delta\lambda := \lambda_{\max} - \lambda_{\min}$. Thus Hoeffding's Inequality yields:

$$\mathbb{P}\Big(\big|\widehat{M}_N - \mathrm{Tr}(\rho M)\big| \geq t\Big) \leq 2\exp\left(-\frac{2Nt^2}{\Delta\lambda^2}\right). \tag{6.13}$$

Suppose now we want an estimator $\widehat{M}_N$ in the range $\mathrm{Tr}(\rho M) \pm \epsilon$ with a $1-\delta$ probability. Hence we want the number of shots to satisfy the following inequality:

$$2\exp\left(-\frac{2N\epsilon^2}{\Delta\lambda^2}\right) \leq \delta \Longrightarrow N \geq \frac{\Delta\lambda^2}{2\epsilon^2}\log\left(\frac{2}{\delta}\right). \tag{6.14}$$

Note that in the bound above $\Delta\lambda$ is the only quantity that depends on the measurement.

Example 6.2.4 (Fidelity)**.** *Recall that $M_A = |0\rangle\langle 0|$ and $M_B = \mathrm{SWAP}$. For M_A we have $\lambda_{\max} = 1$ and $\lambda_{\min} = 0$ thus $\Delta\lambda = 1$. For $M_B = \mathrm{SWAP}$ we have $\lambda_{\max} = 1$ and $\lambda_{\min} = -1$ thus $\Delta\lambda = 2$. Hence $N_B \geq N_A$.*

Example 6.2.5 (Marco's Problem)**.** *Recall that $M_A = Z^{\otimes n}$ and $M_B = \mathbb{I} \otimes Z$. For M_A and M_B we have $\lambda_{\max} = 1$ and $\lambda_{\min} = -1$ thus $\Delta\lambda = 2$. Hence $N_A = N_B$ implying that the convergence is the same.*

Chapter 7

Distinguishing Quantum Systems and Matrix Norms

Suppose we are given a state ρ_0 or ρ_1 and want to know which we actually have and how best to distinguish them. We would in fact like to find the optimal measurement. Another objective is the ability to distinguish them depending on the properties of ρ_0 and ρ_1. In the absence of prior information about whether the state is ρ_0 or ρ_1, and before performing any measurement, the two states are equally likely.
Then suppose we perform a binary POVM measurement $\{M_0, M_1\}$ to answer the question: is the system in state ρ_0 (associated to M_0) or ρ_1 (associated to M_1)?
The probability of guessing the correct output state and assuming that it is equally likely being given ρ_0 and ρ_1 is given by:

$$\mathbb{P}_{\text{correct guess}} = \sum_{b\in\{0,1\}} \mathbb{P}(\text{system in } \rho_b)\mathbb{P}(\text{get outcome } b|\rho_b) = \frac{1}{2}\text{Tr}(M_0\rho_0) + \frac{1}{2}\text{Tr}(M_1\rho_1) = \frac{1}{2}\big(1+\text{Tr}(M_0(\rho_0-\rho_1))\big),$$

where we have used the fact that $M_1 = \mathbb{I} - M_0$. Then, to maximize the probability of distinguishing we choose M_0 to maximize $\text{Tr}(M_0(\rho_0-\rho_1))$.

Example 7.0.1 (Computational basis of states of single qubit)**.** *Let* $\rho_0 = |0\rangle\langle 0|$ *and* $\rho_1 = |1\rangle\langle 1|$*. Then* $M_0 = |0\rangle\langle 0|$*,* $M_1 = |1\rangle\langle 1|$ *and* $\text{Tr}\big(|0\rangle\langle 0|\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big)\big) = 1$*. Then* $\mathbb{P}_{\textit{correct guess}} = \frac{1}{2}(1+1) = 1$*.*

We now present the generalization of the above example with arbitrary single qubit states. To this end, let $\rho_0 = \frac{1}{2}\big(\mathbb{I} + \underline{r}_0\cdot\underline{\sigma}\big)$ and $\rho_1 = \frac{1}{2}\big(\mathbb{I} + \underline{r}_1\cdot\underline{\sigma}\big)$. We can also expand M in the Pauli basis as:

$$M = a\big(\mathbb{I} + bX + cY + dZ\big) = a\big(\mathbb{I} + \underline{x}\cdot\underline{\sigma}\big), \tag{7.1}$$

where $\underline{x} = (b,c,d)^T$ and $\underline{\sigma} = (X,Y,Z)^T$. Thus using the fact that $\text{Tr}(\sigma_i\sigma_j) = 2\delta_{ij}$ we obtain:

$$\begin{aligned}\mathbb{P}_{\text{correct guess}} = \frac{1}{4}\text{Tr}\big(M(\underline{r}_0-\underline{r}_1)\cdot\underline{\sigma}\big) + \frac{1}{2} = \frac{a}{4}\text{Tr}\big(\big(\mathbb{I}+\underline{x}\cdot\underline{\sigma}\big)(\underline{r}_0-\underline{r}_1)\cdot\underline{\sigma}\big) + \frac{1}{2} = \frac{a}{4}\text{Tr}\big(\big(\underline{x}\cdot\underline{\sigma}\big)(\underline{r}_0-\underline{r}_1)\cdot\underline{\sigma}\big) + \frac{1}{2}\\ = \frac{a}{2}\underline{x}\cdot\big(\underline{r}_0-\underline{r}_1\big) + \frac{1}{2}.\end{aligned}$$

Thus we want to pick $\underline{x}$ that maximizes $\underline{x}\cdot\big(\underline{r}_0-\underline{r}_1\big)$. This is the standard dot product between two vectors which is maximized when the two vectors are aligned, i.e. with $\underline{u}\cdot\underline{v} = |\underline{u}||\underline{v}|\cos(\theta) \Longrightarrow$ maximized when $\theta = 0$. Thus we require $\underline{x} \propto \big(\underline{r}_0-\underline{r}_1\big)$, with the norm $|\underline{x}|$ and a are determined by the constraints $M \geq 0$ and $M \leq \mathbb{I}$. However, we know from writing any state ρ as a sum of Pauli operator that this is equivalent to requiring that $|\underline{x}| \leq 1$ and $a = \frac{1}{2}$. Thus:

$$\underline{x}_{\text{optimal}} = \frac{1}{|\underline{r}_0 - \underline{r}_1|}\big(\underline{r}_0 - \underline{r}_1\big) \qquad \text{and,} \qquad M = \frac{1}{2}\big(\mathbb{I} + \underline{x}_{\text{optimal}} \cdot \underline{\sigma}\big). \tag{7.2}$$

As a result:

$$\mathbb{P}_{\text{correct guess}} = \frac{1}{2} + \frac{1}{4}\frac{\big(\underline{r}_0 - \underline{r}_1\big)\cdot\big(\underline{r}_0 - \underline{r}_1\big)}{|\underline{r}_0 - \underline{r}_1|} = \frac{1}{2} + \frac{1}{4}|\underline{r}_0 - \underline{r}_1|. \tag{7.3}$$

What happens if one of the states is more likely to occur than one of the others? This will add an additional consideration to be taken into account when guessing (e.g. the case where it is 99% probability to be ρ_0). A little bit more subtle, this will also effect the optimum measurement to be performed.

Let us now look at how to distinguish more complex (high dimensional) states.

Proposition 7.0.2. *A useful result in this case is given by:*

$$\mathbb{P}^{\textit{optimal}}_{\textit{correct guess}} = \frac{1}{2}\left(1 + \max_{0\le M\le\mathbb{I}} \text{Tr}\big(M(\rho_0 - \rho_1)\big)\right) = \frac{1}{2}\left(1 + \frac{1}{2}\|\rho_0 - \rho_1\|_1\right), \tag{7.4}$$

where the 1-norm of a matrix A is $\|A\|_1 := \text{Tr}\left(\sqrt{A^\dagger A}\right) = \sum_i \sqrt{\lambda_i\big(A^\dagger A\big)}$ with λ_i the eigenvalues of $A^\dagger A$.

This gives an operational meaning to the 1-norm between two states, namely:

$$\frac{1}{2}\|\rho_0 - \rho_1\|_1 = \max_{0\le M\le\mathbb{I}} \text{Tr}\big(M(\rho_0 - \rho_1)\big). \tag{7.5}$$

Proof. We start from $\mathbb{P}^{\text{optimal}}_{\text{correct guess}} = \frac{1}{2}\left(1 + \max_{0\le M\le\mathbb{I}}\text{Tr}\big(M(\rho_0-\rho_1)\big)\right)$ where the trick to optimizing this quantity is to write $\rho_0 - \rho_1$ as $\rho_0 - \rho_1 = \rho_+ - \rho_-$ where ρ_+ and ρ_- are orthogonal projectors onto the positive and negative parts of $\rho_0 - \rho_1$. That is, considering $|\lambda_i\rangle$ be eigenvectors with associated eigenvalues λ_i of $\rho_0 - \rho_1$ we may write:

$$\rho_0 - \rho_1 = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i| = \underbrace{\sum_{\substack{i \\ \text{s.t. } \lambda_i \ge 0}} \lambda_i |\lambda_i\rangle\langle\lambda_i|}_{:=\rho_+} + \underbrace{\sum_{\substack{i \\ \text{s.t. } \lambda_i < 0}} \lambda_i |\lambda_i\rangle\langle\lambda_i|}_{:=-\rho_-}. \tag{7.6}$$

Therefore we have:

$$\mathbb{P}^{\text{optimal}}_{\text{correct guess}} = \frac{1}{2}\left(1 + \text{Tr}\big(M(\rho_+ - \rho_-)\big)\right). \tag{7.7}$$

Now, to maximize this we want to choose M such that $\text{Tr}(M\rho_+)$ is maximized and $\text{Tr}(M\rho_-)$ is minimized. Since $-\rho_-$ only has negative eigenvalues it implies that ρ_- only has positive eigenvalues. Thus, since M is a positive operator, $\text{Tr}(M\rho_-)$ is only negative and hence it is minimized when $\text{Tr}(M\rho_-) = 0$. This is achieved by $M := M_+$ which in turns maximizes $\text{Tr}(M\rho_+)$ and is given by:

$$M_{\text{optimal}} = M_+ = \sum_{\substack{i \\ \text{s.t. } \lambda_i \ge 0}} |\lambda_i\rangle\langle\lambda_i|. \tag{7.8}$$

Thus, for the optimal probability of a correct guess we find:

$$\mathbb{P}^{\text{optimal}}_{\text{correct guess}} = \frac{1}{2}\left(1 + \text{Tr}\big(M_{\text{optimal}}(\rho_+ - \rho_-)\big)\right) = \frac{1}{2}\left(1 + \text{Tr}(\rho_+)\right). \tag{7.9}$$

Now $\|\rho - \sigma\|_1 = \sum_i |\lambda_i| = \text{Tr}(\rho_+) + \text{Tr}(\rho_-)$ and $\text{Tr}(\rho_+) + \text{Tr}(\rho_-) = \text{Tr}(\rho - \sigma) = 0$ thus $\|\rho - \sigma\|_1 = 2\text{Tr}(\rho_+)$, as required. □

Let us now end this chapter by introducing several other matrix norms.

Definition 7.0.3. *A **matrix norm** is a function $\|\cdot\| : K^{m\times n} \longrightarrow \mathbb{R}$, for some space K, that satisfies the following properties:*
(i) $\|A\| \geq 0$ *- Positive valued.*
(ii) $\|A\| = 0 \Longleftrightarrow A = \underline{0}$.
(iii) $\|\alpha A\| = |\alpha|\|A\|$.
(iv) $\|A+B\| \leq \|A\| + \|B\|$ *- Triangle Inequality.*

Definition 7.0.4. *The **Schatten p-norm** is defined as:*

$$\|A\|_p := \left(\sum_{i=1}^{d} |\lambda_i(A)|^p\right)^{\frac{1}{p}} \equiv \left(\sum_{i=1}^{d} \sqrt{\lambda_i(A^\dagger A)}^p\right)^{\frac{1}{p}}. \tag{7.10}$$

In the case where we have non-square matrices, the norm takes the sum of the p-th power of the singular values of A to the power of $\frac{1}{p}$, i.e. $\|A\|_p := \left(\sum_{i=1}^{d} s_i(A)^p\right)^{\frac{1}{p}}$ where $s_i(A)$ is the i-th singular value of A.

We often encounter three types of Schatten norms:

• The 1-norm given by $\|A\|_1 := \sum_i |\lambda_i|$. This is an operationally meaningful norm (as previously discussed) but it could be hard to work with.

• The 2-norm given by $\|A\|_2 := \left(\sum_i |\lambda_i|^2\right)^{\frac{1}{2}} \left[\equiv \sqrt{\mathrm{Tr}(AA^\dagger)} = \sqrt{\mathrm{Tr}(UDU^\dagger UD^\dagger U^\dagger)} = \sqrt{\mathrm{Tr}(DD^\dagger)} = \sqrt{\sum_i |\lambda_i|^2}\right]$. This doesn't have a nice operational interpretation, however it is easy enough to measure on a quantum computer, e.g. $\mathrm{Tr}\big((\rho-\sigma)(\rho-\sigma)^\dagger\big) = \mathrm{Tr}(\rho^2) + \mathrm{Tr}(\sigma^2) - 2\mathrm{Tr}(\rho\sigma)$ where these quantities may be measured in a swap test or in a Loschmidt echo circuit.

• The Infinity norm given by $\|A\|_\infty := \max_i\{|\lambda_i|\}$. Intuitively, it may be **informally** understood as:

$$\|A\|_\infty \equiv \left(\sum_i |\lambda_i|^\infty\right)^{\frac{1}{\infty}} = \big(\underbrace{|\lambda_{\max}|^\infty}_{\text{dominates!}} + |\lambda_{\text{max-1}}|^\infty + ...\big)^{\frac{1}{\infty}} = |\lambda_{\max}|. \tag{7.11}$$

This is an often operationally meaningful norm, e.g. for an arbitrary ρ we have $|\mathrm{Tr}(M\rho)| \leq |\lambda_{\max}| = \|M\|_\infty \equiv \max_\rho |\mathrm{Tr}(M\rho)|$. However it may be hard to compute. For instance to compute $\|M\|_\infty$ if M is an observable we could use $\max \mathrm{Tr}(M\rho(\theta))$, and to compute $\|\rho\|_\infty$ we could use Principle Component Analysis (PCA) on ρ. Note that in the case where A is not square, the infinity norm is $\|A\|_\infty := \max_i(s_i(A))$ (i.e. the largest singular value of A).

Proposition 7.0.5 $(*)$**.** *The Schatten p-norms are unitarily invariant, i.e. $\|A\|_p = \|UAV\|_p$ for any unitary matrices U and V.*

Proof. Consider the singular value decomposition (SVD) of A to be $A = W_1 \Sigma W_2^\dagger$, where W_1 and W_2 are left/right unitaries with Σ being the rectangular diagonal matrix containing the singular values. Thus, we have that the SVD of UAV is given by $UAV = (UW_1)\Sigma(W_2^\dagger V) = (UW_1)\Sigma(V^\dagger W_2)^\dagger$, where UW_1 and $V^\dagger W_2$ are respectively the new left/right unitaries that perform the SVD. However, the rectangular diagonal matrix Σ remains unchanged (i.e. it has the same singular values).

□

Proposition 7.0.6 (∗)**.** *For unitary matrices U and V we have:*

$$\left\|U^N - V^N\right\|_p \leq N\|U - V\|_p. \tag{7.12}$$

Proof. We can use a telescoping series to rewrite $U^N - V^N$ as follows:

$$U^N - V^N = U\left(\sum_{k=0}^{N-1} U^{N-1-k}V^k\right) - \left(\sum_{k=0}^{N-1} U^{N-1-k}V^k\right)V = \sum_{k=0}^{N-1}\left(U^{N-k}V^k - U^{N-1-k}V^{k+1}\right). \tag{7.13}$$

Thus, we may use the Triangle Inequality first and then the unitary invariance of the Schatten p-norm (Proposition 7.0.5) to find:

$$\begin{aligned}
\left\|U^N - V^N\right\|_p &= \left\|\sum_{k=0}^{N-1}\left(U^{N-k}V^k - U^{N-1-k}V^{k+1}\right)\right\|_p \\
&\leq \sum_{k=0}^{N-1}\left\|U^{N-k}V^k - U^{N-1-k}V^{k+1}\right\|_p \\
&= \sum_{k=0}^{N-1}\left\|\left(U^\dagger\right)^{N-1-k}\left(U^{N-k}V^k - U^{N-1-k}V^{k+1}\right)\left(V^\dagger\right)^k\right\|_p \\
&= \sum_{k=0}^{N-1}\left\|\left(U^\dagger\right)^{N-1-k}U^{N-k}V^k\left(V^\dagger\right)^k - \left(U^\dagger\right)^{N-1-k}U^{N-1-k}V^{k+1}\left(V^\dagger\right)^k\right\|_p \\
&= \sum_{k=0}^{N-1}\left\|\left(U^\dagger\right)^{-1}\left(U^\dagger U\right)^{N-k}\left(VV^\dagger\right)^k - \left(U^\dagger U\right)^{N-1-k}V\left(VV^\dagger\right)^k\right\|_p \\
&= \sum_{k=0}^{N-1}\left\|\left(U^\dagger\right)^{-1} - V\right\|_p = \sum_{k=0}^{N-1}\|U - V\|_p = N\|U - V\|_p,
\end{aligned}$$

where in the third line we have used the fact that, for each k, multiplying by the unitary matrices $\left(U^\dagger\right)^{N-1-k}$ and $\left(V^\dagger\right)^k$ was leaving the Schatten p-norm invariant.

□

Proposition 7.0.7 (∗)**.** *For a general matrix Q we have that:*

$$\|Q\|_1 \geq \|Q\|_2 \geq \|Q\|_\infty. \tag{7.14}$$

Proof. • From the definition of the Schatten norm, we have:

$$\|Q\|_1^2 = \left(\sum_i s_i(Q)\right)^2 = \sum_i s_i^2(Q) + \sum_{i\neq j} s_i(Q)s_j(Q) \geq \sum_i s_i^2(Q) =: \|Q\|_2^2, \tag{7.15}$$

where we have used the fact that singular values are positive. Hence by the positivity of the norms it holds that $\|Q\|_1 \geq \|Q\|_2$.

• From the definition of the Infinity-norm, we have:

$$\|Q\|_2^2 = \sum_i s_i^2(Q) \geq \max_i\left(s_i^2(Q)\right) = \left(\max_i(s_i(Q))\right)^2 =: \|Q\|_\infty^2, \tag{7.16}$$

where we have again used the fact that singular values are positive. Hence by the positivity of the norms it holds that $\|Q\|_2 \geq \|Q\|_\infty$.

□

Proposition 7.0.8 $(*)$**.** *For a general matrix Q and for any $p \geq q \geq 1$ we have that:*

$$\|Q\|_p \leq \|Q\|_q. \tag{7.17}$$

Proof. We will use the general fact that for real positive x_i and $r \geq 0$ we have $\sum_i x_i^{1+r} \leq \left(\sum_i x_i\right)^{1+r}$. Now, for $p \geq q$ we set $\frac{p}{q} = 1 + r$ where $r \geq 0$ to find:

$$\|Q\|_q^p = \left(\sum_i s_i^q(Q)\right)^{\frac{p}{q}} \geq \sum_i s_i^p(Q) =: \|Q\|_p^p. \tag{7.18}$$

Since $p \geq 1$ and the norms are positive it holds that $\|Q\|_q \geq \|Q\|_p$.

□

Remark 7.0.9. *Remark that $p > q$ does <u>not</u> imply that $\|Q\|_p > \|Q\|_q$.*

Chapter 8

Entropy and Information

In this chapter, we introduce notions of Entropy and Information for quantum applications. We first recall the classical Shannon's Entropy and related ideas from classical information theory before introducing its quantum counterpart, Von Neumann's Entropy, with some useful notions and properties in relation to quantum information.

8.1 Shannon Entropy

Consider a random variable X and suppose that we want to know about the uncertainty about X before learning it or the amount of information we are given on learning X. This quantity is given by the entropy.

Definition 8.1.1. *The **entropy** is defined as:*

$$H(X) := -\sum_x p_x \log(p_x). \tag{8.1}$$

We use the convention $\lim_{x\to 0^+} x\log(x) = 0$ and $\log \equiv \log_2$.Now suppose that a source is producing data in the form of random variables $X_1, X_2, ...$, where each random variable may take a character x_k with probability p_k. We would like to know what is the minimal physical resources required to store the data produced by the source.

Theorem 8.1.2 (Shannon's Noiseless Coding Theorem)**.** *An n symbol string can be compressed to $nH(X)$ symbols.*

Sketch of Proof. • Let us cover the binary case first, where $X = 1$ with probability p and $X = 0$ with probability $1-p$. Consider now a data string of length n. In the limit of large n, a typical bit string will contain about $n(1-p)$ zeroes and np ones. There are $\binom{n}{np}$ such typical strings. Thus:

$$\log_2\left(\binom{n}{np}\right) = \log_2\left(\frac{n!}{(np)!(n-np)!}\right) = \log_2(n!) - \log_2\big((np)!\big) - \log_2\big((n-np)!\big). \tag{8.2}$$

For large n we may now use Stirling's approximation $\log(n!) \approx n\log(n) - n$ to find:

$$\begin{aligned}\log_2\left(\binom{n}{np}\right) &\approx n\log_2(n) - n - \big(np\log_2(np) - np + n(1-p)\log_2(n(1-p)) - n(1-p)\big)\\ &= -np\log_2(p) - n(1-p)\log_2(1-p)\\ &= nH(p).\end{aligned}$$

Thus, for the binary entropy, the number of typical strings is approximately $2^{nH(p)}$. The compression strategy assigns a positive integer to each of the possible typical bit strings. There are $2^{nH(p)}$ such strings so $2^{nH(p)}$ letters are required, and each letter can be encoded using $nH(p)$ bits.

• Beyond the binary case, if letter k occurs with probability p_k in a string of length n, then each k will typically occur np_k times. There are $\frac{n!}{\prod_k (np_k)!}$ such typical strings, and thus using Stirling once more yields:

$$\frac{n!}{\prod_k (np_k)!} \approx 2^{nH(X)}. \tag{8.3}$$

Hence there is $nH(X)$ possible binary encoding.

□

Remark 8.1.3. *The completely uniform distribution cannot be compressed since:*

$$H\left(\frac{1}{2}\right) = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) = -\log_2\left(\frac{1}{2}\right) = \log_2(2) = 1. \tag{8.4}$$

Thus n bits are encoded in n bits.

Example 8.1.4. *Suppose a source of information produces 1,2,3 or 4 with probabilities* $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ *and* $\frac{1}{8}$ *respectively. Consider the encoding* $1 \equiv 00$, $2 \equiv 01$, $3 \equiv 10$ *and* $4 \equiv 11$. *On average, the length of a string with this encoding is* $2\times\frac{1}{2} + 2\times\frac{1}{4} + 2\times\frac{1}{8} + 2\times\frac{1}{8} = 2$. *Then, we can use the bias to reduce the amount of symbols required to store data from that source by using less characters to store commonly obtained symbols and more to obtain less likely ones. For example one could consider* $1 \equiv 0$, $2 \equiv 10$, $3 \equiv 110$ *and* $4 \equiv 111$. *On average now, the length of a string with this encoding is* $1\times\frac{1}{2} + 2\times\frac{1}{4} + 3\times\frac{1}{8} + 3\times\frac{1}{8} = \frac{7}{4} < 2$. *This new code is more efficient!*

8.2 Joint, Conditional, Relative Entropy and Mutual Information

Consider two random variables X and Y. We would like to know how is the information content of X related to the one of Y. Conditional entropy and mutual information provide some answers.

Definition 8.2.1. *The* ***joint entropy*** *is defined as:*

$$H(X,Y) := -\sum_x \sum_y p(x,y)\log(p(x,y)). \tag{8.5}$$

The joint entropy provides the total uncertainty about X and Y. Suppose now we know the value of Y so that we have gained $H(Y)$ bits of information.

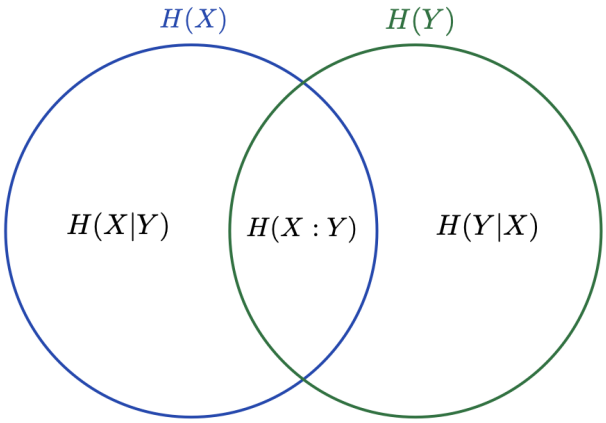
Definition 8.2.2. *The* ***conditional entropy*** *of X given we know Y is defined as:*

$$H(X|Y) := H(X,Y) - H(Y). \tag{8.6}$$

We may also define a quantity that measures the amount of information X and Y have in common, i.e. measures their correlation.

Definition 8.2.3. *The **mutual information** is defined as:*

$$H(X:Y) := H(X) + H(Y) - H(X,Y). \tag{8.7}$$



From the above Venn diagram it is not difficult to see the following properties:

- $0 \leq H(X|Y) \leq H(X)$.
- $H(X|Y) \neq H(Y|X)$.
- $0 \leq H(X:Y) \leq \min(H(X), H(Y))$.

We may finally define a measure of the closeness of two distributions.

Definition 8.2.4. *For two probability distributions $p(x)$ and $q(x)$, the **relative entropy** is defined as:*

$$H(p(x)\|q(x)) := \sum_x p(x) \log\left(\frac{p(x)}{q(x)}\right) = -H(X) - \sum_x p(x)\log(q(x)). \tag{8.8}$$

This measure is also often called the Kullback-Leibler (KL) divergence.

Proposition 8.2.5. *The relative entropy is non-negative: $H(p(x)\|q(x)) \geq 0$.*

Proof. We use the standard inequality $-\log(x) \geq \frac{1-x}{\ln(2)}$ to find:

$$\begin{aligned} H(p(x)\|q(x)) = \sum_x p(x)\log\left(\frac{p(x)}{q(x)}\right) = -\sum_x p(x)\log\left(\frac{q(x)}{p(x)}\right) &\geq \frac{1}{\ln(2)}\sum_x p(x)\left(1 - \frac{q(x)}{p(x)}\right) \\ &= \frac{1}{\ln(2)}\left(\sum_x p(x) - \sum_x q(x)\right) = 0. \end{aligned}$$

□

Finally remark that $H(p(x)\|1/d) = -H(x) - \sum_x p(x)\log(1/d) = \log(d) - H(X)$, thus the Shannon entropy is equivalent to the relative entropy to maximize an uncertain distribution.

8.3 Von Neumann Entropy

We now turn to the quantum counterpart of the entropy of Shannon. Similarly to the classical case, this counterpart $S(\rho)$ quantifies compressibility of quantum information. In fact $\rho^{\otimes n}$ can be compressed to σ that lives on a Hilbert space $\mathcal{H}_c$ with $dim(\mathcal{H}_c) = 2^{nS(\rho)}$.

Definition 8.3.1. *The **Von Neumann entropy** is defined as:*

$$S(\rho) := -\mathrm{Tr}(\rho \log(\rho)) = -\sum_i \lambda_i \log(\lambda_i) \equiv H(\{\lambda_i\}). \tag{8.9}$$

Proposition 8.3.2. *Pure states have zero entropy.*

Proof. Since pure states $\rho = |\psi\rangle\langle\psi|$ only have one eigenvalue equal to 1 and the remaining ones equal to zero, the entropy is just $S(\rho) = -\log(1) = 0$.

□

Proposition 8.3.3. *The Von Neumann entropy is invariant under a change of basis/under unitary transformations. For a unitary matrix U we have:*

$$S(U\rho U^\dagger) = S(\rho). \tag{8.10}$$

Proof. Starting from the LHS we have, by definition, $S(U\rho U^\dagger) = -\mathrm{Tr}\big((U\rho U^\dagger)\log\big(U\rho U^\dagger\big)\big)$. We need to understand the representation of $\log\big(U\rho U^\dagger\big)$. To this end, we look for a matrix A such that $A = \log\big(U\rho U^\dagger\big) \iff \exp(A) = U\rho U^\dagger \iff \rho = U^\dagger \exp(A) U$. By definition of the exponential of a matrix, we find:

$$U^\dagger \exp(A) U := U^\dagger \left(\sum_{n\geq 0} \frac{A^n}{n!}\right) U = \sum_{n\geq 0} \frac{U^\dagger A^n U}{n!} = \sum_{n\geq 0} \frac{(U^\dagger A U)^n}{n!} =: \exp\Big(U^\dagger A U\Big). \tag{8.11}$$

Thus:

$$\rho = \exp\Big(U^\dagger A U\Big) \iff \log(\rho) = U^\dagger A U \iff U^\dagger \log\Big(U\rho U^\dagger\Big) U = \log(\rho) \iff \log\Big(U\rho U^\dagger\Big) = U \log(\rho) U^\dagger.$$

Therefore, we finally obtain:

$$\begin{aligned} S(U\rho U^\dagger) = -\mathrm{Tr}\left((U\rho U^\dagger)\log\Big(U\rho U^\dagger\Big)\right) = -\mathrm{Tr}\big(U\rho U^\dagger U \log(\rho) U^\dagger\big) &= -\mathrm{Tr}\big(U\rho\log(\rho)U^\dagger\big) \\ &= -\mathrm{Tr}\big(U^\dagger U \rho \log(\rho)\big) \\ &= -\mathrm{Tr}(\rho\log(\rho)) = S(\rho). \end{aligned}$$

□

Proposition 8.3.4. *The Von Neumann entropy attains a maximum for the uniform distribution:*

$$\max S(\rho) = S\left(\frac{1}{d}\right) = \log(d). \tag{8.12}$$

Proof. We use a Lagrange multiplier method on a system of dimension d. For a state $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ we need to maximize $S(\rho) = -\sum_i p_i \log(p_i)$ with the constraint $\sum_i p_i = 1$. Then, set the Lagrangian to be

$$\mathcal{L} = -S(\rho) - \lambda_0 \left(\sum_i p_i - 1\right), \tag{8.13}$$

where λ_0 is the Lagrange multiplier. Taking partial derivatives with respect to p_k and λ_0 yields:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial p_k} = \log(p_k) + 1 - \lambda_0 = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_0} = -\left(\sum_i p_i - 1\right) = 0 \end{cases} \tag{8.14}$$

Thus $p_k = e^{\lambda_0 - 1} = constant$ and $\sum_i p_i = \sum_i e^{\lambda_0 - 1} = de^{\lambda_0 - 1} = 1$ which implies that $p_k = e^{\lambda_0 - 1} = \frac{1}{d}$. Overall, $S(\rho)$ is maximized for the uniform distribution $p_k = \frac{1}{d}$ and we finally obtain:

$$\max S(\rho) = -\sum_{k=1}^{d} p_k \log(p_k) = -\sum_{k=1}^{d} \frac{1}{d} \log\left(\frac{1}{d}\right) = S\left(\frac{1}{d}\right) = -\log\left(\frac{1}{d}\right) = \log(d). \tag{8.15}$$

□

Concerning the entropy of measurement, say we measure $M = \sum_j m_j |m_j\rangle \langle m_j|$ with $p(m_j) = \langle m_j | \rho | m_j \rangle$ and the ensemble decomposition $Y = \{|m_j\rangle, p(m_j)\}$ then $H(Y) \geq S(\rho)$. In other words, replacing ρ in any basis with its decohered variant increases entropy.

Lemma 8.3.5 $(*)$**.** *For two states ρ_A and ρ_B, the logarithm of $\rho_A \otimes \rho_B$ is given by:*

$$\log(\rho_A \otimes \rho_B) = \log(\rho_A) \otimes \mathbb{I}_B + \mathbb{I}_A \otimes \log(\rho_B). \tag{8.16}$$

Proof. Consider the eigendecomposition of ρ_A and ρ_B such that $\rho_A = \sum_i \alpha_i |\alpha_i\rangle \langle \alpha_i|$ and $\rho_B = \sum_i \beta_i |\beta_i\rangle \langle \beta_i|$. Note that $\rho_A \otimes \rho_B = \sum_{i,j} \alpha_i \beta_j |\alpha_i\rangle \langle \alpha_i| \otimes |\beta_j\rangle \langle \beta_j|$ is already in diagonal form. Thus:

$$\begin{aligned} \log(\rho_A \otimes \rho_B) = \log\left(\sum_{i,j} \alpha_i \beta_j |\alpha_i\rangle \langle \alpha_i| \otimes |\beta_j\rangle \langle \beta_j|\right) &= \sum_{i,j} \log(\alpha_i \beta_j) |\alpha_i\rangle \langle \alpha_i| \otimes |\beta_j\rangle \langle \beta_j| \\ &= \sum_{i,j} \big(\log(\alpha_i) + \log(\beta_j)\big) |\alpha_i\rangle \langle \alpha_i| \otimes |\beta_j\rangle \langle \beta_j| \\ &= \sum_i \log(\alpha_i) |\alpha_i\rangle \langle \alpha_i| + \sum_j \log(\beta_j) |\beta_j\rangle \langle \beta_j| \\ &= \log(\rho_A) \otimes \mathbb{I}_B + \mathbb{I}_A \otimes \log(\rho_B). \end{aligned}$$

□

Proposition 8.3.6. *The Von Neumann entropy is additive:*

$$S(\rho_A \otimes \rho_B) = S(\rho_A) + S(\rho_B). \tag{8.17}$$

Proof. We use Lemma (8.3.5) and find:

$$\begin{aligned} S(\rho_A \otimes \rho_B) := -\mathrm{Tr}(\rho_A \otimes \rho_B \log(\rho_A \otimes \rho_B)) &= -\mathrm{Tr}\big(\rho_A \otimes \rho_B (\log(\rho_A) \otimes \mathbb{I}_B + \mathbb{I}_A \otimes \log(\rho_B))\big) \\ &= -\mathrm{Tr}\big(\rho_A \log(\rho_A) \otimes \rho_B + \rho_A \otimes \rho_B \log(\rho_B)\big) \\ &= -\mathrm{Tr}(\rho_A \log(\rho_A))\mathrm{Tr}(\rho_B) - \mathrm{Tr}(\rho_A)\mathrm{Tr}(\rho_B \log(\rho_B)) \\ &= -\mathrm{Tr}(\rho_A \log(\rho_A)) - \mathrm{Tr}(\rho_B \log(\rho_B)) \\ &= S(\rho_A) + S(\rho_B). \end{aligned}$$

□

Lemma 8.3.7 (∗)**.** *For a composite system AB in a pure state $|\psi\rangle_{AB}$ it holds that $S(\rho_A) = S(\rho_B)$.*

Proof. When the composite system is in a pure state, the Schmidt decomposition tells us that both subsystems A and B share the same eigenvalues. Thus they also have the same entropy.

□

Proposition 8.3.8 (∗)**.** *The Triangle Inequality with the Von Neumann entropy satisfies:*

$$|S(\rho_A) - S(\rho_B)| \leq S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B). \tag{8.18}$$

*The first inequality is known as the **Araki-Lieb Triangle Inequality** and the second inequality is the **Subadditivity**.*

Proof. We prove each inequality in turns.

• For the second inequality, we use Klein's Inequality $S(\rho\|\sigma) \geq 0$ [Proposition (8.4.2)] with the states $\rho \equiv \rho_{AB}$ and $\sigma \equiv \rho_A \otimes \rho_B$. Then, by Lemma (8.4.7) we obtain:

$$S(\rho_{AB}\|\rho_A \otimes \rho_B) = H(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \geq 0 \Longrightarrow S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B).$$

• For the first inequality, consider a purification ρ_{ABC} of the state ρ_{AB} using a Hilbert space $\mathcal{H}_{ABC}$. From the subadditivity,

$$S(\rho_{AC}) \leq S(\rho_C) + S(\rho_A). \tag{8.19}$$

By construction, $S(\rho_{ABC}) = 0$ since ρ_{ABC} is a pure state. Now, for a composite system ABC in a pure state ρ_{ABC}, by Lemma (8.3.7), it holds that $S(\rho_{AC}) = S(\rho_B)$ and $S(\rho_{AB}) = S(\rho_C)$. Thus, using the subadditivity relation (8.19) we obtain:

$$S(\rho_B) = S(\rho_{AC}) \leq S(\rho_C) + S(\rho_A) = S(\rho_{AB}) + S(\rho_A) \Longrightarrow S(\rho_B) - S(\rho_A) \leq S(\rho_{AB}).$$

By symmetry between the systems A and B we also conclude that $S(\rho_A) - S(\rho_B) \leq S(\rho_{AB})$. Thus, combining the two results yields $|S(\rho_A) - S(\rho_B)| \leq S(\rho_{AB})$.

□

Proposition 8.3.9. *The Von Neumann entropy satisfies a concavity condition:*

$$S\left(\sum_i p_i \rho_i\right) \geq \sum_i p_i S(\rho_i). \tag{8.20}$$

This concavity statement tells us that adding extra randomness only increases uncertainty.

Proof. Consider the state $\rho_{AB} = \sum_i p_i \rho_i \otimes |i\rangle\langle i|$ such that $\rho_A = \sum_i p_i \rho_i$ and $\rho_B = \sum_i p_i |i\rangle\langle i|$. Set also the eigendecomposition of ρ_i to be $\rho_i = \sum_j \lambda_j^i \left|\lambda_j^i\right\rangle\left\langle\lambda_j^i\right|$. Then:

$$\begin{aligned} S(\rho_{AB}) = S\left(\sum_i p_i \rho_i \otimes |i\rangle\langle i|\right) = S\left(\sum_{i,j} p_i \lambda_j^i \left|\lambda_j^i\right\rangle\left\langle\lambda_j^i\right| \otimes |i\rangle\langle i|\right) &= S\left(\sum_{i,j} p_i \lambda_j^i \left|\lambda_j^i i\right\rangle\left\langle\lambda_j^i i\right|\right) \\ &= -\sum_{i,j} p_i \lambda_j^i \log\left(p_i \lambda_j^i\right), \end{aligned}$$

where in this last equality we have used the invariance under a change of basis of the Von Neumann entropy [Proposition 8.3.3]. We now decompose the sum into two parts and apply the fact that $\sum_j \lambda_j^i = 1$.

$$S(\rho_{AB}) = -\sum_{i,j} p_i \lambda_j^i \log(p_i) - \sum_{i,j} p_i \lambda_j^i \log\left(\lambda_j^i\right) = -\sum_i p_i \log(p_i) \sum_j \lambda_j^i + \sum_i p_i S(\rho_i) = H\big(\{p_i\}_i\big) + \sum_i p_i S(\rho_i).$$

On the other hand, $S(\rho_A) = S\left(\sum_i p_i \rho_i\right)$ and $S(\rho_B) = H\big(\{p_i\}_i\big)$. Thus, using the subadditivity of the entropy [Proposition (8.3.8)] we finally find:

$$S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B) \Longrightarrow H\big(\{p_i\}_i\big) + \sum_i p_i S(\rho_i) \leq S\left(\sum_i p_i \rho_i\right) + H\big(\{p_i\}_i\big)$$
$$\Longrightarrow S\left(\sum_i p_i \rho_i\right) \geq \sum_i p_i S(\rho_i).$$

□

Analogously to the classical case we may also define the joint and conditional Von Neumann entropy along with mutual information.

Definition 8.3.10. *The **joint (Von Neumann) entropy** is defined as:*

$$S(\rho_{AB}) := -\mathrm{Tr}\big(\rho_{AB} \log(\rho_{AB})\big). \tag{8.21}$$

Definition 8.3.11. *The **conditional (Von Neumann) entropy** is defined as:*

$$S(\rho_A|\rho_B) := S(\rho_{AB}) - S(\rho_B). \tag{8.22}$$

Definition 8.3.12. *The **(Von Neumann) mutual information** is defined as:*

$$S(\rho_A : \rho_B) := S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \tag{8.23}$$

We remark that the Venn diagram breaks down in this case.

Example 8.3.13 (Conditional entropy can be negative)**.** *Say we consider $\rho_{AB} = \left|\phi^+\right\rangle\left\langle\phi^+\right|$ with $S(\rho_{AB}) = 0$. Then $\rho_A = \rho_B = \frac{1}{2}\mathbb{I}$ such that $S(\rho_A) = -\frac{1}{2}\log\left(\frac{1}{2}\right) - \frac{1}{2}\log\left(\frac{1}{2}\right) = \log(2) = 1$. However, $S(\rho_A|\rho_B) = -1$. The uncertainty in the joint state is less than the one in the reduced state.*

We end this section on a proposition relating entanglement and the conditional entropy.

Proposition 8.3.14 $(*)$**.** *A pure state $|\psi\rangle_{AB}$ is entangled between subsystems A and B if and only if the conditional entropy is less than zero, $S(\rho_A|\rho_B) < 0$.*

Proof. $(\Longrightarrow)$ Consider the definition of the conditional entropy $S(\rho_A|\rho_B) = S(\rho_{AB}) - S(\rho_B)$, where $S(\rho_{AB}) := -\mathrm{Tr}\big(\rho_{AB}\log(\rho_{AB})\big)$ and $S(\rho_B) := -\mathrm{Tr}\big(\rho_B \log(\rho_B)\big)$ with $\rho_B = \mathrm{Tr}_A(\rho_{AB})$ is a reduced state on a subsystem B. When ρ_{AB} is pure, we have that $S(\rho_{AB}) = 0$ and thus $S(\rho_A|\rho_B) = -S(\rho_B)$. Now, if ρ_{AB} is entangled then we have ρ_B to be mixed and $S(\rho_B) > 0$ yielding $S(\rho_A|\rho_B) < 0$.

$(\Longleftarrow)$ Suppose $S(\rho_A|\rho_B) < 0$ then $S(\rho_B) > 0$ implying that ρ_B is mixed. This happens only when ρ_{AB} is entangled. □

8.4 Relative Entropy and the Data Processing Inequality

In this section, we present the Von Neumann version of the relative entropy and we will then focus on an inequality involving channels and entropy (Data Processing Inequality).

Definition 8.4.1. *The **relative (Von Neumann) entropy** is defined as:*

$$S(\rho\|\sigma) := \mathrm{Tr}\big(\rho\log(\rho) - \rho\log(\sigma)\big). \tag{8.24}$$

In particular, if we consider the eigendecompositions of ρ and σ to be $\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$ and $\sigma = \sum_i \mu_i |\mu_i\rangle\langle\mu_i|$ then the relative entropy reads:

$$S(\rho\|\sigma) := \mathrm{Tr}\big(\rho\log(\rho) - \rho\log(\sigma)\big) = \sum_i \lambda_i \left(\log(\lambda_i) - \sum_j |\langle\mu_j|\lambda_i\rangle|^2 \log(\mu_j)\right). \tag{8.25}$$

This expression reduces to the classical relative entropy if we have diagonal states in some basis, however more generally it depends on the overlap between their eigenbasis.

Proposition 8.4.2 ((∗) Klein's Inequality)**.** *The relative entropy is non-negative: $S(\rho\|\sigma) \geq 0$.*

The following proof is taken from Nielsen and Chuang: *Quantum Computation and Quantum Information.*

Proof. Let $\rho = \sum_i p_i |i\rangle\langle i|$ and $\sigma = \sum_j q_j |j\rangle\langle j|$ be the eigendecompositions for ρ and σ. From the definition of the relative entropy, we have:

$$S(\rho\|\sigma) = \sum_i p_i \log(p_i) - \sum_i \langle i|\rho\log(\sigma)|i\rangle. \tag{8.26}$$

Now, since $\langle i|\rho = p_i\langle i|$ and also

$$\langle i|\log(\sigma)|i\rangle = \langle i|\left(\sum_j \log(q_j)|j\rangle\langle j|\right)|i\rangle = \sum_j \log(q_j) P_{ij} \tag{8.27}$$

where $P_{ij} \equiv \langle i|j\rangle\langle j|i\rangle \geq 0$ we have that,

$$S(\rho\|\sigma) = \sum_i p_i \left(\log(p_i) - \sum_j P_{ij}\log(q_j)\right). \tag{8.28}$$

Notice that P_{ij} satisfies the equations $P_{ij} \geq 0$, $\sum_i P_{ij} = 1$ and $\sum_j P_{ij} = 1$. Regarding P_{ij} as a matrix, this property is known as double stochasticity. Furthermore, since $\log(\cdot)$ is a strictly concave function it follows that $\sum_j P_{ij}\log(q_j) \leq \log\left(\sum_j P_{ij} q_j\right) =: r_i$, with equality if and only if there exists a value of j for which $P_{ij} = 1$. Thus,

$$S(\rho\|\sigma) \geq \sum_i p_i \log\left(\frac{p_i}{r_i}\right) \tag{8.29}$$

which has the form of the classical relative entropy. From the positivity of the KL-Divergence (Proposition [8.2.5]) we deduce that $S(\rho\|\sigma) \geq 0$.

□

Proposition 8.4.3 (Non-degeneracy)**.** *The relative entropy is non-degenerate: $S(\rho\|\sigma) = 0 \iff \rho = \sigma$.*

Proof. Following the notation used in the previous proof, $S(\rho\|\sigma) = 0$ if and only if $p_i = r_i$ for all i and P_{ij} is a permutation matrix. To simplify further, note that by relabeling the eigenstates of σ (if necessary), we can assume that P_{ij} is the identity matrix and thus ρ and σ are diagonal in the same basis. The condition $p_i = r_i$ tells us that the corresponding eigenvalues of ρ and σ are identical and therefore $\rho = \sigma$ is the required equality condition.

□

Proposition 8.4.4 (Asymmetry)**.** *The relative entropy is asymmetric: $S(\rho\|\sigma) \neq S(\sigma\|\rho)$.*

Proof. This is a trivial observation. □

Proposition 8.4.5 (Unitary Invariance)**.** *The relative entropy is unitarily invariant: for any unitary matrix U, $S(U\rho U^\dagger\|U\sigma U^\dagger) = S(\rho\|\sigma)$.*

Proof. This property is clear from (8.25), since $S(\rho\|\sigma)$ depends only on the eigenvalues of ρ and σ. □

One may observe, from Proposition (8.4.4), that the relative entropy is not a distance measure. However, from Propositions (8.4.2) and (8.4.3), it can be used to measure the similarity between two quantum states.
We now present (but do not prove) the Data Processing Inequality.

Proposition 8.4.6 (Data Processing Inequality)**.** *For any channel $\mathcal{E}$ we have that,*

$$S(\mathcal{E}(\rho)\|\mathcal{E}(\sigma)) \leq S(\rho\|\sigma), \tag{8.30}$$

where equality holds for unitary evolutions $\mathcal{E}$.

This inequality tells us that there is no channel that we can apply so that it will make ρ and σ more distinguishable. Note also that for any channel $\mathcal{E}$, a similar inequality holds for the 1-norm (but not for the 2-norm):

$$\|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_1 \leq \|\rho - \sigma\|_1. \tag{8.31}$$

Using the above inequality, we may now state and prove an inequality telling us that discarding a quantum system can never increase mutual information. We will use a lemma giving a relationship between the classical mutual information and the quantum relative entropy.

Lemma 8.4.7 (∗)**.** *We have that $H(A:B) = S(\rho_{AB}\|\rho_A \otimes \rho_B)$.*

Proof. We start from the definition of the relative entropy and also use the fact that $\mathrm{Tr}_A(Q_{AB}(\mathbb{I}_A \otimes O_B)) = \mathrm{Tr}_A(Q_{AB})O_B$ along with Lemma (8.3.5). Now:

$$\begin{aligned} S(\rho_{AB}\|\rho_A \otimes \rho_B) &= \mathrm{Tr}(\rho_{AB}\log(\rho_{AB})) - \mathrm{Tr}(\rho_{AB}\log(\rho_A \otimes \rho_B)) \\ &= -S(\rho_{AB}) - \mathrm{Tr}\big(\rho_{AB}(\log(\rho_A) \otimes \mathbb{I}_B + \mathbb{I}_A \otimes \log(\rho_B))\big) \\ &= -S(\rho_{AB}) - \mathrm{Tr}_A\big(\mathrm{Tr}_B(\rho_{AB})\log(\rho_A)\big) - \mathrm{Tr}_B\big(\mathrm{Tr}_A(\rho_{AB})\log(\rho_B)\big) \\ &= -S(\rho_{AB}) + S(\rho_A) + S(\rho_B) = H(A:B). \end{aligned}$$

□

Proposition 8.4.8 (∗). *Discarding a quantum system can never increase mutual information:*

$$H(AB:C) \geq H(B:C). \tag{8.32}$$

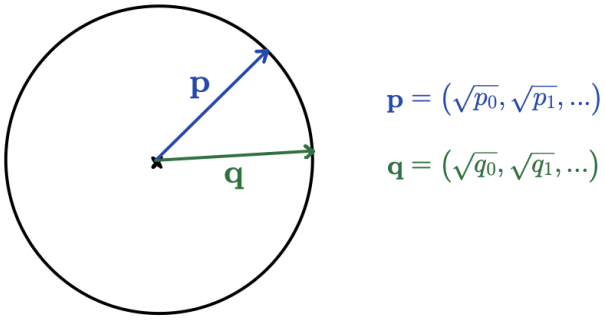
Proof. From Lemma (8.4.7) we first write $H(AB:C) = S(\rho_{ABC}\|\rho_{AB}\otimes\rho_C)$ and $H(B:C) = S(\rho_{BC}\|\rho_B\otimes\rho_C)$. We now use the Data Processing Inequality [Proposition (8.4.6)] with the channel to be the partial trace over the system A, i.e. $\mathcal{E}(\cdot) = \mathrm{Tr}_A(\cdot)$. Setting also $\rho \equiv \rho_{ABC}$ and $\sigma \equiv \rho_{AB}\otimes\rho_C$ in the inequality will yield the result.

□

8.5 Mixed State Fidelity

In this section we present the notion of mixed state fidelity where we first present the classical fidelity. Classically, the fidelity is a measure of similarity between two distributions $\mathbf{p}$ and $\mathbf{q}$. In particular, it is defined as the inner product between the square root of the distribution vectors which lie on the unit sphere. Formally it is given by,

$$F(\mathbf{p},\mathbf{q}) = \sum_i \sqrt{p_i q_i}. \tag{8.33}$$



It is not difficult to see that (i) $F(\mathbf{p},\mathbf{p}) = 1 = \max F(\mathbf{p},\mathbf{q})$, (ii) $F(\mathbf{p},\mathbf{q}) = 0$ if the distributions have no overlapping support, and (iii) $F(\mathbf{p},\mathbf{q}) \geq 0$. We now present two equivalent versions of the definition of the mixed state fidelity.

Definition 8.5.1 (Version 1). *The **mixed state fidelity** is defined as:*

$$F(\rho,\sigma) := \min_{\{M_i\}} \sum_i \sqrt{\mathrm{Tr}(\rho M_i)\mathrm{Tr}(\sigma M_i)}. \tag{8.34}$$

In other words, the POVM that best discriminates two states via the classical fidelity. An equivalent definition is given by a single trace.

Definition 8.5.2 (Version 2). *The **mixed state fidelity** is defined as:*

$$F(\rho,\sigma) := \mathrm{Tr}\left(\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}}\right). \tag{8.35}$$

Proposition 8.5.3 (∗). *The two versions of the definition of the mixed state fidelity are equivalent, i.e.:*

$$F(\rho,\sigma) := \mathrm{Tr}\left(\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}}\right) = \min_{\{M_i\}} \sum_i \sqrt{\mathrm{Tr}(\rho M_i)\mathrm{Tr}(\sigma M_i)}. \tag{8.36}$$

Proof. (Christopher A. Fuchs & Carlton M. Caves - 1994) Consider the polar decomposition $\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}} = \sqrt{\rho}\sqrt{\sigma}U$. To see this decomposition, take Lemma (8.5.5) and let $A = \sqrt{\rho}\sqrt{\sigma}$ to achieve $|A| = \sqrt{AA^\dagger} = \sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}$. Thus we have that $A = |A|U \Longrightarrow AU^\dagger = |A|$, and hence $\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\widetilde{U} = \sqrt{\rho}\sqrt{\sigma}$ with $\widetilde{U} = U^\dagger$ as required. Now, note that,

$$F(\rho,\sigma) = \mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma}U) = \sum_i \mathrm{Tr}\left(\sqrt{\rho}\sqrt{M_i}\sqrt{M_i}\sqrt{\sigma}U\right). \tag{8.37}$$

Thus, Cauchy-Schwarz Inequality gives us the classical fidelity,

$$F(\rho,\sigma) \leq \sum_i \sqrt{\mathrm{Tr}(\rho M_i)\mathrm{Tr}(\sigma M_i)} = F(p_i,q_i), \tag{8.38}$$

where $p_i = \mathrm{Tr}(\rho M_i)$ and $q_i = \mathrm{Tr}(\sigma M_i)$ thus establishing that,

$$F(\rho,\sigma) \leq \min_{\{M_i\}} F(p_i,q_i) = \min_{\{M_i\}} \sum_i \sqrt{\mathrm{Tr}(\rho M_i)\mathrm{Tr}(\sigma M_i)}. \tag{8.39}$$

To have equality, we need to find a POVM $\{M_i\}$ such that the Cauchy-Schwarz Inequality is satisfies with equality for each term in the sum. That is, $\sqrt{M_i}\sqrt{\rho} = \alpha_i\sqrt{M_i}\sqrt{\sigma}U$ for some set of complex numbers $\{\alpha_i\}$. However, $\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}} = \sqrt{\rho}\sqrt{\sigma}U$ and thus for an invertible ρ we have $\sqrt{\sigma}U = \rho^{-\frac{1}{2}}\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}}$. Substituting the later equation, we find that the equality conditions are,

$$\sqrt{M_i}(\mathbb{I} - \alpha_i E) = 0, \tag{8.40}$$

where $E \equiv \rho^{-\frac{1}{2}}\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}}\rho^{-\frac{1}{2}}$. If $E = \sum_i \beta_i |i\rangle\langle i|$ is a spectral decomposition for E then we choose $M_i = |i\rangle\langle i|$ and $\alpha_i = 1/\beta_i$. The case of non-invertible ρ follows from continuity.

□

For special types of density operators ρ and σ we can, for example, recover the classical fidelity or also obtain considerable simplifications in the expressions.

• Suppose that ρ and σ **commute**, then the mixed state fidelity reduces to the classical fidelity. Indeed, considering the eigendecompositions $\rho = \sum_i r_i |i\rangle\langle i|$ and $\sigma = \sum_i s_i |i\rangle\langle i|$ we find:

$$F(\rho,\sigma) = \mathrm{Tr}\left(\sqrt{\sum_i r_i s_i |i\rangle\langle i|}\right) = \mathrm{Tr}\left(\sum_i \sqrt{r_i s_i}|i\rangle\langle i|\right) = \sum_i \sqrt{r_i s_i} = F(\mathbf{r},\mathbf{s}). \tag{8.41}$$

• Suppose that ρ is a **pure state** such that $\rho = |\psi\rangle\langle\psi|$. Then, $\rho^2 = \rho \Longrightarrow \rho^{\frac{1}{2}} = \rho = |\psi\rangle\langle\psi|$. Thus:

$$F(\rho,\sigma) = \mathrm{Tr}\left(\sqrt{|\psi\rangle\langle\psi|\sigma|\psi\rangle\langle\psi|}\right) = \mathrm{Tr}\left(\sqrt{\langle\psi|\sigma|\psi\rangle}\sqrt{|\psi\rangle\langle\psi|}\right) = \sqrt{\langle\psi|\sigma|\psi\rangle}\mathrm{Tr}(\rho) = \sqrt{\langle\psi|\sigma|\psi\rangle}. \tag{8.42}$$

Hence the fidelity between a pure state ρ and a mixed state σ is precisely equal to the overlap.

• Suppose that ρ and σ are **pure states** such that $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$. Then, the mixed state fidelity reduces further to:

$$F(\rho,\sigma) = \big|\langle\psi|\phi\rangle\big|. \tag{8.43}$$

We remark that we are missing a $|\cdot|^2$ from the usual literature in this case. This is a matter of convention, where here we are following the book *Quantum Computation and Quantum Information* by Nielsen and Chuang.
The nice operation in all this would be to find some nice formula for the general case. This is given to us by Uhlmann's Theorem.

Theorem 8.5.4 (Uhlmann's Theorem)**.** *The mixed state fidelity may be given by*

$$F(\rho,\sigma) = \max_{|\psi\rangle,|\varphi\rangle} \big|\langle\psi|\varphi\rangle\big|, \tag{8.44}$$

where the maximum is taken over all the possible purifications of ρ and σ, i.e. $\rho_S = \mathrm{Tr}_R\big(|\psi\rangle\langle\psi|_{RS}\big)$ and $\sigma_S = \mathrm{Tr}_R\big(|\varphi\rangle\langle\varphi|_{RS}\big)$.

To prove this Theorem we first list four lemmas along with their proofs. The first two of these lemmas (with the proofs) are taken from Nielsen and Chuang: *Quantum Computation and Quantum Information.*

Lemma 8.5.5 $(*)$**.** *For any complex square matrix A can be expressed in the polar decomposition $A = |A|U$ where $|A| = \sqrt{AA^\dagger}$ and U is some unitary matrix.*

Proof. We consider the SVD of A which has the form $A = UDW^\dagger$ with D a positive diagonal matrix and U, W some unitaries. Notice that $\sqrt{AA^\dagger} = \sqrt{UD^2U^\dagger} = U\sqrt{D^2}U^\dagger = UDU^\dagger$. Therefore, we may write $A = UDW^\dagger = UD(U^\dagger U)W^\dagger = |A|V$ with $V = UW^\dagger$.

□

Lemma 8.5.6 $(*)$**.** *Let A be any operator and U a unitary. Then $|\mathrm{Tr}(AU)| \leq \mathrm{Tr}(|A|)$, with equality being attained by choosing $U = V^\dagger$ and where $A = |A|V$ is the polar decomposition of A.*

Proof. Equality is clearly attained under the conditions stated. Now observe that

$$|\mathrm{Tr}(AU)| = |\mathrm{Tr}(|A|VU)| = \big|\mathrm{Tr}\big(|A|^{\frac{1}{2}}|A|^{\frac{1}{2}}VU\big)\big|. \tag{8.45}$$

Then, the Cauchy-Schwarz Inequality for the Hilbert-Schmidt inner product $\langle A, B\rangle_{HS} = \mathrm{Tr}(B^*A)$ yields:

$$|\mathrm{Tr}(AU)| \leq \sqrt{\mathrm{Tr}(|A|)\mathrm{Tr}\big(U^\dagger V^\dagger |A| VU\big)} = \mathrm{Tr}(|A|). \tag{8.46}$$

□

Lemma 8.5.7 $(*)$**.** *The purification of a state ρ can be written as $|\psi\rangle = U_R \otimes \sqrt{p}U_S\,|\mathrm{vec}(\mathbb{I})\rangle$, where U_S and U_R are unitaries.*

Proof. Consider the eigendecomposition of a state as $\rho = \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k|$. Any purified state of ρ must respect the following form of the Schmidt decomposition

$$|\psi\rangle = \sum_k \sqrt{\lambda_k}\,|\Phi_k\rangle \otimes |\lambda_k\rangle\,, \tag{8.47}$$

where $\{|\Phi_k\rangle\}$ are orthogonal states in the extended system. We now check that $|\psi\rangle = U_R \otimes \sqrt{p}U_S\,|\mathrm{vec}(\mathbb{I})\rangle$ indeed respects the form of (8.47). By plugging $\sqrt{\rho} = \sum_k \sqrt{\lambda_k}\,|\lambda_k\rangle\langle\lambda_k|$ and $|\mathrm{vec}(\mathbb{I})\rangle = \sum_i |i\rangle \otimes |i\rangle$ we have:

$$U_R \otimes \sqrt{p} U_S \left|\mathrm{vec}(\mathbb{I})\right\rangle = U_R \otimes \left(\sum_k \sqrt{\lambda_k} \left|\lambda_k\right\rangle \left\langle\lambda_k\right|\right) U_S \left(\sum_i \left|i\right\rangle \otimes \left|i\right\rangle\right)$$
$$= \sum_k \sqrt{\lambda_k} \left(\sum_i \left\langle\lambda_k\right| U_S \left|i\right\rangle U_R \left|i\right\rangle\right) \otimes \left|\lambda_k\right\rangle = \sum_k \sqrt{\lambda_k} \left|\widetilde{\Phi}_k\right\rangle \otimes \left|\lambda_k\right\rangle .$$

Remark that $\left\{ \left|\widetilde{\Phi}_k\right\rangle \right\}_k$ are indeed orthogonal states, that is by using the identity $\mathbb{I} = \sum_i \left|i\right\rangle \left\langle i\right|$ we have $\langle \widetilde{\Phi}_k | \widetilde{\Phi}_{k'} \rangle = \langle \lambda_k | \lambda_{k'} \rangle = \delta_{k,k'}$. Thus, it respects the generic form of the purified state.

□

Lemma 8.5.8 (∗)**.** *We have:*

$$\left\langle \mathrm{vec}(\mathbb{I}) \right| A^* \otimes B \left| \mathrm{vec}(\mathbb{I}) \right\rangle = \mathrm{Tr}\big(A^\dagger B\big). \tag{8.48}$$

Proof. We explicitly expand the expression:

$$\left\langle \mathrm{vec}(\mathbb{I}) \right| A^* \otimes B \left| \mathrm{vec}(\mathbb{I}) \right\rangle = \sum_{i,j} (\left\langle j\right| \otimes \left\langle j\right|)(A^* \otimes B)(\left|i\right\rangle \otimes \left|i\right\rangle) = \sum_{i,j} \left\langle j\right| A^* \left|i\right\rangle \left\langle j\right| B \left|i\right\rangle$$
$$= \sum_{i,j} \left\langle i\right| A^\dagger \left|j\right\rangle \left\langle j\right| B \left|i\right\rangle = \mathrm{Tr}\big(A^\dagger B\big),$$

where we have used the fact that $\left\langle j\right| A^* \left|i\right\rangle = \left\langle i\right| A^\dagger \left|j\right\rangle$ and in the last equality we used $\mathbb{I} = \sum_j \left|j\right\rangle \left\langle j\right|$.

□

Proof. (Proof of Theorem 8.5.4 - (∗)) Fix orthonormal bases $\left|i\right\rangle_R$ and $\left|i\right\rangle_Q$ in systems R and Q. Since R and Q have the same dimension, the index i may be assumed to run over the same set of values. Define $\left|\mathrm{vec}(\mathbb{I})\right\rangle \equiv \sum_i \left|i\right\rangle_R \otimes \left|i\right\rangle_Q$. Let $\left|\psi\right\rangle$ be any purification of ρ. By Lemma (8.5.7) we have that

$$\left|\psi\right\rangle = (U_R \otimes \sqrt{\rho} U_Q) \left|\mathrm{vec}(\mathbb{I})\right\rangle , \tag{8.49}$$

for some unitary operators U_R and U_Q on systems R and Q. Similarly, if $\left|\varphi\right\rangle$ is any purification of σ then there exists unitary operators V_R and V_Q such that

$$\left|\varphi\right\rangle = (V_R \otimes \sqrt{\sigma} V_Q) \left|\mathrm{vec}(\mathbb{I})\right\rangle . \tag{8.50}$$

Taking the inner product yields,

$$\left|\langle\psi|\varphi\rangle\right| = \left| \left\langle \mathrm{vec}(\mathbb{I}) \right| \left(U_R^\dagger V_R \otimes U_Q^\dagger \sqrt{\rho}\sqrt{\sigma} V_Q \right) \left| \mathrm{vec}(\mathbb{I}) \right\rangle \right|. \tag{8.51}$$

Using Lemma (8.5.8) we find,

$$\left|\langle\psi|\varphi\rangle\right| = \left| \mathrm{Tr}\left(V_R^\dagger U_R U_Q^\dagger \sqrt{\rho}\sqrt{\sigma} V_Q \right) \right|. \tag{8.52}$$

Setting $U \equiv V_Q V_R^\dagger U_R U_Q^\dagger$ we see that,

$$\left|\langle\psi|\varphi\rangle\right| = \left|\mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma} U)\right|. \tag{8.53}$$

By Lemma (8.5.6) we finally have,

$$\left|\langle\psi|\varphi\rangle\right| \leq \mathrm{Tr}(\left|\sqrt{\rho}\sqrt{\sigma}\right|) = \mathrm{Tr}\left(\sqrt{\rho^{\frac{1}{2}} \sigma \rho^{\frac{1}{2}}} \right) =: F(\rho, \sigma). \tag{8.54}$$

To see that equality may be attained, suppose $\sqrt{\rho}\sqrt{\sigma} = |\sqrt{\rho}\sqrt{\sigma}|V$ is the polar decomposition of $\sqrt{\rho}\sqrt{\sigma}$. Choosing $U_Q = U_R = V_R = \mathbb{I}$ and $V_Q = V^\dagger$ we have that equality is attained.

□

Remark that Uhlmann's formula does not provide a calculation tool for evaluating the fidelity. However, in many instances, properties of the fidelity are more easily proved using Uhlmann's formula. For example, from this formula, it clear that (i) the fidelity is symmetrix in its inputs, $F(\rho,\sigma) = F(\sigma,\rho)$, (ii) the fidelity is bounded between 0 and 1, $0 \leq F(\rho,\sigma) \leq 1$, (iii) if $\rho = \sigma$ then $F(\rho,\sigma) = F(\rho,\rho) = 1$, or (iv) if $\rho \neq \sigma$ then $|\psi\rangle \neq |\varphi\rangle$ for any purifications $|\psi\rangle$ and $|\varphi\rangle$ of ρ and σ and thus $F(\rho,\sigma) < 1$. For explicit computations, one would use the results from Proposition (8.5.3).
We also note that the Data Processing Inequality holds for the mixed state fidelity.

Proposition 8.5.9 $(*)$**.** *Suppose $\mathcal{E}$ is a trace-preserving quantum operation. Let ρ and σ be density operators. Then:*

$$F\big(\mathcal{E}(\rho),\mathcal{E}(\sigma)\big) \geq F(\rho,\sigma). \tag{8.55}$$

Proof. Let $|\psi\rangle$ and $|\varphi\rangle$ be purifications of ρ and σ into a joint system RQ such that $F(\rho,\sigma) = |\langle\psi|\varphi\rangle|$. Introduce a model environment E for the quantum operation $\mathcal{E}$, which starts in a pure state $|0\rangle$ and interacts with the quantum system Q via a unitary interaction U. Note that $U|\psi\rangle|0\rangle$ is a purification of $\mathcal{E}(\rho)$ and $U|\varphi\rangle|0\rangle$ is a purification of $\mathcal{E}(\sigma)$. Then, by Uhlmann's Theorem (8.5.4) it follows that:

$$F\big(\mathcal{E}(\rho),\mathcal{E}(\sigma)\big) \geq |\langle\psi|\langle 0|U^\dagger U|\varphi\rangle|0\rangle| = |\langle\psi|\varphi\rangle| = F(\rho,\sigma). \tag{8.56}$$

□

Example 8.5.10 (Computing the fidelity)**.** *We give two examples for computing the fidelity.*

(1) For $\rho = |\Phi_+\rangle\langle\Phi_+|$ and $\sigma = |00\rangle\langle 00|$ two pure states, we have that $F(\rho,\sigma) = |\langle\Phi_+|00\rangle| = \frac{1}{\sqrt{2}}$.

(2) Let $\rho = \frac{3}{4}|\Phi_+\rangle\langle\Phi_+| + \frac{1}{4}|\Psi_-\rangle\langle\Psi_-|$ and $\sigma = \frac{3}{7}|00\rangle\langle 00| + \frac{4}{7}|{+}{+}\rangle\langle{+}{+}|$. The strategy in this case is to pick ρ in such a way that $\sqrt{\rho}$ is easy to compute, i.e. this is usually a pure quantum state or a state that is already in diagonal form. Thus, we remark that $\rho = \frac{3}{4}|\Phi_+\rangle\langle\Phi_+| + \frac{1}{4}|\Psi_-\rangle\langle\Psi_-|$ is diagonal in the Bell basis and corresponds to $\sqrt{\rho} = \frac{\sqrt{3}}{2}|\Phi_+\rangle\langle\Phi_+| + \frac{1}{2}|\Psi_-\rangle\langle\Psi_-|$. Thus, the fidelity is:

$$\begin{aligned}
F(\rho,\sigma) &= \mathrm{Tr}\sqrt{\left(\frac{\sqrt{3}}{2}|\Phi_+\rangle\langle\Phi_+| + \frac{1}{2}|\Psi_-\rangle\langle\Psi_-|\right)\left(\frac{3}{7}|00\rangle\langle 00| + \frac{4}{7}|{+}{+}\rangle\langle{+}{+}|\right)\left(\frac{\sqrt{3}}{2}|\Phi_+\rangle\langle\Phi_+| + \frac{1}{2}|\Psi_-\rangle\langle\Psi_-|\right)} \\
&= \mathrm{Tr}\sqrt{\left(\frac{3\sqrt{3}}{14}\frac{1}{\sqrt{2}}|\Phi_+\rangle\langle 00| + \frac{2\sqrt{3}}{7}\frac{1}{\sqrt{2}}|\Phi_+\rangle\langle{+}{+}|\right)\left(\frac{\sqrt{3}}{2}|\Phi_+\rangle\langle\Phi_+| + \frac{1}{2}|\Psi_-\rangle\langle\Psi_-|\right)} \\
&= \mathrm{Tr}\sqrt{\frac{3\sqrt{3}}{14}\frac{1}{2}\frac{\sqrt{3}}{2}|\Phi_+\rangle\langle\Phi_+| + \frac{2\sqrt{3}}{7}\frac{1}{2}\frac{\sqrt{3}}{2}|\Phi_+\rangle\langle\Phi_+|} \\
&= \mathrm{Tr}\sqrt{\frac{3}{8}|\Phi_+\rangle\langle\Phi_+|} = \sqrt{\frac{3}{8}}Tr\big(|\Phi_+\rangle\langle\Phi_+|\big) = \sqrt{\frac{3}{8}},
\end{aligned}$$

where we have used the facts that $\langle 00|\Phi_+\rangle = \langle{+}{+}|\Phi_+\rangle = \frac{1}{\sqrt{2}}$ and $\langle 00|\Psi_-\rangle = \langle{+}{+}|\Psi_-\rangle = 0$.

Chapter 9

Entanglement Theory

In this final chapter, we first start by studying Resource Theory by stating the resource theory of entanglement. We will then provide basic majorization identities and bounds, and state the Nielson's Majorization Theorem which is used to determine the type of pure states that can be interconverted via LOCC. After providing an explanation of the Von Neumann entropy as a natural measure of entanglement, we will end the chapter with the notion of entanglement witnesses.

9.1 Resource Theory of Entanglement and Interconvertibility

A **property** may be seen as a Hermitian observable, and/or something that is detectable via a POVM. However, this is a rather restrictive view, and we may be interested in more elusive properties like *entanglement*, *Gaussianity*, *non-classicality*, etc. Resource Theories are all about defining properties by what they are not. When defining Resource Theory, one needs to specify a set F of **free operations**. The states that can be generated by free operations are called **free states**, and those that cannot are called **resource states**. The later case allows one to perform operations which were otherwise impossible to perform. We can then consider two points:
(i) Interconversion $\longrightarrow$ Can we transform $A \to B$ via free operations?
(ii) Quantification $\longrightarrow$ If the answer is yes to interconversion, then A has more resource than B.

Definition 9.1.1. *A **quantum resource theory** is defined by a set of free operations F which is closed under composition and contains the tracing out operation $\rho_{AB} \mapsto \mathrm{Tr}_B(\rho_{AB})$. The subset of free preparations $F_{states} \subset F$ are called **free states**, and the set of states $R = F^c \cap D$ that are not free are called **resource states** where $D := \{\rho \in \mathcal{B}(H) : \mathrm{Tr}(\rho) = 1, \rho \geq 0\}$ [D is the set of bounded linear operators that are positive and normalized].*

A resource state allows us to enlarge the set of possible transformations.

Proposition 9.1.2 (Simulation via resources)**.** *We can simulate $\mathcal{E} \notin F$ using a state σ if there is a free operation $\widetilde{\mathcal{E}} \in F$ such that, for all ρ,*

$$\mathrm{Tr}\big(\widetilde{\mathcal{E}}(\rho \otimes \sigma)\big) = \mathcal{E}(\rho). \tag{9.1}$$

Resource theories induce a **partial ordering** determined by which states can be converted to which other states using free operations. For the remainder of the chapter, if $\rho_A \longrightarrow \rho_B$ via free operations we will write $\boldsymbol{\rho_A \prec \rho_B}$. Note that we may have one-way orderings $\rho_A \longrightarrow \rho_B$ but also two-way orderings $\rho_A \longleftrightarrow \rho_B$. Remark also that we have transitivity in the conversions, such that if $\rho_A \prec \rho_B$ and $\rho_B \prec \rho_C$ then $\rho_A \prec \rho_C$. However, the ordering is only partial as we do not require any two states to be related by $\prec$.

Definition 9.1.3. *A function $M : D \to \mathbb{R}$ is called a **resource measure** if, for all $\mathcal{E} \in F$ and ρ we have,*

$$M(\rho) \geq M\big(\mathcal{E}(\rho)\big). \tag{9.2}$$

In other words, free operations can only decrease the measure of the resource. Let us now shift our interest to the Resource Theory of Entanglement, where we first think about what cannot create entanglement.

• **Local Operations (LO)**, i.e. Alice applies quantum gates to her setup and Bob applies quantum gates to his. These are of the form,

$$\mathcal{E} = \mathcal{E}_A \otimes \mathcal{E}_B. \tag{9.3}$$

• **Classical Communication (CC)**, i.e. between Alice and Bob. Note that classical information is information that is diagonal in the computational basis. Communication involves a relabeling of who owns it, e.g. we have the transformation $\rho_A \otimes |k\rangle\langle k|_A \otimes \sigma_B \longrightarrow \rho_A \otimes |k\rangle\langle k|_B \otimes \sigma_B$. More generally, it is the set of maps of the form,

$$C_{A\to B}(\chi) = \sum_n |n\rangle\langle n|_B \langle n|\chi|n\rangle_A . \tag{9.4}$$

Note that this allows Bob to do something locally that is conditional on the outcome of Alice's measurement. We then have the form, $\rho_{AB} \longrightarrow \sum_i (M_i \otimes U_i)\rho_{AB}\big(M_i^\dagger \otimes U_i^\dagger\big)$. Using controlled unitaries one can show that this operation can be done via a channel of the form (9.4).
As a result, we consider the two previous points as the free operations for the Resource Theory of Entanglement.

Definition 9.1.4 (LOCC - The Resource Theory of Entanglement)**.** *The class **LOCC** consists of local operations and classical communications, and is being generated by finite combinations of LO and CC. The resource theory with $F = LOCC$ is the Resource Theory of Entanglement.*

The **free states of the Resource Theory of Entanglement** are those that can be generated by LOCC and have the form,

$$\rho_{AB} = \sum_k p_k \sigma_A^{(k)} \otimes \sigma_B^{(k)}. \tag{9.5}$$

This is also the set of **separable states $D_{\mathbf{Sep}}$**. Thus, the resource states are then defined to be all non-separable quantum states, i.e. the entangled states.

Let us now think about the **interconvertibility of resources**, where the question at hand is to know when it is possible to transform $|\psi\rangle_{AB} \xrightarrow{\text{LOCC}} |\phi\rangle_{AB}$ deterministically using only LOCC.

Example 9.1.5. *Consider the two states $\big|\phi^+\big\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $|\theta\rangle_{AB} = \cos(\theta)|00\rangle + \sin(\theta)|11\rangle$. Can we transform $\big|\phi^+\big\rangle \xrightarrow{LOCC} |\theta\rangle_{AB}$ via LOCC? The answer is yes, and let us see how!*

• ***Step 1**: Alice performs a measurement,*

$$A_1 = \begin{pmatrix} \cos(\theta) & 0 \\ 0 & \sin(\theta) \end{pmatrix} \quad \textit{and,} \quad A_2 = \begin{pmatrix} \sin(\theta) & 0 \\ 0 & \cos(\theta) \end{pmatrix} \tag{9.6}$$

where we recognize A_1 and A_2 to be Kraus operators, and set the POVM $\mathcal{M}_A = \big\{A_1^\dagger A_1, A_2^\dagger A_2\big\}$.

• ***Step 2**: We now state each outcome. If Alice gets outcome 1 she does nothing, and if she gets outcome 2 she performs a bit-flip.*

• **Step 3**: Now, Alice tells Bob the outcome and conditional on this measurement outcome Bob does nothing ($y1$) or performs a bit-flip ($y2$). Thus, in the case where we have outcome 1 we perform $A_1 \otimes \mathbb{I}$ and for the outcome 2 we perform $A_2 \otimes \mathbb{I}$. Thus,

$$\left|\phi^+\right\rangle \xrightarrow{A_1\otimes\mathbb{I}} (A_1 \otimes \mathbb{I})\left|\phi^+\right\rangle = \frac{1}{\sqrt{2}}\big(\cos(\theta)\left|00\right\rangle + \sin(\theta)\left|11\right\rangle\big) = \left|\theta\right\rangle_{AB}. \tag{9.7}$$

$$\left|\phi^+\right\rangle \xrightarrow{A_2\otimes\mathbb{I}} (A_2 \otimes \mathbb{I})\left|\phi^+\right\rangle = \frac{1}{\sqrt{2}}\big(\sin(\theta)\left|00\right\rangle + \cos(\theta)\left|11\right\rangle\big). \tag{9.8}$$

We remark that in the case of $A_2 \otimes \mathbb{I}$ we do not yet have $\left|\theta\right\rangle_{AB}$. However, applying $X_A \otimes X_B$ yields,

$$\left|\phi^+\right\rangle \xrightarrow{A_2\otimes\mathbb{I}} \frac{1}{\sqrt{2}}\big(\sin(\theta)\left|00\right\rangle + \cos(\theta)\left|11\right\rangle\big) \xrightarrow{X_A\otimes X_B} \frac{1}{\sqrt{2}}\big(\sin(\theta)\left|11\right\rangle + \cos(\theta)\left|00\right\rangle\big) = \left|\theta\right\rangle_{AB}. \tag{9.9}$$

We have thus transformed $\left|\phi^+\right\rangle \xrightarrow{\text{LOCC}} \left|\theta\right\rangle_{AB}$ via LOCC.

• **Other cases:** Can we transform $\left|\phi^+\right\rangle \xrightarrow{\text{LOCC}} \left|\phi\right\rangle_{AB}$, where $\left|\phi\right\rangle_{AB}$ is any 2-qubit state, via LOCC? Yes, since we can transform $\left|\theta\right\rangle \longrightarrow \cos(\theta)\left|00\right\rangle + e^{i\phi}\sin(\theta)\left|11\right\rangle$ via local phase gates.
For transformations in the form $\left|\phi\right\rangle_{AB} \xrightarrow{\text{LOCC}} \left|\phi^+\right\rangle_{AB}$ or $\left|\phi\right\rangle_{AB} \xrightarrow{\text{LOCC}} \left|\psi\right\rangle_{AB}$, it is much harder and in fact not possible in general!

In conclusion, LOCC defines a partial order on the set of pure bipartite quantum states, i.e. given an initial entangled quantum state we can only reach certain other entangled states via LOCC.

9.2 Majorization Theory

In this section, we present an ordering relationship between two vectors $\mathbf{x}$ and $\mathbf{y}$ given by majorization. First, we denote by $\mathbf{x}^{\downarrow} = \left(x_1^{\downarrow}, ..., x_N^{\downarrow}\right)$ the vector $\mathbf{x}$ whose elements are reordered in decreasing order.

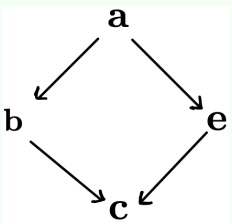
Proposition 9.2.1. *A vector* $\mathbf{x}$ *is majorized by another vector* $\mathbf{y}$*, i.e.* $\mathbf{x} \prec \mathbf{y}$*, if and only if,*
(i) For all $k = \{1, ..., N\}$*,*

$$\sum_{j=1}^{k} x_j^{\downarrow} \leq \sum_{j=1}^{k} y_j^{\downarrow}. \tag{9.10}$$

(ii) We have,

$$\sum_{j=1}^{N} x_j = \sum_{j=1}^{N} y_j. \tag{9.11}$$

Example 9.2.2. *Consider the five vectors* $\mathbf{a} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$, $\mathbf{b} = \left(\frac{1}{6}, \frac{2}{3}, \frac{1}{6}\right)$, $\mathbf{c} = (1,0,0)$, $\mathbf{d} = \left(\frac{1}{2}, \frac{1}{2}, 1\right)$ *and* $\mathbf{e} = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$*. Then* $\mathbf{a}^{\downarrow} = \mathbf{a}$, $\mathbf{b}^{\downarrow} = \left(\frac{2}{3}, \frac{1}{6}, \frac{1}{6}\right)$, $\mathbf{c}^{\downarrow} = \mathbf{c}$, $\mathbf{d}^{\downarrow} = \left(1, \frac{1}{2}, \frac{1}{2}\right)$ *and* $\mathbf{e}^{\downarrow} = \mathbf{e}$*. Using Proposition (9.2.1) it is not difficult to check that* $\mathbf{a} \prec \mathbf{b}$, $\mathbf{b} \prec \mathbf{c}$, $\mathbf{e} \prec \mathbf{c}$ *and* $\mathbf{a} \prec \mathbf{e}$*. Note that* $\mathbf{e}$ *is incompatible with* $\mathbf{b}$ *and that* $\mathbf{d}$ *does not obey (ii).*



More generally, $\left(\frac{1}{N},...,\frac{1}{N}\right) \prec \mathbf{p} \prec (1,0,...,0)$ for all probability distributions $\mathbf{p}$. Also a very useful fact is that for any convex function f we have that $f(\mathbf{x}) \leq f(\mathbf{y})$ if and only if $\mathbf{x} \prec \mathbf{y}$.

Theorem 9.2.3 (Nielsen's Majorization Theorem)**.** *Let $|\phi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ and define the specific set of eigenvalues $\lambda(\phi) := \left\{\text{eigenvalues of } \rho_A = \mathrm{Tr}_B\left(|\phi\rangle\langle\phi|_{AB}\right)\right\}$. Then, $|\phi\rangle_{AB} \xrightarrow{LOCC} |\psi\rangle_{AB}$ if and only if $\lambda(\phi) \prec \lambda(\psi)$.*

In other words, the partial trace induced by LOCC on the set of pure bipartite states coincides with partial order from Majorization Theory for the marginal spectra $\lambda(\phi)$.

Proposition 9.2.4. *The maximally entangled state on $\dim(\mathcal{H}_A) = \dim(\mathcal{H}_B) = d$,*

$$|\Omega\rangle_{AB} = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle_{AB}, \tag{9.12}$$

is majorized by all other states, and thus $|\Omega\rangle_{AB} \xrightarrow{LOCC} |\phi\rangle_{AB}$ for any $|\phi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$.

Proof. Since $\lambda(\Omega) = \left(\frac{1}{d},...,\frac{1}{d}\right)$ and $\lambda(\Omega) \prec \lambda(\phi)$ for all ϕ, by Theorem (9.2.3) we have that $|\Omega\rangle_{AB} \xrightarrow{\text{LOCC}} |\phi\rangle$ for any $|\phi\rangle$.

□

9.3 Asymptotic Conversion Rates and the Measure of Entanglement

As described in the title, in this section we present the asymptotic conversion rate (for a reversible interconversion) along with the definition of the measure of entanglement.

Proposition 9.3.1 (Asymptotic conversion rates)**.** *Given $N \to \infty$ copies of any state $|\phi\rangle_{AB}$, there exists a reversible interconversion,*

$$|\phi\rangle\langle\phi|^{\otimes N} \xleftrightarrow{LOCC} \left|\psi^-\right\rangle\left\langle\psi^-\right|^{\otimes S(\phi_A)N}, \tag{9.13}$$

where ϕ_A is the reduced state on A of a single copy $|\phi\rangle_{AB}$ (i.e. $\rho_{\phi_A} = \mathrm{Tr}_B\left(|\phi\rangle\langle\phi|_{AB}\right)$) and $S(\rho) = -\mathrm{Tr}(\rho\log(\rho))$.

Proof. For simplicity, we will restrict to $|\phi\rangle$ being a qubit state - the generalization to arbitrary dimension does not shed any new light on the matter. We will first look at the distillation protocol $|\phi\rangle^{\otimes N} \xrightarrow{\text{LOCC}} |\psi\rangle^{\otimes k}$ and determine k as a function of N (as N gets large).The argument amounts to a typicality argument, and we identify the typical states obtained under a particular measurement. These typical states span a typical subspace over which the probability becomes more concentrated as $N \to \infty$.

• We start with the total state $|\phi\rangle^{\otimes N}$, where a single copy $|\phi\rangle$ has the Schmidt decomposition $|\phi\rangle_{AB} = \sqrt{1-\lambda}|00\rangle_{AB} + \sqrt{\lambda}|11\rangle_{AB}$. We may expand the total state in the Schmidt basis to obtain,

$$\begin{aligned}
|\phi\rangle^{\otimes N} = (1-\lambda)^{\frac{N}{2}} |0...0\rangle_A \otimes |0...0\rangle_B \\
+ (1-\lambda)^{\frac{N-1}{2}} \lambda^{\frac{1}{2}} \left(|10...0\rangle + |010...0\rangle + ... + |0...1\rangle\right)_A \otimes \left(|10...0\rangle + |010...0\rangle + ... + |0...1\rangle\right)_B \\
+ ... + (1-\lambda)^{\frac{N-x}{2}} \lambda^{\frac{x}{2}} \sum_{\pi\,=\text{permutation}} \underbrace{|\pi(11...10...0)\rangle_A}_{x \text{ ones}} \otimes \underbrace{|\pi(11...10...0)\rangle_B}_{x \text{ ones}} + ...
\end{aligned}$$

We can write this more compactly in the form,

$$|\phi\rangle^{\otimes N} = \sum_k \lambda^{\frac{k}{2}}(1-\lambda)^{\frac{N-k}{2}} \sum_\pi |\pi(11...10...0)\rangle_A \otimes |\pi(11...10...0)\rangle_B \,, \tag{9.14}$$

where each permutation acts to shuffle over all possible arrangements of k ones and $(N-k)$ zeroes. There are $\binom{N}{k}$ such terms for fixed k, which span a charge-k subspace.
We now construct a protocol that achieves the distillation of $S(\phi_A)N$ singlets with probability $p \to 1$ as $N \to \infty$. Thus, for finite N it is technically a probabilistic LOCC transformation, but which becomes deterministic in the asymptotic regime.

- (Local Measurement at A) We define a charge observable $\widehat{X}_A$ that locally counts the number of ones at A. Specifically, we have $\widehat{X}_A |i_1...i_N\rangle = x\,|i_1...i_N\rangle$ with $x = \sum_{k=1}^N i_k$. Alice then performs a local measurement of $\widehat{X}_A$ and obtains some outcome $0 \leq x \leq N$. This means that the total state is collapsed to,

$$|\phi\rangle^{\otimes N} \longrightarrow \sum_\pi |\pi(11...10...0)\rangle_A \otimes |\pi(11...10...0)\rangle_B \,, \tag{9.15}$$

which is a uniform superposition of basis states in the charge-x subspaces for A and B. This outcome happens with the binomial probability $p_x = \binom{N}{x}(1-\lambda)^{N-x}\lambda^x$. Asymptotically, this binomial distribution tends to a Gaussian distribution, peaked at the value $x = \lambda N \pm \sqrt{N(1-\lambda)\lambda}$. As $N \to \infty$ this peak becomes arbitrarily sharp and so we have that the measurement outcomes cluster around $x = \lambda N$. We may thus consider the $x = \lambda N$ outcome, where any x near this will behave essentially the same in the limit. Hence,

$$|\phi\rangle^{\otimes N} \longrightarrow |x = \lambda N\rangle \equiv \frac{1}{\sqrt{\binom{N}{\lambda N}}} \sum_\pi \underbrace{|\pi(11...10...0)\rangle_A}_{\lambda N \text{ ones}} \otimes \underbrace{|\pi(11...10...0)\rangle_B}_{\lambda N \text{ ones}}, \tag{9.16}$$

is the expected outcome state, being a uniform superposition in a $\binom{N}{\lambda N}$-dimensional subspace. Now use the Stirling approximation, where for large m we have that $m! \approx m^m$. Thus it is not difficult to verify that, $\log_2\binom{N}{\lambda N} \approx N\big(-\lambda\log_2(\lambda) - (1-\lambda)\log_2(1-\lambda)\big) = S(\phi_A)N$. Therefore, the measurement prepares from $|\phi\rangle^{\otimes N}$ a uniform superposition of $\binom{N}{\lambda N} \approx 2^{S(\phi_A)N}$ orthogonal product states in the large N limit.

- (Classical Communication) Alice communicates to Bob that she has the $x = \lambda N$ outcome in her measurement.

- (Local Unitaries) Both parties now know what pure entangled state they possess, namely the state $|x = \lambda N\rangle$. This has $2^{S(\phi_A)N}$ orthogonal product vectors, and so wa can tidy up the entanglement onto exactly $S(\rho_A)N$ maximally entangles qubits through the following local unitary, performed at A:

$$|11...100...0\rangle_A \mapsto |00...0\rangle_{A,e} \otimes \underbrace{|0...0\rangle_{A,j}}_{\text{junk qubits}} \tag{9.17}$$

$$|1...1010...0\rangle_A \mapsto |00...1\rangle_{A,e} \otimes |0...0\rangle_{A,j} \tag{9.18}$$

This process continues until we reach,

$$|00...011...1\rangle_A \mapsto |11...1\rangle_{A,e} \otimes |0...0\rangle_{A,j} \tag{9.19}$$

where for the states $|00...0\rangle_{A,e}, |00...1\rangle_{A,e}, ..., |11...1\rangle_{A,e}$ we require $\sim \log_2\binom{N}{\lambda N}$ qubits to label them. We also have an identical transformation performed by Bob. This is effectively a compression of the state which creates the state,

$$\left(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B\right)^{S(\phi_A)N} \otimes |0...0\rangle_{A,j} \otimes |0...0\rangle_{B,j}\,, \tag{9.20}$$

which upon discarding the junk qubits gives us $S(\phi_A)N$ maximally entangled qubits between Alice and Bob. As we know, these may be reversibly converted into singlets under LOCC, which completes the distillation protocol.

• The reverse procedure is possible under LOCC, where $\left|\psi^-\right\rangle_{AB}^{\otimes S(\phi_A)N} \xrightarrow{\text{LOCC}} |\phi\rangle^{\otimes N}$ (i.e. by running the protocol backwards). The idea is then to dilute $S(\phi_A)N$ singlets into N copies of $|\phi\rangle_{AB}$. The protocol relies on *teleportation* and proceeds as follows:

- (Local Operations) Alice creates N copies of $|\phi\rangle_{A_1A_2}$, entirely locally.

- (LOCC) Alice then compresses the state on the N qubits forming the A_2 subsystems to $NS(\rho_A)$ qubits and teleports these $NS(\rho_A)$ qubits perfectly to B, which consumes exactly that number of singlets.

- (Local Operations) Bob then decompresses to N qubits and we obtain the state $|\phi\rangle_{AB}^{\otimes N}$ as required.

Thus, $|\phi\rangle^{\otimes N} \xleftarrow{\text{LOCC}} \left|\psi^-\right\rangle^{\otimes NS(\phi_A)}$ with probability approaching 1 as $N \to \infty$, as required.

□

We now present the **Measure of Entanglement** E. The requirements for E are the following:

- An **LOCC-monotone** measure,

$$|\phi_1\rangle \xrightarrow{\text{LOCC}} |\phi_2\rangle \Longrightarrow E\big(|\phi_1\rangle\big) \geq E\big(|\phi_2\rangle\big). \tag{9.21}$$

- Take the singlet (i.e. maximally entangled qubit) as the **gold standard**, i.e. $E\big(\left|\psi^-\right\rangle\big) = 1$ - Normalization.

- An **extensivity** property, i.e. $E\left(|\phi\rangle^{\otimes k}\right) = kE\big(|\phi\rangle\big)$.

Proposition 9.3.2. *There is a unique (normalized) measure of pure bipartite entanglement that satisfies monotonicity and extensivity - the Von Neumann entropy $S(\rho)$.*

Proof. From Proposition (9.3.1), we know that we can perform the transformation $|\phi\rangle^{\otimes N} \xleftarrow{\text{LOCC}} \left|\psi^-\right\rangle^{\otimes NS(\phi_A)}$ via LOCC. From the LOCC-monotonicity (9.21), we have on the one hand,

$$E\left(|\phi\rangle^{\otimes N}\right) \leq E\left(\left|\psi^-\right\rangle^{\otimes NS(\phi_A)}\right). \tag{9.22}$$

On the other hand we also have,

$$E\left(\left|\psi^-\right\rangle^{\otimes NS(\phi_A)}\right) \leq E\left(|\phi\rangle^{\otimes N}\right). \tag{9.23}$$

Thus, we conclude that $E\left(|\phi\rangle^{\otimes N}\right) = E\left(\left|\psi^-\right\rangle^{\otimes NS(\phi_A)}\right)$. Furthermore, we now use the extensivity property and the gold-standard to find,

$$NE\big(|\phi\rangle\big) = NS(\phi_A)\underbrace{E\big(\left|\psi^-\right\rangle\big)}_{=1} \Longrightarrow E\big(|\phi\rangle\big) = S(\phi_A), \tag{9.24}$$

which is uniquely determined by the requirements. □

9.4 Mixed State Entanglement - Entanglement Witnesses

For mixed states, entanglement entropy is no longer a valid measure since, for example, the states $\frac{1}{4}\mathbb{I}$ and $|\phi^+\rangle$ have the same entropy. Also, for the state $\rho_{AB} = \frac{\mathbb{I}_{AB}}{d_{AB}} = \frac{\mathbb{I}_A \otimes \mathbb{I}_B}{d_A d_B}$, this is a maximally mixed state which is unentangled but has maximum entanglement entropy, e.g. $\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \frac{\mathbb{I}_A}{d_A}$ with entropy $E = \log(d_A)$. We thus need to find a different construction for the entanglement entropy, which will be given by entanglement witnesses.

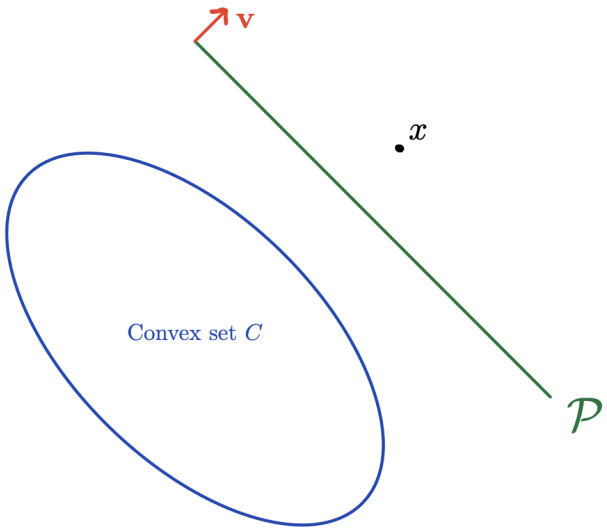
The **set of bipartite separable quantum states** is denoted by,

$$D_{\text{Sep}} = \left\{ \rho_{AB} : \rho_{AB} = \sum_k p_k \sigma_{k_A} \otimes \sigma_{k_B} \right\}, \tag{9.25}$$

such that p_k is a classical probability distribution, and where $\sigma_{k_A} \otimes \sigma_{k_B}$ are classically correlated but not quantum correlated. Furthermore, D_{Sep} forms a convex set (i.e. $p\rho + (1-p)\sigma$ for $\rho, \sigma \in D_{\text{Sep}}$) and is contained in the larger **set of all bipartite quantum states** D.

Proposition 9.4.1 (Hahn-Banach Corollary)**.** *Given a closed convex set C and a point x outside of C, there exists a hyperplane $\mathcal{P}$ that separates x from C.*

Recall, a hyperplane $\mathcal{P}$ is defined by some vector $\mathbf{v}$ orthogonal to the plane such that $\mathcal{P} = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle = 0\}$. One side of $\mathcal{P}$ will be the points $\{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle > 0\}$ and the other $\{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle < 0\}$.

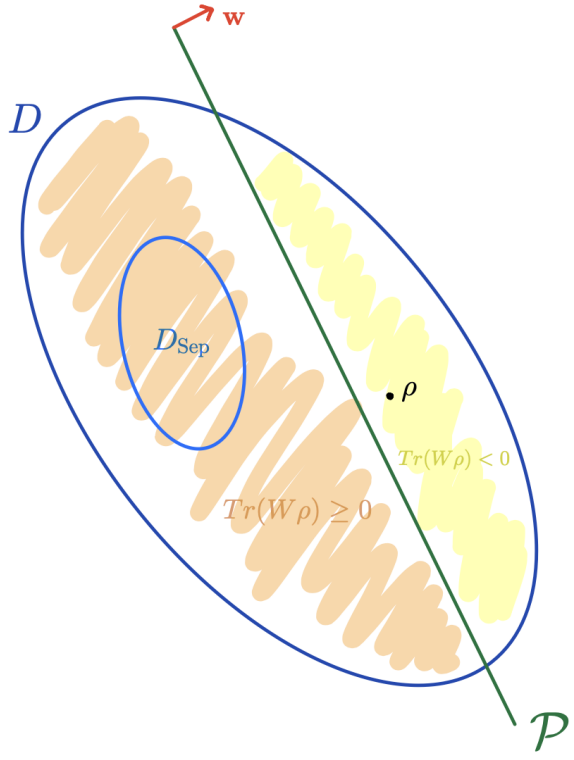


As D_{Sep} is itself a convex set, then the Hahn-Banach Corollary applies to it. The relevant vector space in this case will be all the states and observables that sit in a space of $d^2 \times d^2$ Hermitian matrices. The corresponding inner product is given by the Hilbert Schmidt inner product $\langle X, Y \rangle := \mathrm{Tr}\left(X^\dagger Y\right) = \langle \mathrm{vec}(X) | \mathrm{vec}(Y) \rangle$.

Then, from the Hahn-Banach Corollary, for any state $\rho \notin D_{\text{Sep}}$ (i.e. any entangled state) there exists some plane $\mathcal{P}$ in the space of Hermitian operators that separates ρ from D_{Sep}. This plane is defined by some matrix W such that $\mathrm{Tr}(W\rho_{AB}) \geq 0$ for all separable states but also some entangled states. Turning this idea around, if $\mathrm{Tr}(W\rho_{AB}) < 0$ we know that ρ_{AB} is entangled. In our case, W is called the **Entanglement Witness**.

Definition 9.4.2. *An **entanglement witness** W is a Hermitian observable such that $\mathrm{Tr}(W\sigma_{AB}) \geq 0$ for all $\sigma_{AB} \in D_{Sep}$.*

The picture to have in mind is the following, where the orange part represents $\mathrm{Tr}(W\rho) \geq 0$ and the yellow part is for $\mathrm{Tr}(W\rho) < 0$.



As briefly mentioned before, we reiterate that if we can find such a W (and we can find such a W for any entangled states) satisfying $\mathrm{Tr}(W\rho_{AB}) < 0$ we know that the state is entangled! However, more generally, if we are told that W is an entangled witness such that $\mathrm{Tr}(W\rho_{AB}) \geq 0$ then ρ_{AB} could be entangled or unentangled.

9.5 The Peres-Horodecki Criterion

Recall first the Choi-Jamiolkowski Representation (5.8.5) where any channel can be associated with the positive operator,

$$J(\phi) := \phi \otimes \mathbb{I}\big(|\mathrm{vec}(\mathbb{I})\rangle\langle\mathrm{vec}(\mathbb{I})|\big). \tag{9.26}$$

It turns out that the converse is also true, in the sense that we may associate the positive operators J (Choi-Jamiolkowski Representation) with **superoperators** ϕ,

$$\phi = \mathrm{Tr}_2\big((\mathbb{I} \otimes X^T)J\big). \tag{9.27}$$

We can then use this equivalence to transform an entanglement witness in the form of a Hermitian operator into, what is essentially, a witness in the form of a superoperator. More precisely, we may convert the statement that we can always find a positive operator W such that *if* $\mathrm{Tr}(W\rho_{AB}) < 0$ *then* ρ_{AB} *is entangled*, into a statement where we are able to find some positive operators.

Proposition 9.5.1. *A state σ_{AB} is entangled if and only if there exists a positive (not necessarily completely positive) superoperator Φ such that $(\Phi \otimes \mathbb{I})(\sigma_{AB})$ has a negative eigenvalue.*

Proof. • ($\Longrightarrow$) If a state σ_{AB} is entangled then we know an entanglement witness W exists that detects this entanglement. In other words, $\mathrm{Tr}(W\sigma_{AB}) < 0$. Now swap W (a Hermitian operator) for a superoperator

Φ such that $W = J(\Phi) = \Phi \otimes \mathbb{I}(|\Omega\rangle\langle\Omega|)$. We have that,

$$\mathrm{Tr}\big((\Phi \otimes \mathbb{I})(|\Omega\rangle\langle\Omega|)\sigma_{AB}\big) < 0. \tag{9.28}$$

However, we can always do a Kraus decomposition on the superoperator Φ so that $\Phi(X) = \sum_k A_k X B_k^\dagger$, and make use of the cyclicity of the trace to deduce that,

$$\mathrm{Tr}\big(|\Omega\rangle\langle\Omega|(\Phi^* \otimes \mathbb{I})(\sigma_{AB})\big) < 0 \Longrightarrow \langle\Omega|(\Phi^* \otimes \mathbb{I})(\sigma_{AB})|\Omega\rangle < 0, \tag{9.29}$$

where $\Phi^*(X) = \sum_k B_k^\dagger X A_k$ is also a positive superoperator. Therefore, $\Phi^* \otimes \mathbb{I}(\sigma_{AB})$ cannot be positive and must have a negative eigenvalue.

• ($\Longleftarrow$) Conversely, suppose there is a positive superoperator Φ such that $\Phi \otimes \mathbb{I}(\sigma_{AB})$ has a negative eigenvalue. We establish the result as follows: suppose σ_{AB} is separable. Therefore it can be written as,

$$\sigma_{AB} = \sum_k \tau_k \otimes \rho_k, \tag{9.30}$$

for quantum states $\{\tau_k\}$ for A and quantum states $\{\rho_k\}$ for B, and some probability distribution (p_k). Thus we have that,

$$\Phi \otimes \mathbb{I}(\sigma_{AB}) = \sum_k p_k \Phi(\tau_k) \otimes \rho_k. \tag{9.31}$$

However, since Φ is a positive superoperator, then $\Phi(\tau_k)$ has all its eigenvalues non-negative. Therefore, $\Phi(\tau_k) \otimes \rho_k$ also has non-negative eigenvalues, and since convex mixtures cannot generate negative eigenvalues we must have that $\Phi \otimes \mathbb{I}(\sigma_{AB})$ must have positive eigenvalues if it is separable. Therefore, if it has any negative eigenvalues then it is an entangled state.

□

Remark 9.5.2. *Note that we know that Φ cannot be completely positive, but only positive, since a completely positive Φ would generate a positive W which could not detect entanglement.*

Example 9.5.3 (Positive but not completely positive - the partial transpose)**.** *The partial transpose is positive but not completely positive. Indeed,*

$$\big(|\phi\rangle\langle\phi| \otimes |\psi\rangle\langle\psi|\big)^{T_B} = |\phi\rangle\langle\phi| \otimes \big(|\psi\rangle\langle\psi|\big)^T, \tag{9.32}$$

is a positive state. However, $\big(|\phi^+\rangle\langle\phi^+|_{AB}\big)^{T_B} = \mathrm{SWAP}$ has negative eigenvalues.

Remark 9.5.4. *Note that we may generalize this argument (9.32) to separable states. We have,*

$$\left(\sum_i p_i \rho_i^A \otimes \sigma_i^B\right)^{T_B} = \sum_i p_i \rho_i \otimes \sigma_i^T, \tag{9.33}$$

is still a positive state and does not have negative eigenvalues.

Proposition 9.5.5. *For qubits, we have that every positive superoperator Φ may be written as,*

$$\Phi(X) = \mathcal{E}_1(X) + \mathcal{E}_2\big(X^T\big), \tag{9.34}$$

where $\mathcal{E}_1$ and $\mathcal{E}_2$ are completely positive maps and X^T is the transpose of X.

This result basically tells us that the central difference between single qubit positive and completely positive maps is given by the transpose map (the transpose map is the core example of a map that is positive but not completely positive). We now have the following central result that solves two-qubit (or more) entanglement.

Proposition 9.5.6 (The Peres-Horodecki Criterion)**.** *A quantum state ρ_{AB} is entangled if (if and only if, for 2-qubit states) $(T \otimes \mathbb{I})\rho_{AB}$ has a negative eigenvalue.*

Proof. From Proposition (9.5.5), for qubits we know that $\Phi(X) = \mathcal{E}_1(X) + \mathcal{E}_2\big(X^T\big)$ with $\mathcal{E}_1$ and $\mathcal{E}_2$ completely positive maps. Thus, ρ_{AB} is entangled if and only if there exists a superoperator Φ such that $\Phi \otimes \mathbb{I}(\rho_{AB}) \ngeq 0$ which is true if and only if $(T \otimes \mathbb{I})(\rho_{AB}) \ngeq 0$.

□

Remark 9.5.7. *Note that the operator $(T \otimes \mathbb{I}) \equiv T^A$ is the partial transpose on A.*

Finally remark that we can define the **negativity** of a quantum state as a **measure** of mixed state entanglement,

$$N(\rho_{AB}) = \frac{1}{2} \sum_{\substack{k \\ s.t.\ \lambda_k < 0}} |\lambda_k|, \tag{9.35}$$

i.e. it is the sum of the absolute value of the negative eigenvalues of ρ_{AB}.